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**Roots, templates and the interaction of their meaning: the
case in Modern Hebrew**

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case in Modern Hebrew**

**by
Scott Raymond Spicer**

Dissertation

Presented to the Faculty of the Graduate School of
The University of Texas at Austin
in Partial Fulfillment
of the Requirements
for the Degree of

Doctor of Philosophy

**The University of Texas at Austin
August 2024**

Dedication

To Gail ‘Yaffah’ Korinow ל"י, whose love for the Hebrew language inspired me at a young age, and brought me to where I am today.

Acknowledgments

I struggle even to process the number of people who had a hand in making this dissertation a reality. Anyone who has written a dissertation, or has even been in the orbit of someone writing a dissertation, knows that it may be the work of one person, but it is really the product of the labor of an entire network. I want to make sure that I recognize that network properly, and apologize if I miss someone; after all, it was truly massive.

First, all the credit goes to my advisor, John Beavers, who put up with me for seven years as I pushed my way through this work. I think I was in a unique position among my cohort of working with my first-year advisor throughout my entire graduate school career, and I can say that I would not have had it any other way. I was in fact encouraged to apply to UT in the first place explicitly so I could work with John (thanks to Brady Clark for that encouragement), and, in retrospect, feel very stupid for having considered any other institution. His meticulous advisory work over seven years and scores of projects has helped me grow into a more thoughtful and capable scholar. More importantly, I am so appreciative that John could also be a personal friend as well, and will look fondly back on not just the time we spent together professionally, but socially as well. Wherever I end up after this, I know that this relationship will persist.

I also need to give a nod to all of the friends I made along the way, and in particular, Michael Everdell, Kristie Denlinger and Seyeon Park. There is something to be said about being in graduate school with a group of people who really get what you are going through, and I feel so grateful to have had people that I could not only talk to about work and research, but also who became very close personal friends. In particular, I need to give a nod to Mike: when we are together, in one moment, we are talking about fine points of the syntax/semantics interface, and in the next, making stupid jokes that alienate the rest of the room. I am always happy to be the

peanut gallery backchannel in any Zoom call together, or laugh at a stupid abstract together, and am more than satisfied to be able to call him not just a colleague but a good, close friend.

My committee, Itamar Kastner, Ashwini Deo, Hans Kamp, and Steve Weschler, also deserve my accolades, for giving me such good feedback on this work that I will take with me going forward, and for really challenging and pushing me to improve what I consider to be a first push into a long research program. I also want to recognize the rest of the SynSem students and faculty, who for seven years were present weekly to hear my progress. I want to recognize the contribution of Mira Ariel and Roey Gafter; before coming to UT, they were the first ones to take a chance on me as a young scholar, and I like to think it paid dividends for them. Additionally, I want to thank Eyal Rotbart for being such a good and consistent informant on Hebrew.

Finally, I want to give arguably the largest and most important recognition to my little family of Jordana, Merlin, and Mushy, but in particular, Jordana. Jordana has been there with since before the start, and she has been through all of the highs and the lows of the whole process. She kept me grounded throughout the whole process, and was always there to celebrate my victories and mourn my defeats. I cannot even imagine what graduate school would have been like without her there by my side, and am so grateful that we have chosen to stick together. The last ten years have been great, and I am looking forward to ten more, whatever they may bring.

Abstract

Roots, templates and the interaction of their meaning: the case in Modern Hebrew

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The University of Texas at Austin, 2024

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This dissertation analyzes the meaning inherent in the Modern Hebrew morphological templates, the language's means for sorting verbs into classes, and how that meaning interacts with the meaning of the root, or the means by which a verb represents its underlying idiosyncratic concept. Modern theoretical analyses of the Hebrew templates treat them as primarily argument structural objects which determine how many arguments, subject and object, the verb can take and how those arguments are expressed. My dissertation argues against this view by investigating sets of verbs whose templates are not clearly correlated with argument structure, i.e. those verbs where changing the template does not clearly change the argument structure.

Here, I propose that the template system of Hebrew is guided by a principle where contrasting templates for a single root reflect relative differences in the truth conditional content of each verb. Namely, for a single root occurring in two templates, the template which is more marked, i.e. has more morphological complexity, will generate a verb with strictly stronger truth conditions than one in the less marked template. I formalize this as a constraint on lexicalization called the Verb Complexity Mapping Principle, ensuring that the verb in the more marked template has a set of

lexical entailments which is a strict superset of those of the verb in the less marked template. I investigate three sets of templatic alternations, a roaming motion alternation with HITPA'EL, the locative alternation with HIF'IL, and the causative-inchoative alternation with both HITPA'EL and NIF'AL, and demonstrate that in all three cases, the Verb Complexity Mapping Principle both holds, such that the alternant in the more marked template has a stronger meaning, and can explain the specific templatic choice for both alternants better than an equivalent syntactic analysis. Thus, this dissertation represents an extension of lexical entailment approaches to verb meaning and argument marking (Dowty 1989, 1991; Ackerman & Moore 2001; Beavers 2010; Jerro 2023) into the realm of lexical verb class, expanding the range of empirical phenomena which it can account for.

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Chapter 1: Introduction

1.1 The relation between form and meaning

How might one describe what happens to someone who gets very embarrassed? English has one main lexical option to express this process of a person's cheeks reddening in response to embarrassment; namely, that person might *blush*, as in (1).

- (1) John's mom embarrassed him, so he blushed.

Other verbs may be appropriate in this context, such as *flush*, *redden*, or *burn*, but all of these verbs constitute distinct lexical options, predicated off of different stems or roots. The only relationship these options have to one another is a vague conceptual connection, but one so weak that the three are not even necessarily synonymous.

To contrast, Hebrew has two main and notionally synonymous options, *he'edim* 'CAUS.redden' and *hit'adem* 'REFL.redden', both of which can mean roughly 'turn red':¹

- (2) a. *ha='ima fel yosi hivix-a* *'ot-o*
the-mother POSS Yosi CAUS.embarrass-3SG.F.PST ACC-3SG.M
v=pana-v *he'edim-u*
and=cheek-PL.POSS.M CAUS.redden-3PL.PST
- b. *ha='ima fel yosi hivix-a* *'ot-o*
the-mother POSS Yosi CAUS.embarrass-3SG.F.PST ACC-3SG.M
v=pana-v *hit'adm-u*
and=cheek-PL.POSS.M REFL.redden-3PL.PST
- 'Yosi's mother embarrassed him and his cheeks turned red.' (constructed)

Unlike English, these alternatives are identical, save only for their morphological form, which are additionally related to one another. Namely, the two forms share

¹Hebrew also has *hismik* 'blush', but for expository purposes, I focus on the contrast between these two verbs of reddening, additionally compatible with situations of blushing.

a consonantal root $\sqrt{'DM}$, related to redness, and only differ in their morphological class, or template, which forms the verb. The question then arises of what, if anything, differentiate these two options for expressing a change in cheek color.

Fundamentally, the distinction between these two verbs is predicated on their morphological class membership. On a traditional view (e.g. Ben-Ze'ev 1818), Hebrew mobilizes Semitic root-template morphology, where words are formed through non-concatenative combination of a root and a template. A root is an unpronounceable string of two, three or four consonants. These roots combine with templates, pronounceable frames with open slots for the consonants of the root, to create a word. For example, the root which forms both the verbs of reddening in (2), $\sqrt{'DM}$ can combine with several different templates, as in (3), to form a variety of words across categories.²

- (3) a. $\sqrt{'DM} + _ a _ o _ = 'adom$ 'red'
 b. $\sqrt{'DM} + hi _ _ i _ = he'edim$ 'redde'
 c. $\sqrt{'DM} + _ a _ a _ = 'adam$ 'man'
 d. $\sqrt{'DM} + _ a _ e _ et = 'adem$ 'rubella'

Intuitively, both the root and the template provide structural and semantic information to generate the word with its syntactic and semantic properties. For example, by virtue of the root $\sqrt{'DM}$, all of these words have some conceptual relationship to the concept of redness, either through denoting the state of redness itself in (3a), change in the state of redness in (3b), or a disease typically associated with the color red in (3d). Similarly, the templates provide both the category of the word, and can provide additional meaning. For example, in the root $\sqrt{CHV}$ related to yellowness, the template in (3b) generates another change of state verb in the state of yellowness *hichiv* 'yellow', and (3d) generates a different disease conventionally associated

²For a specifically non-concatenative formal implementation of this kind of system, see e.g. McCarthy (1979, 1986).

with the color yellow, *cahevet* ‘jaundice’. Thus, both the root and the template can contribute some kind of regular meaning to the words which they derive.

In the verbal domain, the set of templates is restricted to seven, which are obligatory in that every verb belongs to one of these seven (see Arad 2005, ch. 2), though not every root is realized in each of the seven below. I provide the seven templates, labeled by their traditional name, with the root $\sqrt{\text{PNH}}$, related to turning, one of the few roots which form verbs in all 7 templates:

- (4) a. PA’AL: *panah* – ‘turn (intrans.)’
 b. NIF’AL: *nifnah* – ‘turn (intrans.)’
 c. PI’EL: *pinah* – ‘vacate, evacuate’
 d. PU’AL: *punah* – ‘be vacated, evacuated’
 e. HIF’IL: *hifnah* – ‘turn (trans.)’
 f. HUF’AL: *hufnah* – ‘be turned’
 g. HITPA’EL: *hitpanah* – ‘turn around (refl.)’

Based just on the paradigm in (4), three main systematic relationships between the verbs in each of templates emerge. First, two templates, PU’AL and HUF’AL, generate passive alternants of the active templates PI’EL and HIF’IL respectively. Second, different templates can combine with the same root to create verbs with different idiosyncratic meanings. Here, PA’AL *panah* contrasts with PI’EL *pinah* to generate two verbs with separate and unrelated meanings, despite the fact that they are morphologically related to one another through their root. These two relationships, passivization and different idiosyncratic meanings, exist on a kind of spectrum, between the strictly syntactic on the side of passivization, to the strictly semantic on the side of idiosyncratic verb formation. This dissertation will put these two phenomena aside as independent phenomena and deal primarily with the third kind of relationship.³

³I do not make the claim here that the three phenomena are entirely independent, in that they

This third kind of relationship between templatic forms exists in between the two extremes of passivization, being a strictly syntactic process, and idiosyncratic verb formation, being a strictly semantic process. Namely, some pairs of templates preserve the idiosyncratic meaning between the two verbs which they form from a single root, but differ in terms of some fine-grained semantic or syntactic properties. Such is the case here of HIF'IL and HITPA'EL, the templates of *he'edim* and *hit'adem* from (2) respectively. With $\sqrt{\text{PNH}}$ in (4), both of these templates contrast with PA'AL *panah* to induce an argument structural effect with a corresponding semantic effect. In the case of HIF'IL, the template produces a causative alternant, with an additional argument and entailments of causation which the PA'AL form does not have:

- (5) a. *ha=raxev panah smola*
the=car turn[3SG.M.PST] rightward
'The car turned right.' PA'AL
- b. *Yosi hifnah et ha=raxev smola*
Yosi CAUS.turn[3SG.M.PST] ACC the=car rightward
'Yosi turned the car to the right.' HIF'IL (constructed)

Similarly, HITPA'EL produces a reflexive alternant, with the entailments of reflexivity not necessarily associated with the PA'AL form:

- (6) a. *ha=raxev panah smola*
the=car turn[3SG.M.PST] rightward
'The car turned right.' PA'AL
- b. *ha=raxev hitpanah (b=acm-o)*
the=car REFL.turn[3SG.M.PST] (in=REFL-3SG.M)
'The car turned around (by itself).' HITPA'EL (constructed)

emerge from completely different morphological, syntactic, or semantic systems. Rather, I consider the three as falling under the same system, that of the Hebrew templates and its morphosyntax, and focus on one here as an entry point to understanding the semantics of the system. I discuss how to extend this approach in chapter 6.

In both of these cases, the core idiosyncratic meaning does not change, in contrast to idiosyncratic verb formation, but the alternation is not strictly syntactic, in contrast with passivization, as there are additional entailments associated with the change in syntax, here a change in argument structure specifically.

In many ways, this sort of alternation reflects differences in event structure, with the morphological template paralleling the event structural template, which in turn determines the argument structure. In general, the morphological roots and templates found in Hebrew can be seen to correspond to the theoretical roots and templates found in theories of word structure and meaning. Specifically, event structural approaches to verb meaning (Dowty 1979; Levin & Rappaport Hovav 1995; Rappaport Hovav & Levin 1998; Ramchand 2008; Rappaport Hovav & Levin 2010; Alexiadou et al. 2015; Beavers & Koontz-Garboden 2020, among many others) presume that a verb divides into a semantic root and a semantic template. The template is a set of generic event types representing consistent, recurring notions across verbs, such as causation, change of state, or action. A root represents idiosyncratic, specific meaning, related to a certain state of affairs in the real world. The root combines with the template to create a full event structure and thus determine the verb's meaning. These event structures are grammatically relevant, such that the specific template and root of the verb determine the grammatical properties of the verb. Most commonly, the semantic event template determines the syntactic argument structure: how many arguments a verb realizes, in what syntactic positions, and by what grammatical means.

he'edim and *hit'adem* in (2) do not differ in their idiosyncratic meaning, as the options for blushing in English do, but based on their morphological differences they may differ in their event structural properties. Based just on the paradigms in (5) and (6), the two verbs would differ primarily in their argument structure, with the *he'edim*'s membership in HIF'IL imposing a transitive, causative argument structure on the verb, and the HITPA'EL imposing an intransitive, reflexive argument structure on the verb. This view corresponds with the traditional generative approach

to the Hebrew verb templates, which treat them as argument structural objects which determine the valency of the verbs which they form and the kinds of arguments they can take, with any semantic effects being an outgrowth of the syntax the template imposes (Doron 2003; Arad 2005; Reinhart & Siloni 2005; Laks 2007; Siloni 2011; Alexiadou & Doron 2012; Kastner 2020).

Yet *prima facie*, the difference between HIF'IL and HITPA'EL as seen in the paradigms in (2), (5) or (6) does not follow this kind of argument structural pattern, where the template uniquely determines the argument structure of the verb it forms. In fact, several different templatic contrasts are conditioned by differences in the meanings of the verbs which they form. Specifically, *he'edim* in (2) is not obligatorily causative, and in fact is able to be both causative and inchoative without any difference in morphology (a labile alternation such as that in English, Haspelmath 1993):

- (7) a. *pana-v* *he'edim-u* *me=m'vuxa*
cheek-POSS.M.PL CAUS.redden-3PL.PST *from=embarrassment*
‘His cheeks reddened from embarrassment.’
- b. *ha=kor* *he'edim* *et* *pana-v*
the=cold CAUS.redden[3SG.PST] ACC cheek-POSS.M.PL
‘The cold reddened his cheeks.’ (constructed)

A labile alternation is fully unanticipated on a one-to-one mapping approach to the Hebrew templates, where each template is a reflex of a single argument structure, and vice versa (such as Doron 2003, *contra* Kastner 2020).

However, in Hebrew, the verb class of degree achievements, change-of-state verbs which vary in their telicity (Hay et al. 1999; Kennedy & Levin 2008), undergo a labile alternation (Lev 2016), such as those in (8).

- (8) a. *hizkin* ‘age’
b. *he'efil* ‘darken’

- c. *hikri'ax* ‘bald’
- d. *hexmic* ‘become (more) sour’
- e. *hexkim* ‘wisen’
- f. *hiskil* ‘become (more) enlightened’ (Lev 2016, p. 66, Table 20)

This effect is unique to the semantic verb class of degree achievements, and other HIF’IL change of state verbs, such as accomplishments like ‘burn’ formed from the root $\sqrt{\text{B’R}}$, do not undergo a labile alternation:

- (9) a. *ha=ya’ar ba’ar-a*
the=forest burn-3SG.F.PST
‘The forest burned.’ PA’AL
- b. *yosi hiv’ir et ha=ya’ar*
Yosi CAUS.burn[3SG.M.PST] ET the=forest
‘Yosi burned (down) the forest.’ HIF’IL
- c. *#ha=ya’ar hiv’ir-a*
the=forest CAUS.burn-3SG.F.PST
(intended) ‘The forest burned.’ HIF’IL (constructed)

Similarly, although HITPA’EL *hitpaneh* in (6) is reflexive, the alternation also constrains the direction of the turning. Where PA’AL *panah* is compatible with an explicit direction, HITPA’EL *hitpaneh* is not:

- (10) a. *ha=rexev panah smola*
the=car turn[3SG.M.PST] rightward
‘The car turned right.’ PA’AL
- b. *#ha=rexev hitpanah smola (b=acm-o)*
the=car REFL.turn[3SG.M.PST] rightward (in=REFL-3SG.M)
‘The car turned around right (by itself).’ HITPA’EL (constructed)

hitpanah lexically specifies for the direction the turning is oriented towards, i.e. to the opposite direction, while *panah* is not, and thus can be specified with a directional

expression like *smola* ‘rightward’. Thus, the HITPA’EL template is not just generating a specific argument structural effect, namely that of reflexivity, but is also producing a fine semantic effect related to the specific verb which the template generates.

1.2 The dissertation, in short

Based on the contrasts in (7) and (10), it is not clear what exactly the boundaries are between the morphology of the Hebrew verb *qua* its template, the argument structure of the verb *qua* the restrictions on the arguments and types of arguments they can take, and the fine grained meaning associated with the verb. For one, a given template does not have a consistent semantics, in that there are a fixed set of lexical entailments associated with it. As in (10), HITPA’EL as a template can have reflexive semantics in some cases, but is additionally able to interact with the specific locative semantics associated with the turning root $\sqrt{\text{PNH}}$. Additionally, changes in templates are able to generate solely semantic differences between verbs based on the same root, without any necessary difference in the argument structure or syntax of the verb; this is the case of MCM verbs such as the contrast between (4a) and (4c). On top of this, a verb is able to shift in argument structure and/or meaning without an actual change in template, like the labile alternation with HIF’IL in (7). Given this wide range of behaviors the Hebrew templates have, both related to the argument structure and the semantics, what are the principles which guide how a Hebrew root’s behavior changes when it shifts templates?

This dissertation will probe the extent to which differences in template reflect differences in event structure, not just by how their arguments differ, but also by fine-grained semantic differences which characterize lexical classes of verbs. Namely, I ask, given a verb, what elements of its meaning can determine its template (the grammatically relevant ontological categories, Rappaport Hovav 2017), and further, how do those meanings correspond to the template which it appears in? Here, I propose that the template system of Hebrew is guided by a principle where contrasting

templates for a single root reflect relative differences in the truth conditional content of each verb, predicated on a contrast in markedness between the templates of the two verbs. Namely, I determine a difference in markedness through pairwise comparisons of templatic forms, based on a combination of phonological and morphological complexity, and frequency in corpus. For a single root occurring in two templates, the template which is more marked, i.e. has more complexity, will generate a verb with strictly stronger truth conditions than one in the less marked template.

I formalize this as a constraint of lexicalization called the Verb Complexity Mapping Principle:

- (11) **Verb Complexity Mapping Principle (informal):** For any two morphologically-realized lexical verb classes, a verb realized in the more marked class will have truth conditions which are a strict superset of a morphologically-related verb realized in the less marked class.

The fundamental correlation being posited here is one between complexity of form and strength of meaning, such that a more complex form will have a more contentful meaning. Here, I do such in the semantics, focusing on how complexity informs the entailed content of a given word, not necessarily how it is used to convey a given message. This approach came up independently starting with Dowty (1989) and the lexical entailment approaches which spawned from that (Dowty 1991; Ackerman & Moore 2001; Beavers 2006, 2010; Jerro 2023, among others). I focus in this dissertation on a set of cases where the contrast in form specifically weighs on the semantic content of the verb, and so I situate myself within that lexical entailment relationship. As such, the Verb Complexity Mapping Principle will constitute a parallel principle for other similar principles constraining verb meaning, such as the Morphosyntactic Alignment Principle from Beavers (2010).

However, in many ways, the Verb Complexity Mapping Principle is a semantic correlate to the pragmatic, competition-based approaches which posit a connection

between formal complexity and informativity. As applied to lexical forms, as I do here, the competition approach falls under the banner of ‘lexical pragmatics’ (e.g. Blutner 1998, 2000, 2006). Under this approach, the choice of word in a given context to be subject to a tension between communicative principles prioritizing efficiency and informativity. More specifically, a simpler word can be ‘blocked’ from conveying a more specific message if a more complex word already exists which can equally convey that message. Here, the effect the VCMP predicts is monotonic strengthening, not necessarily blocking. Though one might expect to see blocking in this domain as in others, it is not always the case (see e.g. Beavers & Nishida 2009, for Spanish), and I will discuss this further below. Just like the VCMP, this sort of approach relies on a unification between formal markedness and semantic markedness where one predicates the other (Horn 1984). The notion of informativity which is mobilized under the lexical pragmatics approach can in principle subsume the lexical entailment approaches by providing a more expansive notion of strength of meaning. I will return to how the two approaches might be related to one another in Chapter 6.

Already, this sort of approach has been applied specifically for Hebrew to explain differences in choice of template in Ahdout & Kastner (2020) and Ahdout (2021). Under this kind of approach, the differences in templatic form derive from an informativity difference, where more complex forms convey a more informative message. In this case, given a choice between two structures *qua* verb forms, a verb in a more complex template is preferred in a less complex template when the speaker conveys a message with more information (cf. Martin & Schäfer 2014, for anticausative marking in French). Two different syntactic structures (each determining a template) compete to realize a given message which could be assigned, in principle, to either. The more complex structure is assigned to the more informative message, which in turn corresponds to the verb in the more complex template being used to convey the more informative message. In this case, specific syntactic structures are competing, and the complexity of the syntax determines the complexity of the verb form, but in both cases, forms compete based on their complexity to determine the strength of

the message it can convey.

This kind of proposal, which centers the meaning of the verb, stands in direct contrast to the construal of the Hebrew templates as primarily syntactic objects, which determine the argument structure of the verbs which they generate, and thus represent differences in the syntactic structure of those verbs (Doron 2003; Arad 2005; Laks 2007; Kastner 2020). I will show through a set of case studies that there are several templatic contrasts which cannot be characterized just in terms of the syntactic differences between the verbs, and instead are better characterized by how the two verbs differ in their meaning. In turn, by focusing on consistent sub-regularities across well-motivated verb classes, I will emphasize the role root semantics plays in explaining the seemingly convergent behavior the templates have. Namely, because the Verb Complexity Mapping Principle is relativized to a root alternating across two templates, the semantic effect of the template is also similarly relativized, allowing for a range of different semantic behaviors which are nonetheless unified under the Verb Complexity Mapping Principle. This dissertation will therefore position the Hebrew templates as independent from, for example, the argument structure of the verbs which a given template can form from a root.

The dissertation is structured as follows. Chapter 2 will overview previous approaches to Hebrew root-template morphology, identify some key deficiencies in those theories, especially those which take argument structure to be at the core of the templates, and lay the groundwork for a semantic theory of templates, culminating in the Verb Complexity Mapping Principle.

Chapter 3 will investigate the syntax and semantics of a set of roaming motion verbs in HITPA'EL. In this alternation, a manner of motion verb in PA'AL alternates into HITPA'EL to generate a motion verb with inferences of aimlessness:

- (12) a. *halax-ti* *l=a=xanut*
 walk-1SG.PST to=the=store
 'I walked to the store.'

- b. *hithalax-ti* *b=a=rexov*
 REFL.walk-1SG.PST on=the=street
 ‘I walked around on the street.’ (constructed)

This alternation generally corresponds with a restriction on appearing with a dative-marked goal argument, thus constituting an argument-structural alternation:

- (13) **hithalax-ti* *l=a=xanut*
 REFL.walk-1SG.PST to=the=store
 (intended) ‘I walked about/roamed to the store.’ (constructed)

However, in this chapter, I will argue that the difference between the PA’AL manner of motion verb and the HITPA’EL roaming verb is not primarily argument structural, in that the two do not in fact differ in their ability to appear with dative marked goals. Instead, I demonstrate that HITPA’EL roaming verbs can appear with dative-marked goal arguments when they do not function as a true final endpoint of the motion event. I derive the behavior of these HITPA’EL roaming motion verbs through a property called antimaximality, which requires them to have no maximal event in their denotation. Because their correlated PA’AL manner of motion verbs do not entail antimaximality, the contrast between the two classes of verbs follows the VCMP.

In chapter 4, I will investigate how the locative alternation is realized by HIF’IL, and demonstrate how the VCMP interacts with other principles of markedness, namely the Morphosyntactic Alignment Principle from Beavers (2010) necessitating more marked arguments to have weaker entailments associated with them relative to less marked arguments. Specifically, I will focus on a set of locative verbs which alternate between PA’AL and HIF’IL, and which appear to limit the range of argument structure frames they can appear with based on their template:

- (14) a. *Noa hit’in-a* *et ha=masa’it b=a=saki-ot*
 Noa CAUS.load-3SG.PST.F ACC the=truck in=the=bag-PL.F
 ‘Noa loaded the truck with the bags.’

- b. *Noa ta'an-a et ha=saki-ot al ha=masa'it*
 Noa load-3SG.PST.F ACC the=bag-PL.F on the=truck
 'Noa loaded the bags onto the truck.' (constructed)

I investigate the distribution of these locative verbs in these two different argument-structural frames and conclude that those in the more marked HIF'IL template have a more restricted distribution, but only when they contrast with a PA'AL correlate. As such, a verb like *he'emis* 'load', which is in the HIF'IL template but has no PA'AL alternant, can appear in either frame:

- (15) a. *Noa he'emis-a et ha=saki-ot al ha=masa'it*
 Noa CAUS.load-3SG.PST.F ACC the=bag-PL.F on the=truck
 'Noa loaded the bags onto the truck.'
- b. *Noa he'emis-a et ha=masa'it b=a=saki-ot*
 Noa CAUS.load-3SG.PST.F ACC the=truck in=bag-PL.F
 'Noa loaded the truck with the bags.' (constructed)

This paradigm derives directly from the verbs in the more marked HIF'IL template having a lexical specification for holistic change which its PA'AL alternant does not have, thus satisfying the VCMP. As such, the VCMP can work in parallel with similar markedness principles like the MAP to constrain the kinds of meanings which a verb can have based on its morphology.

Finally, in chapter 5, I demonstrate that in the causative-inchoative alternation, the different templatic means of realizing the alternation indicate different semantic forms for the inchoative. In particular, I contrast two different alternating pairs of templates, HITPA'EL – PI'EL and NIF'AL – HIF'IL, both of which are realizations of the causative-inchoative alternation:

- (16) a. HITPA'EL inchoative – PI'EL causative:
hitmale 'fill (intrans.)' – *mile* 'fill (trans.)'
- b. NIF'AL inchoative – HIF'IL causative:
ne'elam 'disappear' – *he'elim* 'make disappear'

Specifically, I use an experimental paradigm from Lundquist et al. (2016) which probes whether an inchoative form has reflexive semantics or existential semantics. I demonstrate that HITPA'EL inchoatives prefer contexts where the theme is feasibly construed as the agent in its own event of change:⁴

- (17) *ha=magevet hitkapl-a*
the=towel REFL.fold-3SG.F.PST
‘The towel folded.’ (constructed)
- ✓ A person drops a towel over the top of a chair, which in turn folds over the top of the chair. (External agent only initiates the event)
 - #/? A person intentionally and delicately folds a towel in fourths. (External agent explicitly engaged in the event throughout)

To contrast, NIF'AL inchoatives do not discriminate between contexts where the theme is also feasibly an agent, and those where a clear external agent causes the change of state in the theme:

- (18) *ha=tmuna nifan-a* *'al ha=kir*
the=picture MID.lean-3SG.F.PST on the=wall
‘The picture leaned on the wall.’ (constructed)
- ✓ A person is holding a picture upright, and nonchalantly lets it go, where it falls over and hits the wall, leaning up against it. (External agent only initiates the event)
 - ✓ A person needs to set down a large picture, and slowly lets the picture down until it is leaning up against the wall. (External agent explicitly engaged in the event throughout)

These semantic forms associating with these templates in turn satisfy the VCMP, as inchoatives with reflexive semantics have a stronger meaning than inchoatives with

⁴These judgments represent a relative naturalness of the context as assessed by the experiment in Chapter 5.

existential semantics.

1.3 Preliminaries

Throughout this dissertation, the majority of the Hebrew data comes either from consultation with native speakers of Hebrew, or from the corpus heTenTen 2014. The heTenTen 2014 corpus (Baroni et al. 2009) is a collection of 890 million tokens scraped in 2014 from web sources across genres and tagged for various morphological and syntactic features, such as part of speech and verbal template, the two tags I used throughout the corpus analyses here.⁵ I draw from heTenTen 2014 for both individual data points and sentences, as well as for large patterns of data I take to be representative of the distribution and properties of the relevant verbs. Namely, I will presume that collocational frequency serves as an adequate stand-in for distribution, where differences in collocational frequency indicate a difference in distribution (akin to e.g. Bybee 2006, though I do not commit to an explicitly usage-based model of language). As such, where quantifiable, a statistically significant difference in collocation will indicate a difference in distribution, which I presume derives from a difference in underlying representation. I also draw on data from other work and other languages where relevant, and each example which is not my own will be accompanied with a citation including the number from the original paper. I preserve the glossing standards of the original, except for Hebrew data, where I gloss it following my own conventions.

I represent a Hebrew morphological root as its three consonants in small caps under a root symbol. Similarly, I name templates by their traditional names, and represent them in small caps as well. The labels I give the template should be understood as a descriptive label representing a shared phonological form. Additionally, I repre-

⁵There is an updated version of heTenTen from 2021. However, a portion of the research in this dissertation occurred before the release of heTenTen 2021, and therefore for consistency, I draw from heTenTen 2014 throughout.

sent Hebrew data here through a transliteration scheme which faithfully preserves the root and template. Specifically, I represent all consonants which make up the root, regardless of whether they are phonetically represented or not. In addition, in glossing verb forms, I do not assume a particular decomposition of verb and template, nor any particular derivational relationship between two templatic forms. As such, I gloss the lexical core of the verb (before inflectional morphology) as a single unit. In addition, in order to make evident which template a given verb is, I include the template in the gloss. This gloss is regardless of whether the template is contributing any regular meaning to the verb. As such, the gloss of the template should be understood as a verbal class identifier. I gloss each template using their traditional titles, shown below:

- (19) a. PA'AL: unmarked
 b. PI'EL: INTNS
 c. HIF'IL: CAUS
 d. NIF'AL: MID
 e. HITPA'EL: REFL

I also use the following representations for consonants in Hebrew which do not have a parallel in Latin orthography:

- (20) a. [ts]: *c*
 b. [ʔ]: *'*
 c. [χ]: *x*
 d. [f]: *f*

Additionally, I take the marker of a definite direct object *et* to constitute a case marker (Danon 2006), and as such I gloss it as accusative case with ACC.

Finally, I implement my semantic analyses using a standard typed lambda calculus. As such, I assume a universe of discourse referents U with two major sorts, U_e ,

the set of individuals and U_v , the set of eventualities. Where additional formal mechanisms are required, I will introduce them as necessary. Additionally, I will assume the morphosyntax from Kastner (2020), which I will review in the next chapters. Not much hinges on the specific syntax, though for representational purposes, I assume that a verb does not license its own external argument in the syntax or the semantics, which instead is licensed through a separate functional head (Kratzer 1996). Therefore, I will not include external arguments in my lexical denotations. I do so for consistency with Kastner (2020), though not much depends on whether the lexical verb entails its external argument or not (for arguments to the contrary, see Wechsler 2005b; Müller & Wechsler 2014; Wechsler 2020).

Chapter 2: The Verb Complexity Mapping Principle

2.1 Introduction

In this chapter, I will motivate the Verb Complexity Mapping Principle (VCMP), a principle for relating the morphological class of a verb to its meaning. In general, the VCMP asserts that the more complex the morphological class of a verb in an alternation, the stronger the verb's meaning will be. As this applies to the Hebrew templates, they form verbs which contrast with one another, such that the verbs in templates with more morphological complexity, as measured by a combination of number of segments and complexity in the phonology of those segments, have meanings which are stronger than the ones in templates with less morphological complexity. The VCMP will prove to have more explanatory value than alternative analyses which root the Hebrew templates in the syntax, in that it can capture contrasts between Hebrew verbs which have no necessary connection to the valency of the verb.

This proposal stands in contrast to the view on Hebrew templates as primarily determining the argument structure. Namely, the literature on Hebrew templates has converged on the idea that Hebrew templates determine the argument structure of the verbs which they form. Under this approach, differences in template determine differences in argument structure, such that for a single root, if it forms a verb in two different templates, those two verbs will have different argument structures. I will review this literature, with an eye towards how the morphological template in Hebrew parallels to the event structure of the verbs which they form. Namely, the Hebrew verb comprises a morphological root and a morphological template, which parallels the event structural root and event structural template, respectively. To that end, different templates form verbs with different event structures, such as causative, inchoative, or reflexive, and these event structures in turn manifest in different argument structures, such as transitive or intransitive. However, this generalization connecting the Hebrew templates to argument realization does not capture a wide range of contrasts between

verbs with the same root but a different template. To that end, the Verb Complexity Mapping Principle focuses on the semantic differences between two verbs in different templates with the same root, and predicates those differences off of the complexity of each template.

This chapter is structured as follows. The next section will review the literature on the Hebrew template system, focusing particularly on how they link the Hebrew morphological template to differences in event structure. Section 3 will review lexical entailment approaches to verb meaning, as the backdrop for the investigation of how the Hebrew templates determine the meaning of the verbs which they form. Section 4 will formulate the Verb Complexity Mapping Principle, and will discuss how the Hebrew templates rank relative to one another in their markedness. Finally, section 5 will summarize the ideas here, and, looking forward, present the three case studies making up the bulk of this dissertation, particularly how the Verb Complexity Mapping Principle applies in each of those cases.

2.2 The Hebrew templates as syntactic objects

2.2.1 The argument structure of the Hebrew templates

As discussed in the previous chapter, the Hebrew verb comprises two parts: a consonantal root, and a template representing prosodic information into which that root inserts. This system is pervasive across the Hebrew lexicon, and obligatory in the case of verb formation (Arad 2005, ch. 2). The root typically represents a clustering of conceptual information, and has been noted to play historical, conceptual and acquisitional import within the language.¹ One of the primary functions of the templates is to realize argument alternations between verbs related by a shared root. Such a paradigm is demonstrated in (21) with verbs based on (4).

¹I will not pursue a theory of Hebrew root meaning here, though see Berman (2012) for a discussion of the multidimensional status roots have in Hebrew.

- (21) a. *ha=rexev panah smola*
the=car turn[3SG.M.PST] rightward
‘The car turned right.’
- b. *Yosi hifnah et ha=rexev smola*
Yosi CAUS.turn[3SG.M.PST] ACC the=car rightward
‘Yosi turned the car right.’ (constructed)

In (21), the two verbs differ only by their template, and that difference manifests in an argument-structural difference, where the latter is transitive with a causer subject, the former is intransitive with a theme subject. This function of the templates is one of the most prevalent, being the most frequent across the verbal inventory (Bolzky & Berman 2020, sect. 2.1.1) and the default function in new word formation (Berman 1993).

Following that, a broad swath of literature has developed analyzing the templates as reflecting the underlying event structure of the verbs they form, which in turn determines their argument structure. Under this approach, morphological differences between verbs which share a root are the result of differences in event structure. In its most basic form, HIF’IL *hifnah* in (21) is derived from PA’AL *panah* through applying some form of causativization, such as as in (22) from Horvath & Siloni (2011b).²

- (22) $V < \alpha > \rightarrow \text{CAUS-}V < [+c+m, \alpha'] >$, where α includes a role specified as external.

(Horvath & Siloni 2011b, p. 692, (62))

Under this rule, a verb V with an argument α is input, and the output is a causative version of V , $\text{CAUS-}V$, with an additional argument that is both a causer (+c) and has a mental state (+m). This rule would take *panah* as an input and produce a version of *panah* with an additional external causer, functionally deriving a verb with

²Nothing hinges on the specific theoretical implementation of these rules, which are rooted in theories of Theta Theory and the Generative Lexicon. See Reinhart (2002); Reinhart & Siloni (2005); Siloni (2008, 2011); Reinhart et al. (2016) for specific details.

a causative event structure from one which does not. This derivation also comes with specific morphological changes as well (Laks 2007, 2013, 2014), as reflected by the additional morphological material in the derived causative form *hifnah* over the underived form *panah*, specifically the affix *hi-*. Thus, the HIF'IL template here is intrinsically linked to a causative event structure, where for the root $\sqrt{\text{PNH}}$, HIF'IL conditions the verb to be causative where in another template, it is not.

Similar correlations with other event- and argument-structural patterns hold for the other four non-passive templates under consideration here. As contrasting with HIF'IL in the previous section, HITPA'EL syntactically is intransitive. In (23), the HITPA'EL verb *hitparec* 'burst, bust' cannot appear with an object marked by the accusative case marker *et*, while its PA'AL alternant *parac* can.

- (23) a. *parac-ti* *et* *ha=delet*
burst-1SG.PST ACC the=door
'I busted down the door.' (constructed)
- b. **hitparac-ti* *et* *ha=delet*
REFL.burst-1SG.PST ACC the=door

Accordingly, HITPA'EL realizes a range of intransitive verbs, such as reflexives, reciprocals, and inchoatives:

- (24) Inchoative:
- a. *Ron bifel* *et* *ha=marak b=a=femef*
Ron INTNS.cook[3SG.PST.M] ACC the=soup in=the=sun
'Ron cooked the soup in the sun.'
- b. *ha=marak hitbafel* *b=a=femef*
the=soup REFL.cook[3SG.PST.M] in=the=sun
'The soup cooked in the sun.'
- (25) Reflexive:
- a. *Yicxak 'iper* *et* *Tomi*
Yicxak INTNS.make.up[3SG.PST.M] ACC Tomi
'Yicxak put makeup on Tomi.'

- b. *Tomi hit'aper*
 Tomi REFL.make.up[3SG.PST.M]
 'Tomi put makeup on (himself).'

(26) Reciprocal:

- a. *Yosi xibek et jager*
 Yosi INTNS.hug[3SG.M.PST] ACC Jagger
 'Yosi hugged Jagger.'

- b. *Yosi v=jager hitxabk-u*
 Yosi and=Jagger REFL.hug-3PL.PST
 'Yosi and Jagger hugged.'

(Kastner 2017, p. 3, (4)-(6))

HITPA'EL can go so far as to realize plain activities as well, such as with *hitmate'ax* 'stretch':

- (27) *ha=yelad-im hitmatx-u*
 the=kid-PL REFL-stretch-3SG.PST.PL
 'The kids stretched/did some stretching.' (based on Doron 2003, p. 60, (118))

Appropriately, the one generalization which covers this range of cases is syntactic, namely that HITPA'EL must not permit an internal object *qua* direct object.

NIF'AL is also an obligatorily intransitive template, and functions mainly as a mediopassive. Accordingly, it can form a similar range of verb types as HITPA'EL, namely reflexives, reciprocals, and inchoatives:

- (28) a. Inchoative:

ha=fi'ur nigmar
 the=lesson MID.finish[3SG.M.PST]
 'The lesson ended.'

- b. Reflexive/reciprocal:

Dani ve=Dina nifgef-u
 Dani and=Dina MID.meet-[3PL.PST]

‘Dani and Dina met (each other).’

(Alexiadou & Doron 2012, p. 9, (8-9a))

However, unlike HITPA’EL, NIF’AL can make canonical passives as well, as demonstrated with its compatibility with *al yedey* ‘by’, where a HITPA’EL correlate does not:

- (29) a. *Yosi nirxac al yedey ha='ima shel-o*
Yosi MID.wash[3SG.M.PST] by the=mother POSS-3SG.M
‘Yosi was washed by his mother.’
- b. **Yosi hitraxec al yedey ha='ima shel-o*
Yosi REFL.wash[3SG.M.PST] by the=mother POSS-3SG.M

Accordingly, where HITPA’EL is just intransitive, NIF’AL is specifically a mediopassive intransitive.

Unlike HITPA’EL and NIF’AL, PI’EL is like HIF’IL in that is a robustly transitive template. Specifically, the PI’EL template is the ‘intensive’ template, where the subject has a high degree of involvement in the event. Following Doron (2003, pp. 19-23), this involvement is a weak form of agency, where the external argument roughly needs to be able to exert active force in the event (cf. the notion of ‘action’ in Van Valin & Wilkins 1996, as contrasted with ‘effector’). She evinces this contrast with data like that in (30).

- (30) Context: Scientists discovered that the new vaccine was only effective in the presence of a specific protein which enables the immune system to create the antibodies.

- a. *ha=xelbon yacar b=a=guf nogdan-im*
the=protein create[3SG.M.PST] in=the=body antibody-PL
‘The protein created antibodies in the body.’
- b. #*ha=xelbon yicer b=a=guf nogdan-im*
the=protein INTNS.create[3SG.M.PST] in=the=body antibody-PL
‘The protein manufactured antibodies in the body.’

(Doron 2003, p. 21, (29))

The verbs contrasting in (30) are both formed from the root $\sqrt{\text{YCR}}$, related to creation. In the simple PA'AL template, the root creates the verb *yacar* 'create, produce', while in the intensive template, it creates the verb *yicer* 'manufacture'. As the context indicates, only PA'AL *yacar* is compatible with a reading where the protein is a precondition for the creation of the antibodies, with the immune system actually creating them, while PI'EL *yicer* cannot have such a reading. Instead, *yicer* requires the proteins to create the antibodies directly as the primary force in the event of creation, rather than an intermediary.

This type of agency also can have direct effects on the kinds of external arguments PI'EL verbs can take in contrast with PA'AL verbs. For example, (31) demonstrates a contrast between two verbs formed off the root $\sqrt{\text{JVR}}$, related to breaking.

- (31) a. *ha=yelad-im/ha=tiltul-im* *favr-u* *et* *ha=kos-ot*
the=child-PL/the=vibration-PL break-3PL.PST ACC the=glass-PL
'The children/the vibrations broke the glasses.'
- b. *ha=yelad-im/*ha=tiltul-im* *fibr-u* *et*
the=child-PL/*the=vibration-PL INTNS.break-3PL.PST ACC
ha=kos-ot
the=glass-PL
'The children/*the vibrations broke the glasses.'

(Doron 2003, p. 20, (25))

Here, the animacy difference between the two potential subjects, 'the children' being animate and 'the vibrations' being inanimate, conditions a grammaticality difference between the PA'AL and the PI'EL verb. In the case of the former, the PA'AL verb tolerates both the inanimate and the animate subject. This pattern follows from the generalization of PA'AL as the most unmarked template, which therefore has no restrictions on its meaning and therefore its arguments.

2.2.2 The templates as functional heads

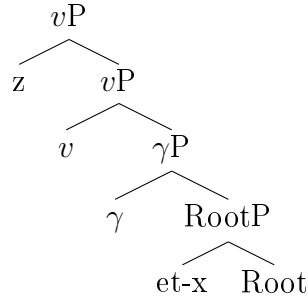
Overall then, these data point towards the Hebrew morphological templates as representing syntactic and semantic information about the verb's arguments, both the number of arguments the verb has (whether it is transitive or intransitive, in the case of HIF'IL and HITPA'EL), as well as what role the arguments hold (such as whether the subject is an actor or a cause). Accordingly, each of the templates have been analyzed as a realization of constellations of functional heads which taken together determine the nature of the arguments of the verb. This analytical approach hearkens back to Doron (2003), who reduces the templates to different types of heads in two separate categories: Agency and Voice. The Agency heads specify the role of the external argument, whether it is just a cause or is also an agent, and the Voice heads specify the realization of that argument (see also Alexiadou & Doron 2012; Doron 2013):

- (32) a. Agency heads:
- i. γ : causative head, $\lambda x \lambda e. \text{Cause}(e, x)$
 - ii. ι : actor head, $\lambda x \lambda e. \text{Actor}(e, x)$
- b. Voice heads:
- i. π : passive head
 - ii. μ : middle head

Under this analysis, each template is a morphological reflex of a combination of these functional heads, such that each combination of heads generates one template.

The causative HIF'IL template is a reflex of γ Merging with a verbalized root with its internal argument introduced:

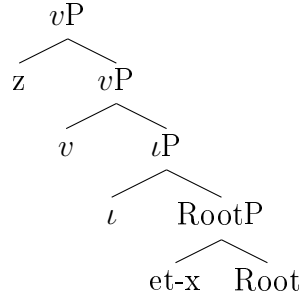
- (33) Causative HIF'IL:



(Doron 2003, p. 43, (74))

γ here restricts the external argument z to holding a Cause role, thus necessitating the verb it forms to have causal semantics and, in this case, be transitive.³ Conversely, PI'EL satisfies its agency requirement through ι entering a similar structure:

(34) Intensive PI'EL:



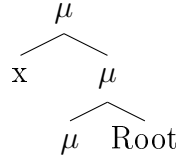
(Doron 2003, p. 42, (71))

In this case, the heads γ and ι determine the morphological templates of the verb, such that a structure with just γ will be realized as a HIF'IL verb and a structure with just ι will be realized as a PI'EL verb.

Conversely, both NIF'AL and HITPA'EL are obligatorily intransitive, and therefore include μ as a middle voice preventing the structure from being transitive. Verbs in NIF'AL template reflect just the presence of μ :

(35) Middle NIF'AL:

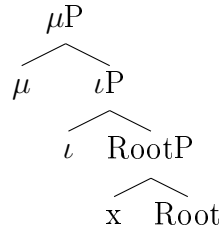
³Formally, transitivity is achieved through the head v , which syntactically introduces the external argument as a generic agent which can be specified either as a Cause or an Actor accordingly. In (33), γ licenses v , and thus the HIF'IL verb is obligatorily transitive, but transitivity does not rely on the HIF'IL template, as the root can also license v . The licensing of v is orthogonal to the phenomenon under issue here, and so I do not pursue it.



(Doron 2003, p. 59, (112))

The reflexive HITPA'EL template is an instance of the voice head μ Merging above a structure with ι to generate a non-active verb out of an active agentive PI'EL verb:

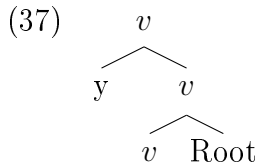
(36) Middle HITPA'EL:



(Doron 2003, p. 59, (113))

In both of these cases, μ directly limits transitivity (through restricting v , see footnote 3), thus capturing the non-active nature of these two templates. The two non-active templates also each correspond to a single set of these functional heads, with NIF'AL reflecting μ and HITPA'EL reflecting ι .

Finally, PA'AL as a template with no necessary restrictions on it derives from a structure with none of these functional heads:



As such, under this system, each template is a reflection of a unique constellation of these functional heads: no heads generate PA'AL, γ generates HIF'IL, ι generates PI'EL, μ generates NIF'AL, $\mu + \iota$ generates HITPA'EL. Accordingly, as the realization of a verb's template is determined by these heads, the templates in turn reflect argument structural information, as each of these heads determine information about the arguments of the verb which they derive.

Reflecting the syntactic functions of the Hebrew templates, recent work has cast the templates as an outgrowth of more universal functional heads which determine syntactic structure. The most notable among these is Kastner (2020), who develops a theory of Hebrew template morphology as reflecting a system of Voice. Specifically, he motivates a series of functional heads which in combination generate the range of syntactic patterns which the verbs in a given template can appear in. These functional heads each are reflected in a different feature of the phonological form of each of the templates, which taken together generate a verb with its templatic form (Kastner 2018). As I will be presuming his syntax here, I will review this system in turn.

Kastner works within a version of Distributed Morphology (Halle & Marantz 1993), a view on morphology which posits no distinction between syntax and morphology. Namely, all word building occurs in the syntax, and thus is subject to the same processes of combination which build sentences. Such an approach was also used by Doron (2003) and Arad (2005), whose syntactic heads combine together in the syntax as in the structures from Doron (2003) to create a Hebrew verb. Where Kastner (2020) differs is in the mobilization of a standard set of independently motivated functional heads, rather than a set of functional heads which are unique to Hebrew.

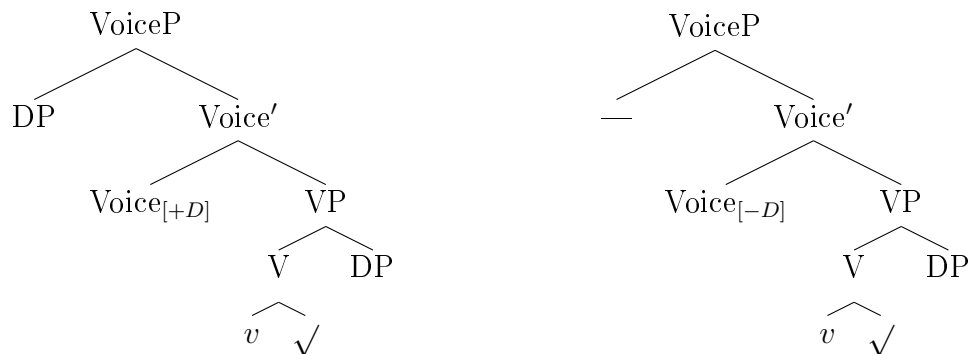
Primary among these heads is Voice, a head which determines the syntactic presence and semantic activity of the external argument of the verb. This system finds its start in Kratzer (1996), who argued for divorcing the external argument from its verb, and placed the responsibility for its realization into the functional head Voice. Initially, the presence or absence of Voice determined the presence or absence of an external argument, and thus determined transitivity of the verb which it forms. However, later work developed a series of featural specifications for Voice. Specifically, work such as Schäfer (2008) and Alexiadou et al. (2015) specified Voice with EPP-features which permit or restrict the introduction of an external argument. Voice_[+D]

requires a DP to Merge in its specifier, and thus requires the verb to have an external argument, while $\text{Voice}_{[-D]}$ specifically prevents a DP from Merging in its specifier.

These structures from Schäfer (2008) are in (38).

(38) a. Transitive structure:

b. Intransitive structure:



Because $\text{Voice}_{[+D]}$ necessitates a DP in its specifier, it generates all of the verb types with an external argument, namely transitive verbs, but also unergative verbs. Similarly, because $\text{Voice}_{[-D]}$ prohibits a DP in its specifier, it generates all verbs without a DP in its specifier, such as unaccusative verbs and verbs with expletive subjects. Additionally, both specifications of Voice determine whether the external argument is semantically active (i.e. represents a participant which bears some sort of thematic role, like agent or cause), or semantically inactive (i.e. is an expletive). Kastner (2020, sect. 2.2) additionally argues for a third specification of Voice, namely unspecified Voice, which has no requirements on its specifier, and thus can generate either transitive or intransitive structures.

In addition to Voice, Kastner includes two other functional heads which determine the realization of a template. First, $p_\emptyset$ licenses the introduction of a prepositionally marked argument (Svenonius 2003; Wood 2014), and generates a figure reflexive (e.g. Talmy 1985) typically representing motion of a ‘figure’ relative to a spatial ‘ground’. Such a head captures the existence of prepositionally marked argu-

ments in several of the generally intransitive templates, such as NIF'AL and HITPA'EL, a fact to be discussed in the next section, but exemplified here:

- (39) a. *Dana nixnes-a* *l=a=kita* *b=bitaxon*
 Dana MID.enter-3SG.F.PST to=the=classroom in=confidence
 'Dana entered the classroom with confidence.'
(Kastner 2020, p. 74, (11a))
- b. *ha=xatul hitnapel* *'al ha=regel fel-i* *b=za'am*
 the=cat REFL.pounce[3SG.M.PST] on the=foot POSS-1SG in=wrath
 'The cat angrily pounced on my foot.'
(Kastner 2020, p. 99, (61a))

Second, he includes an abstract root $\sqrt{\text{ACTION}}$, which provides agency entailments to the verb which it forms. As discussed above, agency entailments have been associated with PI'EL and HITPA'EL verbs (Doron 2003, p. 19-22), and this root generates those agency entailments.

Basically, these functional heads reflect the notion that different verbs place different constraints on their arguments. The various Voice and *p* heads limit the range of argument structures which can be associated with a verb, which in turn generate the argument structural patterns which a verb has. This behavior is language independent, and these heads are meant to be universal, with the Hebrew templates being their language-specific realization. That is, Kastner explicitly links these argument structural patterns to the morphological form of the Hebrew verb, such that a given argument structural pattern as determined by a given set of functional heads will be realized morphologically as a template. Namely, this system explicitly links the argument structure of the verb to the templatic realization, as these heads independently determine the argument structure of the verb, and are realized morphologically as the various templates.

Thus, under Kastner's system, the templates are an outgrowth of the interaction of these three functional heads, Voice, *p*, and $\sqrt{\text{ACTION}}$, which in turn determine

their syntactic properties. I give the specification of each of the five non-passive templates in (40).

- (40) a. PA'AL: Voice
 b. NIF'AL: Voice_[-D]; Voice, $p_{[-D]}$
 c. HIF'IL: Voice_[+D]
 d. PI'EL: Voice, $\sqrt{\text{ACTION}}$
 e. HITPA'EL: Voice_[-D], $\sqrt{\text{ACTION}}$; Voice, $p_{[-D]}$, $\sqrt{\text{ACTION}}$

(based off of Kastner 2020, p. 29, Table 1.6)

Importantly, under this system, the templates do not strictly determine the argument structure of the verb, and vice versa. This stands in stark contrast to the system from Doron (2003), who attributes each template to a unique set of functional heads. However, multiple different argument structures can be realized by the same template, and a single template can realize verbs with different argument structures. For example, NIF'AL verbs can be unaccusative without an external argument if realized through Voice_[-D], as in (28), and a figure reflexive if realized through Voice and $p_{[-D]}$, as in (39a). Similarly, though HIF'IL will always generate an unergative or transitive verb as a reflex of Voice_[+D], PA'AL can as well, as both argument structures are consistent with Voice.

On this approach, the Hebrew templates do not constitute independent morphological categories, but instead are an outgrowth of various language-independent syntactic categories. In all cases, the surface morphological template is not an independent morphological category, but rather a combination of the span of functional heads over the root, primary among them the Voice heads, with the surface morphological root again corresponding to the underlying syntactic root. In particular, unlike the heads in (32), heads like Voice and p are not Hebrew specific, and instead

have been posited crosslinguistically to account for similar transitivity phenomena.⁴ The only thing Hebrew-specific about the templates are the phonological Spell-Outs of these heads, which taken together generate the specific phonological forms of each of the templates (see Kastner 2018 for an explicit morphophonological account).

2.2.3 When the templates are not syntactic

The approaches from above run into issues when the differences between the templates are not ones which can be described in terms of their argument structure. Even Kastner, who attributes no specific status to the templates as morphological objects, cannot predict a difference between templates which cannot be characterized in terms of the argument structures of the resultant verbs. In essence, among the three types of contrasts discussed for the paradigm in (4), this sort of system captures both strictly syntactic differences and syntactic differences with semantic effects, but does not have a theory about contrasts where the only difference is semantic.

Consider, for example, the contrast between PA'AL *patax* 'open' and PI'EL *piteax* 'develop':

- (41) a. *Yosi patax et ha=delet*
 Yosi open[3SG.M.PST] ACC the=door
 'Yosi opened the door.'
- b. *Yosi piteax delet xadaf l=a=fuk*
 Yosi INTNS.open[3SG.M.PST] door new to=the=market
ha=amerika-i
 the=america-ADJ
 'Yosi developed a new door for the American market.' (constructed)

Both *patax* and *piteax* are formed from the root $\sqrt{\text{PTX}}$ related to opening, and only differ in their template. Following both Doron's and Kastner's systems, these two

⁴It is out of the scope of this work to discuss how these heads are realized crosslinguistically, but for a full discussion, see Kastner (2020, ch. 7).

verbs should contrast just in the agency of the subject, through ι in the case of the former and $\sqrt{\text{ACTION}}$ in the case of the latter. However, an analysis focusing just on the generalization misses the fact that the two verbs do not share the same idiosyncratic core. Namely, *piteax* is not just *patax* with additional agency entailments, but rather is a verb with a different meaning altogether.

Such templatic alternations are a case of Multiple Contextual Meaning (MCM) from Arad (2005). MCM alternating verbs share a root and differ in their template, but do not share any of the argument structural effects discussed above. Instead, because the difference is idiosyncratic, Arad (2005) treats them as enumerated. As such they are listed and not derived, and therefore not subject to any necessary regularity. Echoes of this enumeration approach can be found in Kastner (2020) as well, which accounts for differences in the meanings of each of the functional heads through a process of contextual allosemy, where the heads are assigned different meanings based on the roots they appear with (Wood & Marantz 2017). The roots which condition each of the meanings a head can take are similarly enumerated, and such are taken to be memorized and not regular.

However, even though these analyses account for idiosyncratic variability, it does not actually explain how each root is able to express the range of idiosyncratic meanings it can. Each meaning is accounted for in the Encyclopedia, which lists the meanings of each root in each template. Such an analysis is supposed to be sufficient, as idiosyncratic meanings are typically considered to be irregular, memorized, and therefore enumerated. Yet, contrasts such as those between *panah* ‘turn’ and *pinah* ‘evacuate’ discussed the previous chapter, demonstrate that even when verbs do not have any regular semantic relationship to one another, they can often still share the same conceptual space through the root. This in turn suggests that there is a kind of underlying regularity even in the most idiosyncratic alternations.⁵

⁵There are some instances where two words share a root with no seeming semantic relationship to one another, even a notional one, such as *sagar* ‘close’ and *hisgir* ‘extradite’, both formed from

Kastner (2017) provides an initial attempt at systematizing the irregularity found in different idiosyncratic contexts. There, he was addressing the two different types of meanings that HITPA'EL can produce: reflexive in (42a) and inchoative in (42b).

- (42) a. *ha=marak hitbafel* *b=a=femef*
the=soup REFL.cook[3SG.PST.M] in=the=sun
'The soup cooked in the sun.'
- b. *Tomi hit'aper*
Tomi REFL.make.up[3SG.PST.M]
'Tomi put makeup on.' (Kastner 2017, p. 3, (4)-(5))

Given that HITPA'EL in both of these contexts is the reflex of a single Voice₀ which enforces intransitivity, the question becomes how that single head can produce these two disjoint meanings. On his account, the two roots belong to different lexical semantic root classes, for which the transitivity requirements of the template produce the relevant meaning. In (100), the root $\sqrt{\text{BjL}}$ is a change-of-state root, the intransitive of which would be an inchoative. On the other hand, the root $\sqrt{\text{'PR}}$ is a naturally reflexive root, whose intransitive would be a reflexive (see e.g. Kemmer 1993, 1994; Alexiadou & Schäfer 2014; Alexiadou et al. 2014; Beavers & Udayana 2023).

This root class driven analysis may then be able to account for MCM alternations as well, indicating that a semantic characterization of the root-template system might be the most feasible. For example, consider the templatic realizations of the root if $\sqrt{\text{XfV}}$, related to thinking:

- (43) a. PA'AL: *xafav* – 'think'
b. HIF'IL: *hexfiv* – 'take into account'

the root $\sqrt{\text{SGR}}$. It is unclear how prevalent this type of contrast is, but the VCMP is not necessarily construed to capture such cases. However, the VCMP might be able to be expanded to capture them, particularly through adjusting the notion of strength to be able to compare verbs which are not necessarily related to one another. I discuss this in chapter 6.

- c. PI'EL: *xifev* – ‘calculate’
- d. HITPA'EL: *hitxafev* – ‘consider, be considerate’

If $\sqrt{XfV}$ is by hypothesis a psych root, the meaning of *hitxafev* might then be reducable to intransitivity operating on the specific psych semantics of the root. However, the set of lexical semantic root classes and their definitional properties have not been thoroughly explicated. Though Kastner (2017) discusses ‘naturally reflexive’ roots, he only defines it notionally as actions which are typically done to oneself, without any testable diagnostic properties distinguishing it from other roots. On the other side, with a thorough understanding of the semantics of the templates, many of the exceptions to their behavior may also be systematized.

Namely, what is missing to account for these types of alternations is a theory of what the Hebrew root and template can mean, not just what argument structures they associate with. Specifically, a lexical semantic analysis would highlight the status of differences in meaning to reflect differences in structure or form (Levin 1993; Rappaport Hovav & Levin 1998, 2010; Rappaport Hovav 2017; Beavers & Koontz-Garboden 2017, 2020; Beavers et al. 2021), and such a relationship between the two could be mobilized to explain when templates deviate from their syntactic generalizations. Naturally, as a system of word generation, it would not be out of the ordinary to expect a degree of irregularity or idiosyncrasy, and no theory of Hebrew grammar has ever claimed to impute any regularity onto these types of examples. Gesenius (1910), arguably the best known grammar of Hebrew, noted how untenable it would be to maintain consistent functions for each template (and, in fact, rejected the template as a valid morphological structure for Hebrew), and, working within modern linguistic frameworks, Schwarzwald (1973) noted how rarely the generalizations typically attached to the templates actually hold. However, where a difference in template indicates a difference in meaning, idiosyncratic or otherwise, there may be a regular relationship between the meanings of the two verbs which corresponds with the difference in template. Here, I will focus on how templatic membership indicates differences

in entailments for the verbs they form, and review the basis for that approach in the next section.

2.3 Lexical entailments and markedness

2.3.1 The linking problem

Throughout this dissertation, I will be adopting a lexical entailment approach to verb meaning (Dowty 1989, 1991), and will be focusing on what kinds of differences in sets of lexical entailments correspond to a difference in template. By focusing on lexical entailments specifically, I will be able to track differences in fine-grained meaning as a root forms a verb in one template for another. In particular, I will be assessing whether a difference in template for a root or root class indicates a difference in the lexical entailments associated with the verbs those templates form from that root. As I am focusing on morphological differences, and thus asymmetries in the morphological forms of the templates, I will draw parallels to the lexical entailment approaches to argument mapping. In particular, I will review how morphological or prominence asymmetries in the morphological strategies for marking an argument reflect asymmetries in the lexical entailments of the verb, especially as it relates to those arguments.

The main question behind argument mapping is the linking problem, which asks how arguments of a verb map to syntactic positions or marking strategies (Fillmore 1968, 1970). In general, linking is a two-pronged phenomenon: syntagmatically, how are two co-arguments of a single verb mapped to different argument positions, and paradigmatically, for two realizational options for a single argument, how are each of those options chosen. (44) demonstrates the main syntagmatic issue.

- (44) (In a context where the children are the ones eating and the cake is being eaten.)
- a. The children ate the cake.

b. ~~#~~The cake ate the children.

In (44), the children are the agent and the cake is the theme in the event of eating. Accordingly, *the children* maps to the subject position and *the cake* maps to the object position. The reverse mapping, where the theme is the subject and the patient is the object, is infelicitous. The question here is what principles ensure this connection between argument position and thematic role.

To contrast, the English conative alternation in (45) (Levin 1993, sect. 1.4) demonstrates the paradigmatic question, where the object ‘the cake’ is mapped either to a direct object position or an oblique:

- (45) a. The children ate the cake. (The whole cake is eaten.)
b. The children ate at the cake. (Only a part of the cake is eaten.)

Here, the object ‘the cake’ differs in its marking between the two sentences, as a direct object in (45a) and as an oblique with *at* in (45b). That marking conditions a difference in the meaning associated with that argument. Namely, the difference in marking conditions a ‘holistic affectedness’ reading, where the cake is completely eaten in (45a) but potentially partially eaten in (45b) (Anderson 1971). Here, I review how a lexical entailment approach handles both the syntagmatic and the paradigmatic portions of the linking problem. Specifically, the lexical entailments associated with an argument determines when that argument is mapped to a given argument position, and which position it is mapped to.

2.3.2 Thematic roles and the Argument Selection Principle

In (44), the feature which determines which argument is assigned to which position is the thematic role each of the arguments hold. Namely, the agent must be assigned to the subject position, while the theme must be assigned to the object position. Thus, the question of how to determine which of two co-arguments are

mapped to which position comes down to how the thematic roles of the two arguments contrast. Dowty (1989) defines a thematic role as the set of lexical entailments associated with a particular argument position of a predicate represented by a verb, thus constituting an individual thematic role, relativized to a single verb:

- (46) Given an n -place predicate δ and a particular argument x_i , the **individual thematic role** $\langle \delta, i \rangle$ is the set of all properties α such that the entailment $\Box[\delta(x_1, \dots, x_i \dots, x_n) \rightarrow \alpha(x_i)]$ holds. (Dowty 1989, p. 76, (4))

A thematic role here is an individual thematic role, relative to a single predicate and a single argument position. This definition of a thematic role relies on what Dowty called ‘lexical entailments’, or, roughly, entailments associated with a given verb which hold of the sentences headed by the verb. Specifically, these include all entailments which the verb says must be true of its arguments (cf. Dowty 1989, p. 75). Compositionally, lexical entailments are introduced by the verbal predicate itself, and thus must be below any operators introduced higher in the syntax, such as negation, modality, or quantification.

In order to evince the lexical entailments associated with a given verb, sentences containing that verb must be controlled for such operators. As such, a sentence with a single clause headed by the verbal predicate, with definite DP arguments, and no modality or negation, will entail the lexical entailment, in the sense of a relationship between two sentences or propositions. Therefore, wherever possible, I only consider sentences in corpus without any of these features, and in constructed contexts, I do not include them. Additionally, when comparing two sentences to evince or verify the existence of a lexical entailment, I maintain the same context and obligatory inflection (e.g. tense and aspect), and do not modify the sentence in any way except where necessary to maintain grammaticality (e.g. if one verb requires one of its arguments to be marked by a preposition but the other verb does not). Here, I will be using the term ‘entailment’ to mean lexical entailment in this sense, consistent

with Dowty (1989) and following literature, and wherever another sense of entailment is appropriate, I will specify as such.

A thematic role type is a generalization over individual thematic roles, such that for a set of predicates and an argument position, a thematic role type is the intersection of all the individual thematic roles of those predicates for that position:

- (47) Given a set T of pairs $\langle \delta, i_\delta \rangle$, where δ is an n -place predicate and i_δ the index of one of its arguments (possibly a different i for each verb), a **thematic role type** τ is the intersection of all the individual thematic roles determined by T . (Dowty 1989, p. 77, (6))

Thus, a thematic role type is the set of all entailments shared by a set of verbs for one of their arguments as determined by the verb. Further, a subset of these, such as agent or theme, are linguistically relevant; these are the L-thematic roles.

Dowty (1991) develops a theory of syntagmatic mapping based on the specific entailments associated with each participant in an event. Argument mapping operates over these thematic roles, specifically the thematic role types which the individual thematic roles contain. Dowty (1991) conceptualizes linking in terms of proto-roles, namely proto-agent and proto-patient, which a given argument of a predicate can hold to greater or lesser degrees (see also Blume 1998; Primus 1999; Ackerman & Moore 2001; Primus 2002; Beavers 2010; Grimm 2011 for various implementations of the proto-properties). The proto-properties which Dowty proposes are as in (48).

- (48) a. **Proto-agent properties:**
- i. volitional involvement in the event or state
 - ii. sentience (and/or perception)
 - iii. causing an event or change of state in another participant
 - iv. movement (relative to the position of another participant)
 - v. exists independently of the event named by the verb

b. **Proto-patient properties:**

- i. undergoes change of state
- ii. incremental theme
- iii. causally affected by another participant
- iv. stationary relative to movement of another participant
- v. does not exist independently of the event, or not at all

(Dowty 1991, p. 571, (27)-(28))

For Dowty, argument mapping is accomplished by determining how many proto-properties each argument holds, and mapping the one where the thematic role type has proto-agent properties to the subject position, and the one where the thematic role type has more proto-patient properties to the object position.

This process is captured by the Argument Selection Principle, which governs how proto-properties associated with an argument determine the mapping of that argument:

- (49) **Argument Selection Principle:** In predicates with grammatical subject and object, the argument for which the predicate entails the greatest number of Proto-Agent properties will be lexicalized as the subject of the predicate; the argument having the greatest number of Proto-Patient entailments will be lexicalized as the direct object.

(Dowty 1991, p. 576, (31))

As this applies to (44), the children hold only proto-agent properties, i.e. those of volitional involvement, sentience, and causing a change of state in another participant, while the cake only holds proto-patient properties, namely undergoing a change of state. Therefore, the former is realized as the subject, while the latter is realized as the object.

Correlatedly, if two arguments share the same amount of proto-patient or proto-agent properties, either argument can be mapped to either position (see Dowty

1991, p. 576, Corrolary 1). Such is the case for e.g. symmetric verbs like those of comparison, as in (50).

- (50) a. John resembled James.
b. James resembled John

In this case, both John and James hold the same amount of proto-agent properties, namely none. As such, both arguments are able to map to either argument position, characteristic of symmetric verbs. Therefore, the thematic role that a given participant holds for a verb determines their grammatical realization. Based on the entailments a particular argument holds and which constitutes its thematic role, that argument will either be realized as the subject or the object. This general approach, of comparing entailments between arguments and conditioning their grammatical realization on how they differ, also applies across realizations of a single argument, the paradigmatic approach which I review next.

2.3.3 Paradigmatic mapping and the Morphosyntactic Alignment Principle

In some cases, single arguments of a verb are able to alternate into different argument structural patterns, differing in their marking. This is exemplified with the English conative alternation in (45), repeated here.

- (51) a. The children ate the cake. (The whole cake is eaten.)
b. The children ate at the cake. (Only a part of the cake is eaten.)

Here, the object ‘the cake’ differs in its marking between the two sentences, as a direct object in the former and as an oblique with *at* in the latter. That marking conditions a difference in the meaning associated with that argument. Namely, the difference in marking conditions a ‘holistic affectedness’ reading, where the cake is completely eaten in the former but only partially eaten in the latter (Anderson 1971). The question then is how this semantic difference in holistic affectedness connects to

the object marking, and why each reading attaches to the argument marking strategy it does.

More broadly, this paradigmatic issue within the linking problem addresses the conditions under which one argument can be realized several different ways. Namely, for one argument with two different options for its realization, what are the conditions which determine what option is chosen. Building off of Dowty (1991), Ackerman & Moore (2001) complements the syntagmatic Argument Selection Principle with a separate paradigmatic mapping system along these lines: instead of determining to which position each co-argument of a single predicate is mapped, they used proto-properties to determine how one argument in the same position across related predicates is realized.

This relationship constitutes their Paradigmatic Argument Selection Principle, which relates the proto-properties associated with two realizations of a single argument:

- (52) **Paradigmatic Argument Selection Principle:** Let $P(\dots, arg_i, \dots)$ and $P'(\dots, arg'_i, \dots)$ be related predicates, where arg_i and arg'_i are corresponding arguments. If arg_i and arg'_i exhibit different grammatical encodings and arg_i is more prototypical with respect to a particular proto-role than arg'_i , then arg_i 's encoding will be less oblique than arg'_i 's encoding.

(Ackerman & Moore 2001, p. 67, (13))

The Paradigmatic Argument Selection Principle relates two predicates connected by some semantic rule deriving one from the other, and for the argument in some position in both, if one has more proto-properties than the other, then that one will be marked using a less oblique grammatical strategy than the other one.

Thus, for two different realizations of the same argument, the one which is more oblique, i.e. lower on a case hierarchy or realized with grammatical means that

is more marked, will have fewer proto-properties associated with it. To exemplify this, they demonstrate how object case marking in Finnish correlates with telicity:

- (53) a. *Terttu luki kirjaa (tunni-n)*
 Terttu read-PAST.3SG book-PART (hour-GEN)
 ‘Terttu was reading the book (for an hour).’
 b. *Terttu luki kirjan (tunni-ssa)*
 Terttu read-PAST.3SG book-ACC (hour-IN)
 ‘Terttu read the (whole) book (in an hour).’

(Finnish, Ackerman & Moore 2001, p. 84, (1))

In (53), the two sentences differ in which case the object is marked with, which in turn conditions a telicity difference. When the object is marked with the genitive case, the sentence is atelic, as demonstrated with the durative adverb *tunni-n* ‘for an hour’. Comparatively, when the object is marked with the accusative case, the sentence is telic, as demonstrated with the time span adverbial *tunni-ssa* ‘in an hour’. They attribute this difference in telicity to an additional proto-patient property which they term Bounding Entity, namely, whether the object determines the end of the event or not. In the case where the object has the property of Bounding Entity, the object bounds the event and the sentence is telic, and if it is not, the sentence is atelic.⁶

Accordingly, the case marking of the object differs based on whether it has that proto-property or not. The version of the argument which has the additional proto-role will have more proto-patient properties than the version of the argument which does not. As such, the former is more prototypical and, following the Paradigmatic Argument Selection Principle, gets marking which is less oblique than the marking of the latter.⁷ Accordingly, the telic sentence, with its bounding entity, gets an object

⁶Beavers (2006) demonstrates that how an object serves as a bound to an event derives directly from the reference of the noun itself, and so Bounding Entity as a proto-property is not necessary.

⁷Ackerman & Moore technically determine this mapping through ‘thematic loading’, where proto-agent and proto-patient entailments factor together into a single value for each argument, which is then used to determine which argument takes which position. However, just comparing greater or fewer proto-entailments suffices to explain the simple case here.

marked with the less oblique accusative case, while the atelic sentence, lacking a bounding entity, gets an object marked with the more oblique partitive case.

If one were to extend this analysis to the conative alternation in (45), the version of the internal argument marked with the oblique *at* in (45b) is not a bounding entity, thus leading to the lack of a holistic reading. This contrasts with the version realized as a direct object in (45a), which is a bounding entity, and therefore has a holistic reading. The difference in marking between the two versions of the same argument is conditioned by the difference in proto-properties between the two, where the alternant which entails more proto-properties bears less oblique *qua* less marked marking.

However, Beavers (2010) provides a more motivated analysis of paradigmatic alternation which does not rely on an *ad hoc* notion of a bounding entity. He develops a system of mapping based on monotonic contrasts between thematic role types across a set of fixed hierarchies. Namely, he proposed the Morphosyntactic Alignment Principle for direct/oblique alternations:

- (54) **Morphosyntactic Alignment Principle (MAP):** When participant x may be realized as either a direct or oblique argument of verb V , it bears L-thematic role R as a direct argument and L-thematic role $Q \subseteq_M R$ as an oblique. (Beavers 2010, p. 848, (69))

For two different realizations of a single participant, the direct object will have a thematic role which is minimally stronger than the thematic role which the oblique has. Minimal contrast is a relationship between two thematic roles as sets of entailments such that one is monotonically larger than the other. This notion is defined in (55).

- (55) **Minimal Contrast:** Q is minimally weaker than R ($Q \subseteq_M R$) on a hierarchy of L-thematic roles iff $Q = R$ or $Q \subset R$ and there is no role P on the hierarchy such that $Q \subset P \subset R$. (Beavers 2010, p. 848, (68))

The MAP holds relative to the entailments associated with the alternating argument, such that the thematic roles being compared are those which just the alternating argument holds in both cases. This approach circumvents the main issue of Ackerman & Moore (2001), namely that their thematic loading approach (see fn. 6) allowed for wide comparison of sets of entailments which are not necessarily related, and as such predicted alternations which were not attested or extremely unexpected. Instead, the MAP constrains alternations to occur relative to a single role with a fixed set of entailments, which a participant can have more or fewer of.

Additionally, the MAP operates over constrained hierarchies of thematic roles, such that alternation from direct object to oblique involves a monotonic decrease in entailments for that argument along a rankable set of thematic roles where each is monotonically larger than the other. Importantly, these hierarchies are not stipulated, but rather derive directly from implicational relationships between sets of lexical entailments. For example, the conative alternation in (45) involves a decrease in affect-ness, or specificity of change (Beavers 2011), which roughly determines how much change occurs to a given theme.⁸ The contrast between the direct and oblique arguments in the conative alternation in (44) is one between quantized and non-quantized change (Hay et al. 1999). Quantized change entails that a theme reaches a predicate-specific degree of some property over the course of a change, while non-quantized change entails that a theme reaches some nonspecific degree of some property over the course of a change. The latter is minimally weaker than the former, as if a verb entails that its theme reaches a lexically specified endpoint, it will also necessarily entail that that theme reaches some nonspecific endpoint as well. As such, the alternant of *eat* with a direct object entails quantized change, namely that the cake is fully eaten, while the alternant with an oblique entails non-quantized change, namely that the cake need only be partially eaten. Thus, the MAP holds, as the oblique alternant holds a minimally weaker role than the direct object alternant.

⁸I review the affectedness hierarchy in full in chapter 3.

Importantly, because the MAP is relativized to a single argument of a single verb, the conative alternation can differ based on the verb which undergoes the alternation (cf. Beavers 2010, p. 839, Table 3). For example, the conative alternation can apply to *cut*, as in (56).

- (56) a. John cut the paper.
b. John cut at the paper.

Here, *cut* with a direct object in (56a) entails non-quantized change for its object, as it requires the paper to have undergone some amount of cutting, but not necessarily a specific amount. *Cut* with an oblique in (56b) instead entails potential for change, as there is no requirement for an actual cutting to occur to the paper. The latter is minimally weaker than the former, as actual change entails potential change in the theory of affectedness which Beavers outlines.

While the difference in entailments between the direct and oblique alternants is not the same between the conative alternation with *eat* in (45) and the conative alternation with *cut* in (56), the alternants in both cases are still related by minimal contrast, where the oblique alternant is minimally weaker than the direct alternant. In addition, both instances of the conative alternation involve a contrast along the same hierarchy of thematic roles, the affectedness hierarchy, and only differ by which levels of the hierarchy are being contrasted. Thus, the MAP casts paradigmatic argument alternation as both relative to the entailments of the individual alternating verbs, and constrained to operate over certain generalized hierarchies of thematic roles, such as affectedness.⁹ Because these hierarchies relate thematic roles to each other monotonically, it importantly rules out relationships between alternating arguments which do not involve a monotonic increase or decrease in entailments, as well as those which do not follow the hierarchical relationships between the thematic roles. Overall,

⁹See e.g. Beavers (2010, pp. 851-855, (85)-(87), (90)) for a discussion of a potential traversal hierarchy for motion verbs or a recipient hierarchy for verbs of transfer.

because the MAP compares sets of lexical entailments that are related to one another through a paradigmatic relationship, and are constrained to independently motivated hierarchies of thematic roles, it holds more power to describe and predict argument alternations.

Extending this approach past argument marking, Jerro (2023) connects strength of entailment to verb alternations, where marking on the verb can induce stronger truth conditions associated with an argument of the verb. Jerro focuses on applicativization, a process which, parallel to causativization adding a new subject, typically adds a new object, as exemplified in the Bantu language Kinyarwanda. Namely, Jerro argues that the main function of applicativization is to increase the entailments associated with the thematic role of an internal object. Specifically, applicativization is constrained by an output condition on the meaning of the applied form, the Applicativization Output Condition:

- (57) **Applicativization Output Condition (AOC):** In alternations between applied and non-applied forms of a verb,
- a. the applied variant has at least one internal non-Theme argument, and
 - b. the truth conditions associated with that internal argument are a strict superset of those associated with the non-Theme argument (if present) of the non-applied variant. (Jerro 2023, p. 1443, (30))

The AOC allows for three general classes of verbs to occur as a result of applicativization. First, if the verb does not entail any internal non-Theme argument, one will be added, satisfying the first condition:

- (58) a. *Uwase a-ri ku-vug-a*
 Uwase 1S-be INF-talk-IMP
 ‘Uwase is talking.’ (Kinyarwanda, Jerro 2023, p. 1449, (38a))
- b. *Uwase a-ri ku-vug-ir-a mu n-zu*
 Uwase 1S-be INF-talk-APPL-IMP 18 9-house
 ‘Uwase is talking in the house.’

(Kinyarwanda, Jerro 2023, p. 1449, (41a))

Here, the verb *kuvuga* ‘talk’ is affixed with the applicative *-ir*, resulting in the addition of an argument representing a location where the talking takes place. Before the addition of the applicative, there was no internal non-Theme argument, and so, following the AOC, one is added. Definitionally, this process represents an increase in the truth-conditional strength associated with that argument position, as there were no entailments associated with it at all before the addition of the applied argument, and now there are some.

Second, if the verb entails the existence of a participant with a particular thematic role but does not select for it syntactically, a new argument is added which holds that thematic role. Such is the case with verbs of motion, as in (59).

- (59) a. *Karemera y-Ø-ambuts-e in-yanja*
Karemera 1S-PST-cross-PRFV 9-ocean
‘Karemera crossed the ocean.’ (Kinyarwanda, Jerro 2023, p. 1451, (47a))
- b. *Karemera y-Ø-ambuk-iy-e i Mombasa (mu) in-yanja*
Karemera 1S-PST-cross-APPL-PRFV 28 Mombasa (18) 9-ocean
‘Karemera crossed the ocean from Mombasa.’

(Kinyarwanda, Jerro 2023, p. 1452, (50a))

Here, the addition of the applicative to the verb *kwambuka* ‘cross’ results in the addition of an argument representing the source of the motion. By virtue of being a verb of motion along a path, *kwambuka* entails that the motion has a source where the path begins, a route comprising the intermediary points, and a goal where the path ends (cf. Jerro 2023, p. 1450, (42)). The applicative here adds an argument which corresponds to one of the three parts of the path. In essence, the applicative is naming a specific participant, and by naming the participant, more is entailed of it.

Finally, if the verb entails the existence of a realized non-Theme argument, applicativization can add entailments to the truth conditions associated with that argument. Such is the case with the ditransitive verb of transfer in (60).

- (60) a. *Habimana y-a-tey-e* *Karekezi i-buye*
 Habimana 1S-PST-throw-PRFV Karekezi 5-rock
 ‘Habimana threw a rock at Karekezi.’
- b. *Habimana y-a-ter-ey-e* *Karekezi i-buye*
 Habimana 1S-PST-throw-APPL-PRFV Karekezi 5-rock
 ‘Habimana threw a rock to Karekezi.’

(Kinyarwanda, Jerro 2023, p. 1454, (54))

Here, the addition of the applicative to the ditransitive verb *guteru* ‘throw at’ increases the entailments associated with the preexisting internal argument *Karekezi*. In the non-applied case, there is an entailment of potential arrival of the rock at Karekezi, while in the applied case, Karekezi as a participant gains an entailment of coming into the possession of the rock. Thus, the entailments associated with that argument are monotonically greater, where Karekezi need not come into possession of the rock in the non-applied case, but must gain possession of the rock in the applied case.

Thus, the AOC extends the connection between marking and truth conditional strength from marking on the arguments to marking on the verb which takes those arguments. Applicative markers on a verb induce a truth conditional strength increase in one of arguments associated with that verb, relative to the truth conditions of the verb which it applies to. Lexical entailments and the thematic roles which they form are therefore sensitive to the marking on the verb, at least in the case of derivational morphology, which applicative morphology is a part of. I will use the case of Hebrew here to extend this approach past derivational morphology and to relate the complexity between two non-derived forms based on their truth conditions. Specifically, this contrast will be guided by a parallel principle to the MAP and the AOC which operates over lexical verb class membership indicative of what meaning is lexicalized by that form, the Verb Complexity Mapping Principle.

2.4 Verb Complexity Mapping Principle and markedness in the Hebrew templates

2.4.1 The Verb Complexity Mapping Principle

The approach in the last section converged on the idea that differences in marking, whether it is a prominence asymmetry in argument marking (direct-oblique alternations) or through derivational morphology as in the case of applicativization, correspond to differences in truth conditional strength. Thus, principles like the MAP or the AOC derive a principled link between the morphology of a form and its semantics. These in turn predict possible meanings of the forms which they apply to. Namely, in general, between two morphologically related forms, where one has more marking, its meaning is stronger than the meaning of the one with less marking.

To that end, I propose here that Hebrew verbal morphology operates on the same axis. Verbs in Hebrew are related to one another through their root (cf. the paradigm in (4)), and the templatic morphology which derives the verb from the root constitutes the marking on the root. These templates, through the differences in their phonological forms, exist in various markedness asymmetries with one another, and over those markedness asymmetries can be stated a principle which relates the meanings of those forms. The principle I propose here is the Verb Complexity Mapping Principle, which I state informally in (61).

- (61) **Verb Complexity Mapping Principle (informal):** For any two morphologically-realized lexical verb classes, a verb realized in the more marked class will have truth conditions which are a strict superset of a morphologically-related verb realized in the less marked class.

Although I call this a principle (in line with prior works in the lexical semantics which discuss the relationship of form to meaning), the Verb Complexity Mapping Principle is a generalization over the Hebrew data considered here, and fundamentally constitutes a hypothesis which predicts specific kinds of relationships between

the meaning of two verbs based on their morphological form. The Verb Complexity Mapping Principle relies on a relation between two notions, morphological markedness asymmetries between lexical verb forms and strength differences between the meanings of those verbs. This principle explicitly focuses on non-derived forms, and so excludes potential morphological markedness differences between derivational or affixal morphology, such as tense, aspect or mood.

In addition, I am focusing here on verbs which exist in an actual semantic paradigmatic relationship with one another. Namely, I am presuming that templatic alternation is operating over a shared core root idiosyncratic meaning represented by the morphological root, and that differences in template condition differences in entailments layered on top of that root. As such, I am intentionally excluding differences in template which condition different idiosyncratic meanings, or alternations which generate Multiple Contextual Meanings like that in (41). This restriction corresponds to the notion that the morphological template in forming the Hebrew verb corresponds to the event structural template, which operates independently of the idiosyncratic root. Thus, differences in strength between two verbs will be measured in terms of a monotonic increase of entailments, where one verb will have all the entailments of the other, plus additional entailments.

To that end, I parallel the AOC in mobilizing a more constrained notion of strength I term *strict strength*, where the meaning of a verb is strictly stronger than the meaning of another if the set of lexical entailments, closed under entailment, which make up truth conditions of the former are a strict superset of the latter.

- (62) **Truth conditional strength:** A predicate P is **strictly stronger** than a predicate Q ($P >_s Q$) iff the set of entailments associated with P , $T_P = \{X_P | P \rightarrow X_P\}$, is a superset of the entailments associated with Q , $T_Q = \{X_Q | Q \rightarrow X_Q\}$, i.e. $T_P \supset T_Q$.

Unlike the MAP and the AOC, this notion of strength is not relativized to a single

participant, nor does it utilize minimality like that in (55).¹⁰ Importantly, the VCMP relates the whole lexical meaning of two verbs which are morphologically related to one another. Just as the MAP and AOC concern the thematic content of an alternating argument with realizations through two different morphological means related to that argument, the VCMP concerns the content of the verb itself as derived through the morphological alternation of its verbal class. The VCMP then will predicate differences between the core lexical meaning of the verb, which includes specifications on both the event itself and the participants of that event.¹¹

Therefore, the informal statement of Verb Complexity Mapping Principle in (61) can be stated in full as in (63).

- (63) **Verb Complexity Marking Principle (VCMP):** For two n morphologically-realized verb classes α_1, α_2 , which contrast in their markedness such that $\alpha_1 \ll \alpha_2$, and which derive morphologically-related verbs $V_{\alpha_1}, V_{\alpha_2}$, $\llbracket V_{\alpha_2} \rrbracket >_S \llbracket V_{\alpha_1} \rrbracket$.

Fundamentally, the VCMP constitutes a hypothesis predicting what the relationship between a verb's form and its meaning should be, and the kind of patterns it predicts derive from a lexical restriction on what verb meanings can be given a verb form. Namely, based solely on the morphological class of a verb, its meaning must be stronger or weaker depending on the relative markedness of that verb class to another, relative to the same root. Therefore, a verb which has a more marked class has

¹⁰Note that also unlike the MAP, strict strength does not allow for two predicates to have the same set of entailments. As such, the VCMP would not allow for two verbs to have the same meaning but contrast in their morphology. There may be some utility in weakening the VCMP in this way, such as to capture passives, which would appear in a more marked template but have identical entailments associated with them. But as I put phenomena like passives aside in this dissertation, I will not be pursuing that route.

¹¹Another way to accomplish this formally is through requiring the stronger predicate to have more restricted domain rather than a more restricted range, as comparing entailments would result in. In absence of any strong evidence favoring one approach over the other, I choose to compare entailments here, for consistency with the lexical entailment literature.

a stronger meaning than a verb with the same morphological root in a less marked class.¹²

As might be expected on parallel with the lexical pragmatic approaches discussed in the previous chapter, this formulation of the VCMP creates the exact conditions where one might expect blocking to occur between the two verb forms. Namely, the less marked form will have a more vague meaning than the more marked form, in that the latter will have all of the entailments of the former plus more. As such, when that more specific meaning needs to be conveyed, though the less marked, more vague is compatible with it, the more marked, more specific form is used.

Blocking is relatively common in the lexical domain considered here, especially in cases where the existence of a word with a more specialized meaning blocks that meaning in another less specified word. Such is the case with, for example, count and mass nouns in English, as in (64) from Blutner (1998).

- (64) a. I ate pork/?pig.
b. I like beef/?cow.
c. The table is made of wood/?tree. (Blutner 1998, p. 129, (17))

Here, the existence of the mass noun (*pork*, *beef*, *wood*) blocks a mass interpretation for the count noun. However, the cases I review here do not demonstrate blocking, and instead all show the more vague form being used in contexts where the more specific form is possible.

Important to note is that blocking, as part of lexical pragmatics, is fundamentally a pragmatic process, and so it is not necessary that it must occur. In fact, there are many situations exactly like the ones presented here where blocking does

¹²This relationship is predicted to hold for any meaning assigned to a verb with its class, and so where a verb might be ambiguous, the VCMP would constrain all of its possible interpretations. However, I do not address any ambiguous cases here.

not occur despite an asymmetry in markedness. (65) shows an example of the English conative alternation (cf. (45)) from Beavers (2010).

- (65) a. Marie ate the cake.
b. Marie ate at the cake. (Beavers 2010, p. 830, (21))

With the conative alternation, the more marked oblique alternant in (65b) only entails that some of the cake is eaten, in contrast to the less marked direct alternant in (65a), which entails that all the cake is eaten. If blocking were anticipated to hold here, the conative alternant, as the less informative or weaker meaning, should be dispreferred in cases where the message being conveyed is that the whole cake is eaten. However, as demonstrated in (66), it is fully compatible with the whole cake being eaten, and can be used as such.

- (66) Marie ate at the cake, and (eventually) ate the whole thing.

Thus, even when blocking is anticipated, it does not need to occur, and it is fully possible that in spite of its level of informativity, a more vague form will be used.

Even further, Beavers & Nishida (2009) equivalently demonstrated for Spanish dative alternations that blocking occurs irregularly, with some cases showing blocking, and others not. In (67), the dative clitic *le* is blocked in sentences of insertion (with *en* ‘in’) due to the existence of an equivalent sentence without it.

- (67) a. *Juan metió la fruta en el frigo*
 John inserted the fruit in the fridge
 ‘John put the fruit into the fridge.’
 b. #*Juan le metió la fruta al frigo*
 John CL inserted the fruit to.the fridge
 (Beavers & Nishida 2009, p. 11, (29))

However, this pattern of blocking does not always hold for the dative clitic, as with alternations with *para* ‘for’.

- (68) a. *Juan construyó una casa de campo para su hijo primogénito*
 John built a country house for his son first-born
 ‘John built a country house for his first son.’
- b. *Juan le construyó una casa de campo para su hijo primogénito*
 John CL built a country house for his son first-born
 ‘John built his first son a country house.’

(Beavers & Nishida 2009, p. 12, (32))

Thus, despite the fact that the VCMP predicts exactly the conditions where blocking should occur, it need not and in fact does not. Why blocking does not occur in situations where it might be anticipated is a large open question, but I will not be discussing it further here.

2.4.2 Markedness in the Hebrew templates

Finally, in order to fully apply the VCMP and understand which verbs are predicted have stronger meanings, there must be a separately motivated set of markedness relationships between the various morphological classes of a language. In Hebrew, the templates constitute the classes, and so depending on their phonological or morphological complexity, some templates are more marked than others. Throughout this dissertation, I will be performing primarily pairwise comparisons, and so I will need to know, for any two templates, which one is more marked than the other. To do such, I will compare three main properties, frequency of occurrence, prosodic prominence, and morphological complexity, which, taken together, will suggest the partial ordering in (69).

(69) **Markedness hierarchy for the Hebrew templates:**

PA’AL << HIF’IL, PI’EL << NIF’AL << HITPA’EL

Because all of the case studies here involve only pairwise comparisons, the hierarchy in (69) should not be construed as an absolute hierarchy, but instead serves as an *a priori* basis to determine which template, between two in a pairwise comparison, is

more marked. Thus, the hierarchy in (69) emerges from the combination of all possible pairwise comparisons.

Phonologically, the Modern Hebrew verb is built off of three types of phonological material: the root, the rhyme, and prefixal material. The root consists of typically three consonants into which the rhyme, a pattern of vowels, insert. For some templates, additional affixal material applies to derive the full verb form. Thus, for the HIF'IL verb *hitbil* 'dip (as into a liquid or food)', the phonological form is derived as in (70):

(70) *hitbil*:

- a. Root: tbl
- b. Rhyme: $\emptyset$ - i
- c. Affix: hi-

As every Hebrew verb is derived from a root, the differences in complexity in the templates can be assessed based on their rhyme and affix. The following table shows the rhyme and affix of each of the templates under consideration. This verbal mor-

Template	Rhyme	Affix
PA'AL	a - a	$\emptyset$
NIF'AL	$\emptyset$ - a	ni-
HIF'IL	$\emptyset$ - i	hi-
PI'EL	i - e	$\emptyset$
HITPA'EL	a - e	hit-

Table 2.1: Templates by rhyme and affix

phology is accompanied by a system of tense and aspect marking which is primarily affixal; I treat this as orthogonal to the template system here and do not consider it, as it is not relevant to the lexical morphological class of the verb.

From the perspective of the phonology, Laks (2013) motivates a notion of markedness based primarily on faithfulness. Under this view, a template is less marked

if it retains a uniform phonology under inflection. For example, consider the tense paradigm for the PI'EL template, represented by the verb *kines* 'gather' as compared to the PA'AL template, represented by the verb *sagar* 'close'.

		PI'EL	PA'AL	
	past	<i>kines</i>	<i>sagar</i>	
(71)	present	<i>mekanes</i>	<i>soger</i>	(Laks 2013, p. 360, (13), (14))
	future	<i>yekanes</i>	<i>yisgor</i>	
	infinitive	<i>lekanes</i>	<i>lisgor</i>	

In the PI'EL template, the vocalic rhyme remains relatively fixed, with the only alternation across inflectional categories being in the past, with [a] changing to [i]. Comparatively, the PA'AL template changes the vocalic rhyme across each form, in some cases dropping the initial [a], in others alternating [a] with [o]. Thus, because PI'EL preserves its phonological structure across forms, it is less marked than PA'AL. HITPA'EL retains the same level of high faithfulness as PI'EL, while NIF'AL has a similar level of prosodic variation within its forms as PA'AL, and HIF'IL exists somewhere in the middle.

The markedness hierarchy which this heuristic generates is as in (72).

(72) **Markedness hierarchy (faithfulness):**

PI'EL, HITPA'EL << HIF'IL << PA'AL, NIF'AL (Laks 2013, p. 361, (15))

Markedness predicated on this kind of faithfulness would constitute a kind of markedness based on phonological difficulty (Haspelmath 2006, p. 26, Table 1, (5)). Namely, the more marked forms undergo more phonological processes in derivation or paradigmatic relation from form to form, and thus constitute a more phonologically marked class.

However, this notion of markedness is not the one I will pursue here (though it can be compatible with what I propose, in principle), as the VCMP requires a measure of markedness which can apply to compare the templates only. Phonological faithfulness is a measure which can be applied across category and across word

type. Different categories of words undergo different types of phonological processes, but the measure of faithfulness only cares about how much of a word's form is preserved as they undergo those processes. As such, establishing a markedness hierarchy based on faithfulness alone would necessarily build in comparisons to forms which are not related to one another at all. For the templates, it would generate markedness comparisons which are not relevant to how the templates relate to one another, and therefore is less meaningful than other measures which relate the templates based on their own structural properties.

For example, nouns in Hebrew have two basic stem forms, a free form, and a construct state form. The construct state form is a morphologically complex form which is utilized when the noun enters a compounding construction, which can serve also as a type of possessive construction. (73) gives examples of three types of construct states which nouns can take.

	Noun	Construct state	Example
(73)	<i>rofe</i> 'doctor'	<i>rofe</i>	<i>rofe</i> <i>nafim</i> 'gynecologist'
	<i>tmuna</i> 'picture'	<i>tmunat</i>	<i>tmunat ha=’oman</i> 'a picture of the artist'
	<i>bayit</i> 'house'	<i>bet</i>	<i>bet xolim</i> 'hospital'

First, for *rofeh* 'doctor', the free state and the construct state are the same, indicating a null derivation between the two states. Second, in the case of *tmunah* 'picture', the noun becomes affixed with [t] in the process of deriving the construct state from the free state. Finally, *bayit* 'house' undergoes a process of vowel fusion in deriving the construct state, modifying the stem itself.

Thus, the three types of construct states in (73) rank relative to each other on the basis of their faithfulness to their base noun, establishing a potential ranking as in (74).

(74) **Construct state markedness hierarchy:**

rofeh-type << *tmuna*-type << *bayit*-type

rofeh-type nouns preserve the form of the noun fully between the free form and the

construct state form, and therefore are entirely faithful with respect to construct state derivation. *tmuna*-type nouns are partially faithful, as they preserves the stem of the free form, but is not identical to the free form, and is instead derived by affixation. Finally, *bayit* is entirely unfaithful, with the stem itself changing between the free form and the construct form.

Accordingly, because nouns are marked differently based on their faithfulness with respect to construct state derivation, they can equally be ranked relative to the templates with respect to their derivational morphology. Namely, templates which are more faithful with respect to derivational morphology would be less marked than nouns which are less faithful with respect to construct state derivation. That is, the most faithful template PI'EL is less marked than the least faithful noun type, those like *bayit*:

(75) PI'EL << *bayit*-type nouns

Crucially, a ranking like the one above indicates that faithfulness is not necessarily a measure which ranks the templates relative to one another, but rather ranks all types of word forms relative to one another. Given the VCMP requires a markedness hierarchy of verb classes, faithfulness would not necessarily qualify to rank the templates, as it also ranks the templates relative to other word classes, like nouns.¹³ As the VCMP crucially is sensitive to contrast between verb forms in particular, I instead will pursue one predicated on complexity and asymmetry between features of the templates.

To do so, I will draw on comparisons between the morphophonological and morphosyntactic features of each of the templates to asses which are more complex,

¹³Another way to think about the shortcomings of this approach is that faithfulness compares paradigms, and the paradigm which involves the least amount of morphophonological processes to produce one form from another is the least marked (Ashwini Deo, p.c.). As such, faithfulness does not rank individual verb forms on their own and therefore does not qualify as an appropriate metric to determine markedness for the VCMP.

and therefore more marked. This sort of comparison would be over markedness based on formal complexity (Haspelmath 2006, p. 26, Table 1, (3)), where a more marked form is one with additional morphological material. For example, recall from above that the Hebrew verb comprises three main elements, the root, the vocalic rhyme, and any affixal material. These three features can all vary in their complexity in order to generate rankings which order the templates by how marked they are.

For example, templates differ in whether they have affixal material at all. Namely, NIF'AL, HIF'IL, and HITPA'EL all include affixal material where PA'AL and PI'EL do not. Accordingly, the former three templates are more morphologically complex than the latter two, establishing the provisional markedness ranking below:

$$(76) \quad \text{PA'AL, PI'EL} \ll \text{HIF'IL, NIF'AL, HITPA'EL}$$

While it is straightforward to quantify complexity with respect to the affix, the vocalic rhyme does not have a clear means to compare which is more complex. However, consider the contrast between PA'AL and PI'EL from section 2.2. Morphologically, these two templates differ just in their vocalic rhyme. This difference in morphological template additionally generates a contrast between the two templates, where the latter has entailments of agency that the former does not. As such, the latter template is more complex, where the morphological difference in rhyme indicates a more restrictive verb. This ranking updates the provisional markedness ranking as below:

$$(77) \quad \text{PA'AL} \ll \text{PI'EL} \ll \text{HIF'IL, NIF'AL, HITPA'EL}$$

In fact, under the morphological system from Kastner (2020), the kinds of restrictions which the templates place on their arguments are a direct outgrowth of the heads which comprise the template, each of which correspond to a particular morphophonology (e.g. Kastner 2018). For example, the formal specification of PI'EL includes a root $\sqrt{\text{ACTION}}$ which provides the agency entailments. This stands in contrast with PA'AL, which is generated only by unspecified Voice shared by the two templates. Additionally, $\sqrt{\text{ACTION}}$ specifically generates the vocalic rhyme unique to

PI'EL, and therefore indicates that its rhyme is more complex than that of PA'AL. Accordingly, it is possible to use the specifications which the templates place on their arguments to assess their complexity and therefore how marked they are.

As a quantifiable stand-in for these specifications, I mobilize the set of functional heads used in Kastner (2020) to assess how morphologically specified each template is. Nothing about the markedness hierarchy hinges on this specific formal system, but under this system, each functional head both represents a syntactic restriction which the templates provide, as well as phonological complexity (e.g. Kastner 2018). These heads simply build in the intuition that a greater amount of syntactic specification indicates a greater amount of morphological complexity, such as the fact that HIF'IL requires an external argument (through $\text{Voice}_{[+D]}$), or that PI'EL requires an agent (through $\sqrt{\text{ACTION}}$).

I repeat the templatic specifications from Kastner (2020) from (40) here.

- (78)
- a. PA'AL: Voice
 - b. NIF'AL: $\text{Voice}_{[-D]}$; Voice, $p_{[-D]}$
 - c. HIF'IL: $\text{Voice}_{[+D]}$
 - d. PI'EL: Voice, $\sqrt{\text{ACTION}}$
 - e. HITPA'EL: $\text{Voice}_{[-D]}$, $\sqrt{\text{ACTION}}$; Voice, $p_{[-D]}$, $\sqrt{\text{ACTION}}$

(based off of Kastner 2020, p. 29, Table 1.6)

For the sake of exposition, I will assume here that different types of featural specification are ordered, creating strata of markedness which build on another. First, templates with unspecified Voice are less marked than those with specified Voice. Already, the provisional hierarchy in (77) builds in this distinction, between those templates which restrict their arguments and those which do not (i.e. those with unspecified Voice, and those with a specified Voice).

Second, the addition of $\sqrt{\text{ACTION}}$ creates a more marked template than those without it, corresponding to having a more marked rhyme. This ranking is captured in

(77) through PI'EL being more marked than PA'AL. However, this ranking also ranks HITPA'EL as the most marked template. HITPA'EL has additional agency specifications which the other unranked templates do not through the presence of $\sqrt{\text{ACTION}}$:

$$(79) \quad \text{PA'AL} \ll \text{PI'EL} \ll \text{HIF'IL}, \text{NIF'AL} \ll \text{HITPA'EL}$$

Finally, templates which emerge from multiple different sets of functional heads are more marked than those which are not. This parallels a kind of markedness through syntactic complexity, as a single template corresponds to multiple different options for argument structure. Specifically, the compatibility with NIF'AL with figure reflexives (through $p_{[-D]}$) positions it as more complex than those which are not ranked relative to it and are not compatible with figure reflexives, in this case, HIF'IL. After iteratively applying these three rankings, (80) is the final markedness hierarchy based on morphological specification.

(80) **Markedness hierarchy (morphological specification):**

$$\text{PA'AL} \ll \text{PI'EL} \ll \text{HIF'IL} \ll \text{NIF'AL} \ll \text{HITPA'EL}$$

The hierarchy in (80) directly reflects the frequency of the templates across the verb inventory of Hebrew. Haspelmath (2006) specifically argues for a correlation between markedness and frequency, where more frequent forms typically are less marked (see also Haspelmath 1993; Haspelmath et al. 2014; Haspelmath 2016). In strict markedness terms, this kind of markedness constitutes a form of textual markedness, where markedness is measured relative to how rare or common it is in a given text or corpus. Here, I assessed the frequency of each template in heTenTen 2014, measured as the number of tokens of each verb in the corpus, sorted by the template of each of the verbs. This count therefore represents the frequency of each template as the total number of times a verb of that template appears in the corpus. This count is represented in Table 2.2. The frequencies as measured in Table 2.2 therefore generate the hierarchy in (81).

Template	Count
PA'AL	3,993,042
HIF'IL	1,944,972
PI'EL	1,737,459
NIF'AL	895,922
HITPA'EL	721,447

Table 2.2: Count of verbs by template in heTenTen 2014

(81) **Markedness hierarchy (frequency):**

PA'AL << HIF'IL << PI'EL << NIF'AL << HITPA'EL

The only difference between this hierarchy and the one in (80) is the relative ordering of HIF'IL and PI'EL. In the absence of any additional strong evidence of one over the other, I choose here not to order them.¹⁴

Therefore, the final hierarchy, based on morphological complexity and frequency of occurrence, is as in (82).

(82) **Markedness hierarchy for the Hebrew templates (final):**

PA'AL << HIF'IL, PI'EL << NIF'AL << HITPA'EL

As noted above, this hierarchy is just emergent from all possible pairwise comparisons, and should not be taken as an absolutely hierarchy. Though, as a hierarchy, it could potentially predict tripartite distinctions or larger, this dissertation only deals in pairwise comparisons, and as such the hierarchy in (82) represents the basis on which I compare the templates in pairs in each case study. Therefore, the VCMP predicts truth conditional strength differences between verbs based on their templatic membership along this hierarchy, with verbs in templates higher on hierarchy having stronger meanings than those lower on the hierarchy.

¹⁴For my purposes here, a lack of ordering does not substantively affect any of the case studies which I investigate, as none of them involve a direct contrast between PI'EL and HIF'IL.

2.5 Conclusion

In this chapter, I have proposed a principle for linking verb forms with meanings, the Verb Complexity Mapping Principle. Under this principle, more marked verbs have stronger truth conditions than less marked verbs, relative to a hierarchy which orders the possible verb forms in a given language. For Hebrew, I will use the VCMP to demonstrate how differences in templates determine differences in meaning between verbs which share a root. In turn, the results which I will present here will indicate that the templates do have a semantic function, contra the prevailing argument structural view. Namely, the templates determine the strength of the verbs which they derive relative to the other templates.

I have also motivated a markedness hierarchy for the Hebrew templates based on a combination of the morphological specification of each of the templates as well as their frequency in corpus. The VCMP will operate over this hierarchy and predict contrasts between verbs in those templates. Verbs in templates higher on the hierarchy will have more specific meanings than those lower on the hierarchy following the definition of strict strength in (62). This approach stands in stark contrast to the syntactic approach to the Hebrew templates presented in this chapter, which posits that the main difference between the templates are the syntactic restrictions they place on the verbs they form. The remainder of this dissertation will assess case studies where the VCMP predicts a semantic contrast, and demonstrate that those verbs in a more marked template will have stronger meanings than those in the less marked template, and thus forward a semantic view on the templates. In the next chapter, I will present the first of these, a roaming alternation involving HITPA'EL, the most marked template, and demonstrate that the VCMP both predicts the semantic contrast between the HITPA'EL roaming verb and a non-roaming PA'AL correlate, and explains the difference between the two forms better than the argument structural analyses presented here.

Chapter 3: Derived roaming verbs and HITPA'EL: templatic alternation as independently meaningful

3.1 Introduction

In the overview of the Hebrew template system from the previous chapter, the majority of the literature converged on the idea that templates are syntactic objects. They are anticipated to affect properties of the verb's syntax, namely its argument structure, and the extent to which they are correlated with elements of meaning is only where a given syntactic structure has a certain meaning. As such, ample attention has been given to how templatic membership participates in alternations such as the causative-inchoative alternation or the reflexive alternation, each of which have explicit arity effects which in turn affect the verb's meaning. In this chapter, I will present data suggesting that this view of the templates cannot capture all of the types of regular alternations they produce, and instead, that templatic alternation can operate only to change the meaning of the verb for a given root. I will suggest that the templates can contrast with one another morphologically to indicate a strengthening relationship between two verbs in different templates.

I will be presenting an analysis of a roaming alternation where PA'AL alternates with HITPA'EL, exemplified here in (83).

- (83) a. *halax-ti* *l=a=xanut*
 walk-1SG.PST to=the=store
 'I walked to the store.'
- b. *hithalax-ti* *b=a=rexov*
 REFL.walk-1SG.PST on=the=street
 'I walked around on the street.' (constructed)

The most canonical of these are given in (84), along with their related non-roaming motion forms in PA'AL.

- (84) a. *hithalex* ‘walk about’; *halax* ‘walk’
 b. *hitrocec* ‘run about’; *rac* ‘run’
 c. *hitnoe* ‘a’ ‘roam about’; *na* ‘roam’
 d. *hitnoded* ‘wander about’; *nad* ‘wander’
 e. *hit’ofef* ‘fly about’; *’af* ‘fly’

These verbs do not allow the *l=* marked argument possible with its non-roaming counterpart (which I will call ‘dative marking’):

- (85) **hithalax-ti* *l=a=xanut*
 REFL.walk-1SG.PST to=the=store
 (intended) ‘I walked about/roamed to the store.’ (constructed)

On its face, this alternation does seem strictly syntactic, preventing dative marking as well the internal goal argument, and in turn the verb has a roaming meaning. In fact, this kind of analysis would accord well with HITPA’EL’s general function as a canonical intransitive and reflexive template, unified under some process of detransitivization (e.g. Reinhart 2002; Reinhart & Siloni 2005, or the Voice_[−D] head from Schäfer 2008; Kastner 2020).

Instead, I will demonstrate that these roaming verbs can appear with dative-marked arguments, including goals. Specifically, following a corpus analysis, I will show that HITPA’EL roaming motion verbs are indistinguishable from PA’AL non-roaming motion verbs with respect to how frequently they appear with a dative-marked argument. Instead, I will attribute the ungrammaticality of some datives with HITPA’EL roaming motion verbs to a property I term *antimaximality*, which ensures that these verbs can never be maximal, or rather, do not have a largest event in their denotation.

In the next section, I will overview the literature on event lexicalization, specifically focusing on how verbs express portions of a motion event, and introduce the notion of goal resistance as characterizing these roaming motion verbs. In section 3, I

will present a corpus study on the distribution of dative-marked arguments with these roaming motion verbs, which will in turn demonstrate that HITPA'EL roaming motion verbs can appear with a dative-marked goal. Section 4 will introduce the notion of antimaximality as characterizing these roaming verbs, and show that the kinds of dative-marked goals that can appear with HITPA'EL roaming motion verbs are exactly those which are consistent with antimaximality. Finally, section 5 will return to the issue of templatic membership, and will discuss how the contrast between PA'AL and HITPA'EL can be characterized in terms of a truth-conditional strength difference.

3.2 How verbs express motion

3.2.1 Event framing

A key factor in determining the properties of the HITPA'EL roaming motion verb, especially as it relates to its ability to express a goal through dative marking, is an understanding of how verbs can express the various elements of a motion event. Intuitively, a motion event involves at least three components: a figure (or theme) that undergoes motion, a path which that motion occurs on (which may include a goal, a source, or a route between them), and a manner which that motion is performed by. A great deal of work has explored how those components of motions are expressed in different languages, starting with the seminal work from Talmy (1985, 2000) (see also Aske 1989; DeLancey 1989; Folli 2002; Slobin 2003; Folli & Harley 2006, 2020; Croft et al. 2010; Zlatev et al. 2021, among many others). Specifically, this question weighs on the issue of how these verbs lexicalize these elements, through what grammatical strategies they can be expressed, and under what conditions those grammatical strategies are mobilized.

A basic distinction in how languages express an event of motion is how they frame the event, or rather, what elements of that event are lexicalized in the verb, and what elements are introduced by grammatical elements external to the verb. Talmy (1985, 2000) proposed there are two types of languages regarding how the path is

expressed. In *verb-framed* languages the path is part of the meaning of the verb, while in *satellite-framed* languages the path is in some satellite to the verb. Manner, the other major component of motion (beyond the moving figure), is expressed wherever path is not, so in the verb in satellite framed languages and in some satellite in verb-framed languages. Consider the examples in (86).

- (86) a. *Je suis entré dans la maison en boitant*
 I am entered in the house in limping
 ‘I entered the house limping.’
 b. I limped into the house.

(French, English, Beavers et al. 2010, p. 333, (1a-b))

In (86a), the oblique expression *en boitant* ‘limping’ represents the manner of motion, while the verb *entré* represents directed motion, and thus lexicalizes the path, the goal of which is *dans la maison* ‘in the house’. In contrast, in (86b), the verb *limp* only represents the manner of motion, while the path is introduced obliquely with the prepositional phrase *into the house*. On Talmy’s initial classification, these two classes are meant to be complementary, such that the verbs of a language should lexicalize only one of path or manner, a kind of manner-result complementarity (e.g. Rappaport Hovav & Levin 2010).

The two classes of languages in the Talmyan typology divide languages based on which component between the path and the manner is expressed by the verb, and which is expressed by the satellite. However, many languages permit expressions which do not neatly fall into this binary. Such a language might be those which permit serial verb constructions, where the manner and the path are both expressed by main verbs. Such is the case for Thai, exemplified in (87).

- (87) *chán dǝən khâam thanǎn khâw paj naj sǎan*
 I walk cross road enter go in park
 ‘I walked across the road and into the park.’

(Thai, Zlatev & Yangklang 2004, p. 160, (1))

In (87), both the manner and the path are expressed using main verbs, *dəən* ‘walk’ in the case of the former, and *paj* ‘go’ in the case of the latter. As such, a language with serial verb constructions like Thai cannot fit into Talmy’s binary typology. Instead, such languages have been analyzed as falling into a third category of *equipollently-framed* languages, where both manner and path are lexicalized in main verbs (Slobin 2003; Zlatev & Yangklang 2004; Chen & Guo 2010; van Putten 2017, among others, though see Talmy 2009 to the contrary).

The Talmyan typology, even with the addition of equipollent framing, further breaks down from the fact that languages regularly eschew their typological class, with verb-framed languages permitting satellite-framed expressions, and vice versa. Slobin (2003) classifies Hebrew as a verb-framed language (see also Doron & Dubnov 2017; Rappaport Hovav 2019), as exemplified in (88).

- (88) *hu nixnas* *l=a=xanut* *b=rica*
he MID.enter[3SG.M.PST] to=the=store in=running
‘He entered the store running.’ (constructed)

However, Hebrew also permits satellite-framed structures, as shown in (89).

- (89) *halax-ti* *l=a=xanut*
walk-1SG.PST to=the=store
‘I walked to the store.’ (constructed)

Just like Hebrew, many languages eschew their Talmyan class, allowing both path-framed and satellite-framed structures. Therefore, the notion of a language level classification of verb framing lacks utility, as it cannot accurately describe the range of expressions available to a given language or the conditions under which those expressions might arise.

Beavers et al. (2010) approach the question not from the perspective of which typological class a given language may fall into, but rather from the motion-independent

grammatical strategies a language might have to express notions like manner and path. Namely, a language would have a preference for either satellite framing or path framing based on what verbs in those languages lexicalize. So, French in (86a) prefers structures where the manner is expressed obliquely because its verbs lexicalize a path, while English in (86b) prefers structures where the path is expressed obliquely because its verbs lexicalize a manner. These preferences constitute language-specific lexicalization tendencies, such that these verbs specify for one or the other as a part of their meaning. Additionally, these two languages differ in their lexical options for satellite lexicalization, such that English has a generally-applicable goal expression in *to*, while French does not.

However, languages also have other generic expressions of manner or result which can be used in motion contexts to express manner of motion or path (a subtype of result) respectively. Where such elements exist, languages may mobilize them even if such an expression is not compatible with how a language lexicalizes path or manner in the verb. A key example of this is the case of delimiting expressions (Beavers 2008; Gehrke 2008), where expressions that represent a general limit (e.g. of an event participant) can take on the interpretation of a goal in the context of a motion event. Such is the case of Japanese *-made* ‘until’ in (90).

- (90) *John-wa kishi-made oyo-ida*
 John-TOP shore-until swim-PST
 ‘John swam until shore. (Beavers et al. 2010, p. 346, (23d))

Typically, Japanese is a verb-framed language, and as such typically expresses the path as lexicalized in the verb. However, in (90), *-made* is able to mark a path as part of an oblique, contrary to its general pattern. In fact, *-made* contrasts with another goal marker *-ni* ‘to’, in that *-ni* is fully ungrammatical when marking a spatial goal:

- (91) ??*John-wa kishi-ni oyo-ida*
 John-TOP shore-to swim-PST
 (intended) ‘John swam to the shore’ (Beavers et al. 2010, p. 342, (19b))

Where *-made* and *-ni* differ is that *-made* is not confined to spatial interpretations, and instead is able to express boundaries in many other domains, such as the temporal or numerical domains:

- (92) a. *Ohiro-made kore-o shide-kudasai*
 noon-until this-ACC do-please
 ‘Please do this by noon.’
- b. *Kono hooru-wa nisen-nin-made haieru*
 this hall-TOP 2000-CLF.PEOPLE-until hold
 ‘This hall can hold 2000 people.’ (Beavers et al. 2010, p. 346, (24a,c))

Thus, *-made* serves as a general delimiter expressing a boundary on some dimension, and how that limit which it marks is interpreted is dependent on the dimension which it is predicated over. This includes the spatial domain in (90), where the limit of a path is its goal. Thus, because Japanese has a generic delimiter, it is able to mobilize that grammatical marker to mark a goal, and therefore generate satellite-framed structures which are otherwise uncharacteristic of Japanese.

Thus, if a language has the grammatical means of expressing path or manner, it should be able to do so. This both includes motion-specific elements, such as goal makers, as well as more general elements which are merely compatible with motion. In particular, the verb has a special status, being both obligatory in its clause and able to express either path or manner, and as such, it will tend to be the default way to do so, generating preferences for verb framing or satellite framing. However, more complex expressions of path or manner, such as limiting expressions, can also be used, but are dispreferred due to their complexity.

3.2.2 Goal resistance

On the backdrop of Beavers et al.’s framework for understanding variability in motion expressions, the ungrammaticality of the dative-marked goal in (85) could reasonably come down to an incompatibility of the meaning of the roaming motion

verb with a goal expression. This case in Hebrew has a direct parallel in Spanish, where Bassa Vanrell (2013) analyzed a similar set of motion verbs she termed *goal resistant*. Goal resistance is a grammatical restriction on path expression where a verb is ungrammatical with the language’s canonical goal marker. This property is demonstrated for Spanish in (93), with the canonical goal marker *a* ‘to’.

- (93) # *Juan deambuló a la tienda*
 Juan wandered to the store
 ‘Juan wandered to the store.’ (Bassa Vanrell 2013, p. 34, (36a))

Goal resistance exists on a continuum, where certain verbs which typically resist a goal can allow one in certain contexts where a goal is presupposed:

- (94) a. # *Juan patinó a la tienda*
 Juan skated to the store
 ‘Juan skated to the store.’
 b. (How did Juan get to the store?)
 patinó a la tienda
 skated to the store
 ‘[He] skated to the store.’ (Bassa Vanrell 2013, p. 35, (38))

However, even verbs which are completely goal resistant can still permit expressing a goal if it is not marked with *a*. *Hasta* ‘until’ typically expresses temporal or numerical extent, as in (95). In some motion contexts, *hasta* is also able to express a goal, even with verbs that are completely goal resistant, as in (96).

- (95) a. *Juan correrá hasta mañana*
 Juan will.run until tomorrow
 ‘Juan will run until tomorrow.’
 b. *Juan correrá hasta tres metros*
 Juan will.run until three meters
 ‘Juan will run up to three meters.’ (Bassa Vanrell 2013, p. 19, (16a-b))

- (96) a. **La chica bailó al otro lado de la habitación*
the girl danced at.the other side of the room
‘The girl danced to the other side of the room.’

(Bassa Vanrell 2013, p. 3, (4b))

- b. *La chica bailó hasta el otro lado de la habitación*
the girl danced until the other side of the room
‘The girl danced to/as far as the other side of the room.’

(Bassa Vanrell 2013, p. 3, (5b))

In order to capture these three facts about goal resistance in Spanish, Bassa Vanrell first distinguishes between argument and adjunct expressions of goal, following Beavers (2008). The dative marker *a* syntactically patterns like an argument, while prepositions like *hasta* pattern like an adjunct. Second, the denotations for *a* and a goal-permissive verb, here *subir* ‘go up’, are in (97).

- (97) a. $\llbracket \textit{subir} \rrbracket = \lambda g \lambda e. \exists p [go.up(e) \wedge path'(e, p) \wedge goal''(e, g)]$

(Bassa Vanrell 2013, p. 40, (41))

- b. $\llbracket a \rrbracket = \lambda P.P$

(Bassa Vanrell 2013, p. 43, (44))

Because *a* functions as an argument marker, it does not semantically introduce the goal. Instead, a goal-permissive verb itself selects for a goal in its denotation (*g* in (97a)) and the dative marker denotes the identity function, passing up the argument to compose with the verb itself and saturate the goal. Only verbs which explicitly select for goals can therefore accept an *a*-marked goal. Otherwise, the goal will be passed up by *a* but there is no open argument for it to saturate.

Third, due to their status as an adjunct, other prepositions which can mark goals are independently contentful. Specifically in the case of *hasta*, its wide range of functions can be sufficiently generalized as marking a limit, shown in the denotation in (98).

$$(98) \quad \llbracket hasta \rrbracket = \lambda g \lambda P \lambda e. \exists x [MAX_E(P)(e) \wedge Limit'(e, g, x)]$$

(Bassa Vanrell 2013, p. 41, (42a))

In this context, a spatial limit is a goal, mapped onto the largest event as determined by the maximalization operator MAX_E . g here is the limit of some scale of measurement represented by x . As the event e is an event of motion, the limit is construed as a spatial limit. The maximalization operator MAX_E then picks out the largest event which satisfies the predicate P , which here represents an event type (e.g. *dance'*, *run'*, *skate'*). Because e is the largest such event of its type, the limit g marks the endpoint of the event e , thus creating a goal. As such, even verbs which do not explicitly have an open goal argument can occur with one when introduced prepositionally, as they do not rely on their own argument selection to support a goal, but rather can support one through the preposition. Finally, context-dependent overriding of goal resistance as in (94) is derived when a goal-resistant verb is coerced into selecting for a goal. In a context where a goal is presupposed or strongly supported by the context, as in (94b), where the goal is introduced earlier in the discourse, the verb can undergo a process of coercion which alters its denotation to one which can accept a goal.

3.3 The syntax (or lack thereof) of Hebrew goal resistance

3.3.1 The syntactic approach to goal resistance

Based just on the data from (83), Hebrew goal resistance would follow exactly the same lines as Spanish goal resistance as analyzed in Bassa Vanrell (2013). The dative clitic $l=$ functions as an argument marker, which passes up the goal and tries to saturate an open goal argument of the HITPA'EL roaming verb, which does not exist. Fundamentally then, goal resistance is a form of enforced intransitivity, where the verb does not permit an argument in the syntax.

This kind of enforced intransitivity accords with the syntactic analysis of the HITPA'EL template as a reflex of Voice_[-D] from Kastner (2020). In general, HITPA'EL

serves as a canonical intransitive template, forming verbs such as inchoatives, reflexives and reciprocals:

(99) Inchoative:

- a. *Ron bifel* *et ha=marak b=a=femef*
 Ron INTNS.cook[3SG.PST.M] ACC the=soup in=the=sun
 ‘Ron cooked the soup in the sun.’
- b. *ha=marak hitbafel* *b=a=femef*
 the=soup REFL.cook[3SG.PST.M] in=the=sun
 ‘The soup cooked in the sun.’

(100) Reflexive:

- a. *Yicxak ’iper* *et Tomi*
 Tomi INTNS.make.up[3SG.PST.M] ACC face=PL.3POSS
 ‘Yicxak put makeup on TOmi.’
- b. *Tomi hit’aper*
 Tomi REFL.make.up[3SG.PST.M]
 ‘Tomi put makeup on (himself).’

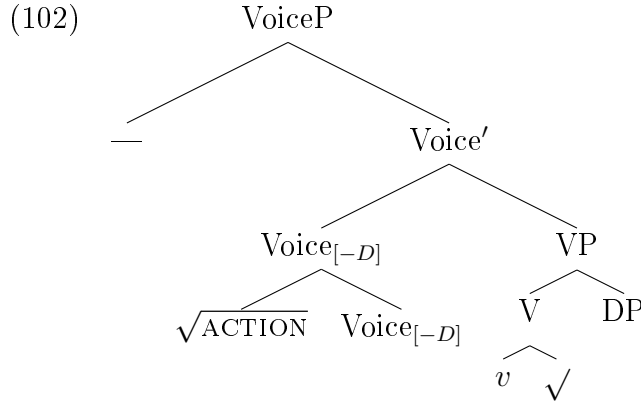
(101) Reciprocal:

- a. *Yosi xibek* *et jager*
 Yosi INTNS.hug[3SG.M.PST] ACC Jagger
 ‘Yosi hugged Jagger.’
- b. *Yosi v=jager hitxabk-u*
 Yosi and=Jagger REFL.hug-3PL.PST
 ‘Yosi and Jagger hugged.

(Kastner 2017, p. 3, (4)-(6))

Kastner (2020) captures the intransitivity of HITPA’EL by placing specifier restrictions on the heads which reflect the morphological form of the template. Specifically, for the intransitive forms above, Voice_[−D] restricts a external argument *qua* canonical subject from Merging in its specifier, thus leaving the verb with just its single, internal argument, which then raises to subject position. This structure is given schematically

in (102).¹



(based off of Kastner 2020, p. 105, (78c))

Despite their different semantics, this one structure is intended to derive all three different classes of intransitive verbs. Kastner (2017) attributes this structure's ability to do so to an interaction between the intransitivity enforced by $\text{Voice}_{[-D]}$ and the lexical semantics of the root.² Specifically, based on which root enters the structure in (102), the resultant verb will differ based on the lexical semantic class of that root. A root representing a state will generate an inchoative, naturally reflexive or self-oriented root will generate a reflexive verb, and an other-oriented root will generate a reciprocal.

On this principle, HITPA'EL roaming motion verbs would derive their meaning through an interaction between the intransitivity of $\text{Voice}_{[-D]}$ which characterizes the HITPA'EL template and the fact that the roots forming these verbs are all motion roots. The roaming meaning would be dependent on the intransitive syntax; without it, there would be no way to distinguish a roaming verb from another type of manner

¹Kastner's derivation of HITPA'EL also involves the introduction of an $\sqrt{\text{ACTION}}$ root which corresponds to some agency entailments. As they are not relevant to the issue of HITPA'EL's intransitivity, I put it aside here.

²Kastner (2019) notes a similar connection between syntactic requirements and root class which factors into the realization of inchoatives in HIF'IL, potentially indicating a general principle guiding the realization of roots in different syntactic contexts.

of motion verb on the basis of the syntax. The syntactic analysis of HITPA'EL and the goal resistance of its motion verbs therefore predicts that the roaming verbs in (84) should all fully resist dative-marked goals, as excluding one would be a precondition for the roaming meaning to arise.

3.3.2 Datives and HITPA'EL: a corpus study

Here, I am going to demonstrate that this prediction from the syntactic approach to the templates does not hold. Specifically, I am going to show through a corpus analysis of the distribution of datives with the verbs in (84) that they collocate with a significant amount of dative-marked nouns, and that they can accept dative-marked goals much like their non-roaming counterparts in PA'AL. To motivate this approach, consider the sentence in (103) with the verb *hitrocec*.

- (103) *maca-ti et acm-i mitrocec l=a=supermarket,*
 find-1SG.PST ACC REFL-1SG REFL.run[PRS.SG.M] to=the=supermarket
l=a=bet merkaxat, l=a=hor-im, l=a=bayit, l=a=bet xolim,
 to=the=pharmacy to=the=parent-PL.M to=the=house to=the=hospital,
v=xozer
 and=return[PRS.SG.M]
 'I found myself running around to the supermarket, the pharmacy, my parents,
 home, the hospital, and back.' (corpus)

Here, a HITPA'EL roaming verb not only appears with several conjoined dative-marked nouns, but all of those dative-marked nouns are goals, as each serves as a place where motion putatively ends. As such, there are clearly some conditions under which the HITPA'EL roaming verbs are able to appear with dative marking, indicating that the syntactic approach would not be tenable.

It is important to note that both Bassa Vanrell (2013) and Kastner (2017) would allow for a degree of coercion within their analyses, under which a typically goal-resistant verb could eschew their typical syntactic property of being intransitive. Bassa Vanrell includes a process of coercion which generates a goal-permissive verb

out of a goal-resistant verb in the case where a goal is presupposed or supported by the context. Similarly, Kastner (2017) notes that context is able coerce the root into another class, but provides little explication as to the process. Regardless, a coercion process of some kind would permit the HITPA'EL goal resistant verbs to take a dative-marked goal in contexts where, for instance, one would be particularly salient, such as the long string of goals essentially constituting a path in (103).

To assess whether data like (103) is in fact reflective of the broader distribution of these verbs with dative marking, I conducted a corpus study comparing the collocative frequency of motion verbs in HITPA'EL to those in PA'AL. Specifically, I conducted a frequency analysis of dative collocation comparing HITPA'EL and PA'AL, using the heTenTen 2014 corpus in Sketch Engine (Baroni et al. 2009). In this study, both HITPA'EL and PA'AL are represented by 30 non-caused motion verbs within their respective template. PA'AL was selected as the template for comparison as all of the canonical goal-resistant verbs in (84) have a counterpart in PA'AL, and PA'AL is the most unmarked template. The set of verbs in both templates are being compared to a sample of 100 verbs pulled randomly from the 1000 most frequent verbs in Hebrew, not including verbs represented in the test groups.³ This is meant to represent a baseline of collocational frequency with $l=$ in Hebrew, i.e. how likely is a verb to collocate with $l=$ for just being a verb of Hebrew. For each of these sets of verbs, I calculated the average number of tokens which appeared directly adjacent to a noun marked with $l=$ ⁴ and divided it by the total number of tokens for that verb in the corpus to generate a mean collocative frequency.⁵

Thus, if the intransitivity of HITPA'EL is truly conditioning the roaming mean-

³All verbs and their frequencies can be found at <https://osf.io/rt74x/>.

⁴While the choice to limit the range of collocation to just adjacent to the verb will artificially exclude some legitimate instances of $l=$, it also creates an conservative estimate of instances of means of collocation for which there is no *a priori* reason to believe will differ by template. As such, the results will not be affected.

⁵This method generates a mean collocational frequency for the templates which are not weighted by the raw frequencies of the individual verbs, to control for the effect of an individual verb within the class having an outsized effect on the behavior of the whole class.

ing of the HITPA'EL roaming verbs, then they should have a significantly smaller mean collocation with the dative clitic $l=$ than that of the motion verbs in PA'AL. This result will in turn indicate that HITPA'EL with a motion root does sufficiently constrain dative marking, and data such as (103) are likely outside the normal distribution of a HITPA'EL roaming verb. However, if HITPA'EL and PA'AL motion verbs do not differ significantly in their mean collocation with $l=$, then the intransitivity of HITPA'EL cannot be the determining factor in deriving the HITPA'EL roaming verbs, as they are not sufficiently intransitive or goal resistant. Figure 3.1 shows the means of collocation

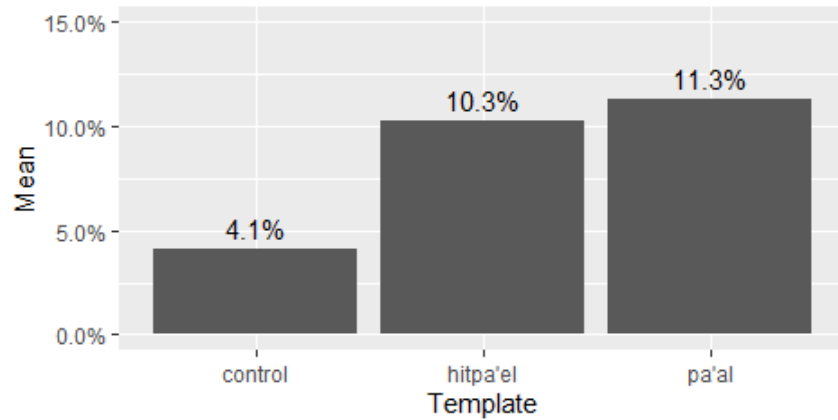


Figure 3.1: Means of dative marking by template

with $l=$ within 0 tokens of the verb for each of the three groups. Both templates have substantially higher rates of collocation than the baseline group. This result is anticipated, as motion verbs would naturally mobilize the dative in more cases to mark parts of the motion event, like goals, while non-motion verbs would be restricted to more adjunctive uses of the dative. The templates are not substantially different from one another, with a difference of only 1%, indicating that there is no real collocational difference by template.

To confirm these two patterns, I fit a quasibinomial logistic regression estimating the effect of template on collocation with $l=$, weighted by the frequency of each verb, and with the baseline group as the reference group. The positive significant

	Log odds	Std. Error	t-value	p-value
(Intercept)	-3.1432	0.1908	-16.48	0.0000***
template HITPA'EL	0.9785	0.4006	2.44	0.0157 *
template PA'AL	1.0866	0.2087	5.21	0.0000***

Table 3.1: Quasibinomial logistic regression estimating effect of template on proportion of $l=$. Effect size represents how much more likely a verb in its template is to appear with $l=$ from the baseline.

effects ($\log \text{ odds} > 0$; $p \leq 0.05$) of both templates indicate that motion verbs in both templates do collocate with $l=$ more than an average verb of Hebrew. However, just as the means are close to one another, the effects of each template are of roughly the same magnitude. Post-hoc estimated marginal means analysis reveals that the difference between the two templates is not significant ($\log \text{ odds} = -0.108$; $p = 0.95$), confirming that HITPA'EL and PA'AL mobilize roughly the same amount of dative marking.

3.3.3 Exception 1: variability in motion roots

One potential confound in the data from the previous section is the fact that the verbs were included in the sample based just on whether they fell under the broad class of motion verbs. Levin (1993) identifies at least five different classes which fall under ‘verbs of motion’ for English (sect. 51). In fact, the verbs which alternate into HITPA'EL to create a roaming verb are all manner of motion verbs (sect. 51.3). Therefore, it is possible that the sample of HITPA'EL motion verbs considered above was heterogenous, including several different subclasses of motion verbs with different semantic properties which would not lead to goal resistance or roaming.

One such class would be the class of HITPA'EL inherently directed verbs, the most canonical of which are in (104).

- (104) a. *hitkarev* – ‘approach’
b. *hitkadem* – ‘move forward’

- c. *hitraxek* – ‘distance’
- d. *hitparec* – ‘burst’
- e. *hitaleh* – ‘go up’
- f. *histalek* – ‘be ejected, get out’
- g. *hidarder* – ‘go downwards’
- h. *hitganev* – ‘‘steal’ into’
- i. *hitgale’a* – ‘erupt’
- j. *hiftalfel* – ‘descend on a rope’

In direct opposition to the HITPA’EL roaming verbs, these verbs obligatorily require a prepositionally-marked goal or source, and are ungrammatical without one:

- (105) a. *hitparac-ti* *l=a=dira*
 REFL.burst-1SG.PST to=the=apartment
 ‘I burst into the apartment.’
- b. *hitparac-ti* *me=ha=xeder* *fel-i*
 REFL.burst-1SG.PST from=the=room POSS-1SG
 ‘I burst out of my room.’
- c. **Iris hitparc-a* *etmol*
 Iris REFL.burst-3SG.PST.F yesterday
 (Intended) ‘Iris burst out yesterday.’ (constructed)

Such verbs, which require a goal or a source, are equivalent to the class of inherently directed motion verbs from Levin (1993, sect. 51.1). This class of HITPA’EL inherently directed verbs stand in direct contrast to the HITPA’EL roaming motion verbs. While the former require a prepositionally-marked argument to be grammatical, the latter at the very least have several contexts which do not permit them.

Therefore, the inclusion of subclasses of motion verbs which do not share properties with the HITPA’EL motion verbs into the sample from the previous analysis might have skewed the mean collocation with the dative clitic *l=*. If such were the

case, then treating the verbs from (84) as a class separate from other classes of HITPA'EL motion verbs should control for the variability in motion verb classes which are realized in HITPA'EL. To that end, I compared the mean collocational frequency of $l=$ for the HITPA'EL roaming verbs in (84) to the mean collocational frequency of $l=$ for the HITPA'EL inherently directed verbs in (104). Following the same methodology from the previous analysis, for each set of verbs, I calculated the number of tokens of each verb which appeared directly adjacent to a noun marked with $l=$ and divided it by the total number of tokens of each verb in the corpus.

In this analysis, the syntactic approach to the templates would predict that the HITPA'EL roaming verbs should have a mean collocation with $l=$ indistinguishable from 0. On that approach, the intransitive syntax of the roaming verbs determines their roaming meaning, and because of that connection, belonging to the class of roaming verbs should correlate to rejecting a dative-marked argument. Toward that end, Figure 3.2 shows the mean collocational frequency with $l=$ for both the HITPA'EL roaming motion verbs and the HITPA'EL inherently directed verbs. The 9.4% difference

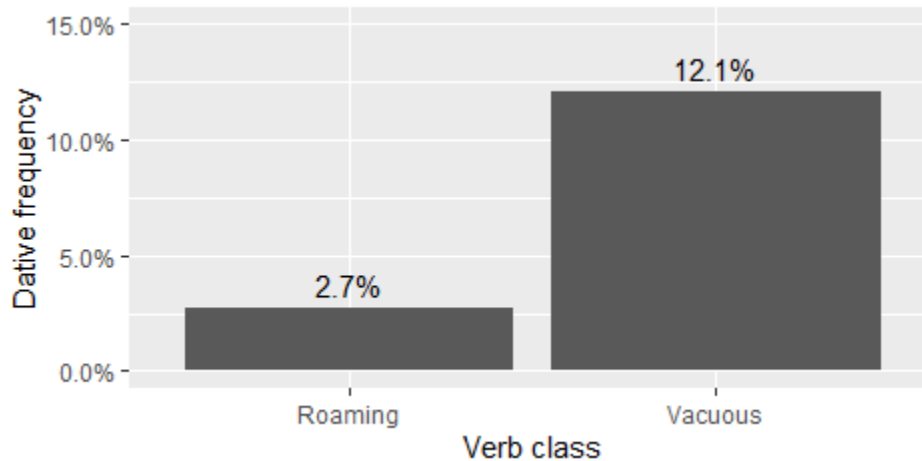


Figure 3.2: Frequency of adjacent datives by motion verb class

between the dative collocational frequency of roaming verbs and inherently directed verbs is significant following a two-sample t-test ($t = -2.393$, $p = 0.045$), indicating that the two classes do mobilize dative marking differently. This result is not

unanticipated, as the two sets of verbs do belong to different classes, with obviously different distributions with respect to dative marking. However, the 2.7% frequency of adjacent datives with the roaming verbs is not the 0 or near-zero anticipated by the syntactic approach, and a one-sample t-test confirms that this mean frequency is significantly higher than the anticipated mean of 0 ($t = 3.225$, $p = 0.032$). As such, the HITPA'EL roaming verbs do significantly mobilize dative marking just as the inherently directed verbs do, and significantly enough to indicate that HITPA'EL roaming verbs can grammatically permit datives.

3.3.4 Exception 2: other types of datives

The question then remains of whether that 2.7% dative collocational frequency actually represents dative-marked goals, as would contradict a syntactic analysis of the goal resistance of the HITPA'EL roaming verbs. Dative marking serves a variety of functions in Hebrew, and Ariel et al. (2015) identifies six distinct functions for dative marking. First, verbs can lexically specify for their direct object to be marked with a dative:

- (106) *ani e-ezor l-ax maxar*
 I 1SG.FUT-help to-2SG.F tomorrow
 'I will help **you** tomorrow.' (Ariel et al. 2015, p. 260, (3))

Second, datives can mark the possessor in a predicative possession construction introduced with *yef* 'have/there is':

- (107) *ze ma she=yef l-i*
 this what COMP=POSS to-1SG
 'This is what **I** have.' (Ariel et al. 2015, p. 260, (4))

Third, datives mark the recipient of a transfer verb such as *natan* 'give':

- (108) *az nata-ta l-a et ha=telefon shel maya*
 so give-2SG.M.PST to-3SG.F ACC the=telephone POSS Maya
 'So you gave **her** Maya's phone number?' (Ariel et al. 2015, p. 260, (5))

Fourth, datives can mark a possessor in an external possession construction, where the possessor is not introduced by canonical possession markers such as *shel* ‘POSS/of’ or *yef*:

- (109) *hi lo roc-a she=ye-xatet-u l-a b=a=xayim*
 she NEG want-3SG.F.PRS COMP=3PL.FUT-poke-PL to-3SG.F in=the=life
 ‘She doesn’t want people to rummage around in **her** life.’

(Ariel et al. 2015, p. 261, (6))

Fifth, datives can mark non-core affectees such as benefactors:

- (110) *hexlaf-ti l-ax ot-o*
 CAUS.change-1SG.PST to-2SG.F ACC-3SG.M
 ‘I changed him **for you**.’ (Ariel et al. 2015, p. 261, (7a))

Finally, a dative can co-refer with one of the other participants in the sentences to provide a sense of nonintentionality or casualness to the sentence:

- (111) *ve=’ata ka=ragil roe le-xa srat-im*
 and=you.2SG.M as=normal watch.PRS.M to-2SG.M movie-PL
 ‘And you, as usual are (**leisurely**) watching movies?’

(Ariel et al. 2015, p. 261, (8))

The crucial property which Ariel et al. (2015) adjudicated is how much of an argument or adjunct each of these datives are. They placed each of these functions of dative marking onto a continuum from most argument-like to most adjunct-like, depending on the diagnostic. Of all of these functions of dative marking in Hebrew, the category into which dative-marked goals fall are governed datives, as they fail the Hebrew *do so/’asa oto davar* anaphora test, which distinguishes governed datives as the most argument-like compared to the rest of the functions of dative marking:

- (112) * *Yosi halax l=a=xanut ve=dana as-ta*
 Yosi walk[3SG.M.PST] to=the=store and=Dana do-3SG.F.PST
oto davar l=a=mis'ada
 same thing to=the=restaurant
 (intended) ‘Yosi walked to the store and Dana did the same thing (walked
 to the restaurant).’ (constructed)

The question then becomes whether the datives which HITPA’EL motion verbs can appear fall into any of these other, less argument-like categories, which are not goals.

To assess this, I collected the top ten most frequent dative-marked nouns which collocate with the HITPA’EL roaming motion verbs and are directly adjacent. If the datives which do appear with HITPA’EL roaming verbs are not regularly goals, then the nouns which appear with them most frequently should not be those readily interpretable as a goal. Table 3.2 shows these nouns. While a few of these nouns

Noun	Translation	Count
<i>'orex</i>	length	64
<i>kivun</i>	direction	57
<i>clil</i>	sound	27
<i>kecev</i>	pace	26
<i>cad</i>	side	22
<i>hana'a</i>	pleasure	17
<i>merxak</i>	distance	14
<i>rega'</i>	moment	14
<i>derex</i>	path	14
<i>makom</i>	place	13

Table 3.2: Top 10 most frequent dative-marked nouns collocating with HITPA’EL roaming verbs

are not typical motion participants, such as *clil* ‘sound’ and *hana'a* ‘pleasure’, both nouns which can convey a manner of motion when marked as a dative, many do in fact represent parts of a motion event. *kivun* ‘direction’ indicates a direction of motion as well as a goal towards which the motion oriented. Both *'orex* ‘length’ and *derex* ‘path’ indicate a path. Most notably is *makom* ‘place’ which naturally functions as a

goal, which I exemplify below.

Based on these data, HITPA'EL motion verbs do in fact mark goals and other parts of the motion event as datives, and as such, the frequency which HITPA'EL roaming verbs collocate with dative-marked nouns from Figure 3.2 does successfully represent a frequency of collocation with goals.

3.3.5 Discussion

All of the data presented in this section points towards rejecting the syntactic approach to the templates and goal resistance of HITPA'EL. Under that approach, the roaming meaning of the HITPA'EL roaming verbs is dependent on their inability to accept dative-marked goals. But, the data indicate that the HITPA'EL roaming verbs do mobilize dative marking, and in particular, dative marked goals. HITPA'EL motion verbs are indistinguishable from non-HITPA'EL motion verbs in their collocation of datives. Even controlling for variability in the motion roots, HITPA'EL roaming motion verbs still accept a significant number of datives, and those datives mark parts of the motion event, up to and including actual goals.

As such, the HITPA'EL roaming motion verbs are not syntactically goal resistant, and so their roaming meaning cannot be attributed to their intransitivity. However, the question still remains as to why some dative-marked goals are not grammatical with HITPA'EL roaming motion verbs (cf. (85)) and some are. To address this question, I will take the alternative approach to understanding these verbs, that there is a property of the meaning of the HITPA'EL motion verbs which restricts certain types of dative-marked goals and permits others. I will develop this approach in the next section.

3.4 Antimaximality

3.4.1 Event cessation, maximality, and antimaximality

Having established in the previous chapter that HITPA'EL roaming verbs can accept dative-marked goals, I turn now here to the question of what can describe the distribution of dative-marked goals with these verbs. Despite the fact that these verbs do not fundamentally restrict dative marking through their intransitivity, they still do not always accept them either. Here, I am going to investigate the range of dative-marked goals which can appear with the HITPA'EL roaming motion verbs to assess what unifies them. I will identify three classes of goals – cycles, indefinite goals, and periphrastically expressed goals – and indicate that what unifies the three of them is that motion oriented towards each of them is fundamentally unbounded. Specifically, each type of goal is consistent with motion that could potentially continue, even though it has stopped. As such, I will analyze the HITPA'EL roaming verbs as entailing a property called *antimaximality*, which roughly requires there to be no maximal event in the denotation of these verbs.

This discussion will be couched in the framework of mereology, which concerns itself with how individual parts join together to create larger structures. Specifically, I will be assuming the framework developed in Krifka (1998). Under such a framework, a path is a type of part structure, where various subpaths join together to create larger and larger paths. Events decompose similarly, where a given event comprises various subevents which map onto a runtime through a temporal trace function which returns the time at which a given subevent occurs. On this approach, a motion event is tracked by progress on the path, where parts of the event occurring one after the other in linear time map to parts of the path adjoined spatially until the end of both is reached. This mapping constitutes a homomorphism between the event and the path which tracks the progress of the event by how far along the path the theme is at a given point in the event. The final point of the path traversed over the course of a motion event is therefore the goal. However, for consistency, I will continue to refer to

the dative-marked nominals under discussion here as ‘dative-marked goals’, with the understanding that they represent points along a path which might not necessarily be the endpoint.

I will suggest that what unifies the three types of dative-marked goals which HITPA’EL roaming verbs can appear with is that they are all kinds which cannot represent a final endpoint in the event of roaming motion as represented by the roaming verb. Namely, the motion itself is unbounded because the path does not have a final point which bounds it and therefore the event. Intuitively, these verbs, as verbs of roaming, involve motion without any intention of going to a particular place, with no goal in mind and no necessary path or locations which must be visited over the course of the roaming. Formally, this intuition translates to a complete incompatibility with a final endpoint. As such, HITPA’EL roaming motion verbs can appear with a dative-marked goal as long as that dative-marked goal does not represent a final semantic goal *qua* final endpoint of the event. I propose that this requirement takes the form of antimaximality, a property of predicates of events such that there is no maximal event which satisfies that property, i.e. there is no motion which reaches a goal for which that goal is the last possible point which could be reached and after which no motion could continue.

In more formal terms, the dative-marked goals of a HITPA’EL roaming motion verb are points where an event can cease, but can never be a point at which an event is a maximal event. I quickly review the concepts underpinning these terms before providing a formal definition in (118). An event ceases when it ‘ceases to develop’ in the sense of Landman (1992). On that system, an event unfolds in ‘stages’, roughly, subparts of an event which together make up the whole event and which can be tracked through a temporal trace function which determines at what point in time the subpart of the event falls, and ceases when it reaches its final stage, i.e. the point at which the event no longer continues in time. In essence, an event ceases when it no longer continues in time. To contrast, an event is maximal when it reaches a stage

after which no other possible stages could occur. As such, when an event is maximal, there is no possible continuation of that event.

Event maximality has therefore been analyzed as a condition over event parts such that it is the largest of the events of its type, i.e. there is no proper subevent which includes it. In its most basic form, it just requires a condition like that in (113) which Koenig & Muansuwan (2000) established for Thai perfectives.

- (113) $\forall e, P_{<v,t>} [MAX(e, P) \leftrightarrow P(e) \wedge \neg \exists e' [P(e') \wedge e < e']]$
 ‘MAX holds of an event e and a property of events P iff there does not exist an event e' of which P holds and of which e is a proper subpart.’

(Koenig & Muansuwan 2000, p. 169, (37))

This definition captures the intuition that a maximal event is one that no longer develops, in that e above is the largest event of which P holds.

This kind of definition for maximality builds in the notion that at the end of the event, the property which describes the event holds. In short, this notion of maximality presumes that the event must culminate. Following Zucchi (1999), culmination relates an event to a property and holds when that property is true of that event (Parsons 1990; Kratzer 2004). Notionally, maximal events will also culminate, as when an event has reached its necessary end, it will successfully have become an event of the type represented by the verb. This leads to culmination entailments whenever maximality is entailed, such as in languages like Czech with perfective aspect, typically analyzed as involving maximality (e.g. Filip 2008):

- (114) *Oblékl se, ??/*ale nezcela*
 dressed.PERF REFL but not.completely
 ‘He dressed, but not completely.’ (Filip 2008, (30))

Namely, the perfective aspect here, in enforcing maximality, ensures that the event of dressing culminates, in that the dressing is completed to its maximal amount, i.e. the

theme is fully dressed. It therefore is infelicitous to assert that the dressing is only partially done, due to this culmination entailment.

However, maximality in this sense does not always correspond to an entailment. Such is the case of languages with ‘weak perfectives’ (see e.g. Altshuler 2014; Martin 2019; Martin & Demirdache 2020; Martin & Gyarmathy 2019), which do not require culmination. This is illustrated by Hindi, where the verb with a perfective aspect marker does not entail that the event culminated (Singh 1998):

- (115) *us ne ciTThii paRhii par puurii nahĩĩ kii*
 he ERG letter read-PERF but complete NEG do-PERF
 ‘He read a letter but did not complete it.’ (Singh 1998, p. 184, (28))

Even though the verb has the perfective marking in (115), there is no entailment of culmination, because it is possible to deny that the reading of the letter is completed, and so the property of the letter having been read does not hold.

Yet, despite the lack of maximality, the event still meaningfully stops, in that there a point in time after which the reading stops. Altshuler (2014) contrasts maximality with this notion of cessation, where an event can stop without culmination occurring, as is required by maximality. In both cases, the event no longer develops, in that it is no longer progressing in time; in this way, both events which cease and events which are maximal contrast with events which are progressive, in that the latter still develops and continues in time. Only in the latter case could there never be a further development of the event. Thus, the distinction between cessation and maximality is whether the event could possibly continue, regardless of the fact that it did not. In (115), the perfective marking imposes cessation on the event, but not maximality.

Thus, I follow Nadathur & Filip (2021) in formalizing this notion of possibility as a modal property operating over possible worlds, where a potential continuation is a continuation which occurs in a possible world. As such, an event can be the

largest event of its kind just in the world of evaluation, producing event cessation (cf. Bohnemeyer & Swift 2004):

- (116) $\forall e, P, w [CESS(w, e, P) \leftrightarrow P(e, w) \wedge \forall e' [(P(e', w) \wedge e \leq e') \rightarrow e' = e]]$
 ‘A P -event e ceases in world w iff for all e' in w , if e' is a P -event in w and e is a part of e' , then e' must be equivalent to e , i.e. e is not a part of e' .’

(cf. Nadathur & Filip 2021, p. 6, (12a))

Cessation is a kind of ‘local maximum’, where an event which ceases is the largest event in its world, but not necessarily in all worlds. This notion subsequently casts event maximality as a ‘global maximum’, where a maximal event is the largest event in all worlds:

- (117) $\forall e, P [MAX(e, P) \leftrightarrow \forall w [CESS(w, e, P)]]$
 ‘A P -event is maximal iff it ceases in all worlds.’

(cf. Nadathur & Filip 2021, p. 6, (13a))

Though an event may cease in the world of evaluation, that event might not be the maximal event, because there is a possible continuation of that event in another possible world. Because of the clear formal relationship between cessation and maximality under this system, I adopt these definitions here.

Because cessation and maximality are modal properties, even if a given event has no continuation in the world in which it is occurring, if there is a possible continuation for that event which did not happen but could happen, the event will not be maximal. However, for my purposes here, I remain agnostic as to what the modal base could be to determine relevant possible worlds for those continuations to hold in, as which worlds are potentially under consideration are not important to the principle I am investigating here, namely that of antimaximality. What is important is that event cessation and maximality as a modal property implies that an event which may

cease in one world could continue in another world, and this potential for continuation will be an important factor in the calculation of antimaximality. For now, I will leave it open to any possible worlds with the understanding that this might create an unboundedly large set of events over which cessation and maximality can be computed. Instead, I will focus on the fact that a possible worlds analysis of cessation and maximality allows the two to be related to one another in terms of potential continuation.

Thus, I model antimaximality as I have described it thus so far as a property of predicates with an event argument where no event which satisfies that property is maximal. In particular, I define an operator which derives an antimaximal property from one which is not by restricting it to be true only for events which are not maximal. Effectively, this operator removes all maximal events from the denotation of the predicate. I give this definition formally in (118).

$$(118) \quad \forall P \forall x_1 \dots \forall x_n \forall e [ANTIMAX(P(x_1, \dots, x_n, e)) \quad \leftrightarrow \quad [P(x_1, \dots, x_n, e) \wedge \neg MAX(e, \lambda e_1 [P(x_1, \dots, x_n, e_1)])]]$$

‘A property of P is antimaximal if for all events for which P is true, none of those events are maximal.

As formulated here, a predicate which is antimaximal can cease in the world of evaluation, but as long as no event in all possible worlds is the maximal event, antimaximality will be satisfied. That is, as long as another potentially larger event can be found in some possible world which both satisfies the property and is also not maximal, antimaximality will hold.

I propose that HITPA’EL roaming motion verbs are antimaximal, and as such, the goals which they take should not be compatible with a maximal event. As a typical goal marked as a dative would be a final endpoint and therefore produce maximality, those are incompatible with predicates that are antimaximal, and thus derive their incompatibility with HITPA’EL roaming motion verbs. Instead, because

maximality and cessation are modal, even if the event ceases, there should be a possible continuation for it. As such, this analysis predicts that the three types of dative-marked goals which can appear with HITPA'EL roaming motion verbs should allow for unbounded motion and therefore always leave open a potential continuation regardless of where the event actually ceases. I review how each of the three satisfy this prediction in turn.

3.4.2 The infinite continuation of goals of roaming motion

3.4.2.1 Cycles

First, HITPA'EL roaming motion verbs can appear with dative-marked goals which together constitute a type of looping path which I will refer to as a cycle:

- (119) *ha=nohag-im al masa'i-ot ma-'amis-im sxora*
 the=driver-PL on truck-PL PRES-CAUS.haul-M.PL cargo
u=mit-rocec-im l=po u=l=fam
 and=PRES-REFL.run-M.PL to=here and=to=there
 'Truck drivers haul cargo and run about back and forth.' (corpus)

Here, the two dative-marked goals *l=po* 'to here' and *l=fam* 'to there' combine to create a two point path which the subject of the sentence on from one end to the other. As such, a cycle is similar to loop in that both types of paths have the same start and end point.

The key property of a cycle as contrasted with a loop is that a cycle is a more general kind of loop. In both cases, the theme moves between a source and a goal, but with no constraints on how many trips. In a loop, it is possible to limit the number of trips along the circular path through an expression like a count phrase, as in (120).

- (120) *ha=nohag-im al masa'i-ot ma-'amis-im sxora v=nos'-im*
 the=driver-PL on truck-PL PRES-haul-M.PL cargo and=drive-PRS.M.PL
l=kol ha=makol-ot b=a=fxuna me'a p'am-im
 to=all the=market-PL in=the=neighborhood one.hundred time-PL

‘Truck drivers haul cargo and drive to all of the stores in the neighborhood 100 times.’ (constructed)

Here, the loop in (120) *kol ha=makol-ot b=a=fxuna* ‘all of the stores in the neighborhood’ is able to be counted, demonstrated by its felicity with the count phrase *me’a p’am-im*. The interpretation is one where the drivers visit all of the stores in the neighborhood 100 times each, after which the event ends.

Cycles, unlike loops, are fundamentally unrestricted, and thus cannot be constrained in the number of trips along the path. As such, they resist phrases which count the number of times which the path has been traversed:

(121) #*ha=nohag-im al masa’i-ot ma-’amis-im sxora*
the=driver-PL on truck-PL PRES-haul-M.PL cargo
u=mit-rocec-im l=po u=l=fam me’a p’am-im
and=PRES-REFL.run-M.PL to=here and=to=there one.hundred time-PL
(intended) ‘Truck drivers haul cargo and run about back and forth 100 times.’ (constructed)

Here, the only interpretation available for the count phrase *me’a p’am-im* ‘one hundred times’ is one where it is counting the number of individual instances the subject does an arbitrary number of traversals of the path, here *l=po u=l=fam* ‘here and there/back and forth’. It cannot count the number of times the subject traverses the path itself.

Therefore, a cycle, as contrasting with a loop, is fully agnostic to where the motion actually stops, and motion along a cycle can continue arbitrarily long, not being bound by a limit on number of trips along the cycle, i.e. a count phrase limiting the number of trips. This in turn leads to a definition for cyclical motion where a cycle consists of motion along a looping path, in that the path has the same start and endpoint, which includes at least one loop, and could continue for one more loop.

I therefore define cyclical motion as in (122).

$$(122) \quad \forall e, p, \theta [CYCLE(e, p, \theta) \leftrightarrow \forall y [\theta(y, e, p) \rightarrow \exists x \exists w [CESS(w, e, \lambda e_1 [\theta(y, e_1, p)]) \wedge \\ GOAL(x, e, p) \wedge SOURCE(x, e, p) \wedge \exists w' \exists e' [e < e' \wedge CESS(w, e, \lambda e_1 [\theta(y, e_1, p)]) \wedge \\ GOAL(x, e', p) \wedge SOURCE(x, e', p)]]]]]$$

‘A relationship θ of an event e occurring on a path p with a theme y represents a cycle iff

1. e ceases in a world w and x is both the goal and source of p in e , and
2. there is a world w' and an event e' in which e is a part of e' , e' ceases in w' , and for which x is the goal and source of p in e' .

Essentially, cyclical motion involves two components. First, it requires motion along a loop where at least one loop has been accomplished, in that the motion begins and ends at the same point along the path. Second, it requires there to be a potential continuation of the motion, in that there is a world where the motion continues with one more trip along the loop. This definition of cyclical motion is fundamentally unbounded, in that there no necessary final loop along the cycle after which no other loops are possible.

Formally, cyclical motion ensures that if an event along a cycle ceases, there will always be another potential continuation of that event which continues the cycle. In addition, the path which cyclical motion occurs on must always start and end at the same point. No matter how many loops a theme makes as part of a cycle, that theme ends up in the same exact place where they started. As such, the point where the cyclical motion ceases cannot be identified as a necessary endpoint for the event of motion. Specifically, if a theme is at the end of a path of motion, it is not necessarily the case that the event of motion has ceased, because the point of the path which constitutes the end of a loop is also the beginning.

Therefore, the start of an event of motion is fundamentally indistinguishable from the end, and the end of a loop is the beginning of another one. While this is also true of loops, looping motion must also involve an element which can track

the number of loops to permit felicity with count phrases. As such, it can identify the bounding point of the looping motion event, namely, when the number of loops meets the number specified by the count phrase. This allows for a maximal event of looping motion, namely the event which constitutes the loop corresponding to the count phrase. There is no such specification for a cycle, and as such, no way to bound it by number of loops. As such, every loop traversed as part of an event of cyclical motion implies a possible continuation of that cyclical motion constituting another loop.

3.4.2.2 Indefinite goals

Second, HITPA'EL motion verbs can appear with dative-marked goals which taken together imply a path which is agnostic to its extent, being compatible with an arbitrary number of locations reached throughout the course of the motion:

- (123) *hithalx-u* *l=mkom-ot she='od lo xakar-tem*
 REFL.walk-IMP.PL to=place-PL COMP=still NEG investigate-2PL.M.PST
 'Walk about to places that you have not yet looked into.' (corpus)

Here, the goal *mkom-ot she='od lo xakar-tem* 'places you have not yet looked into' represents multiple different points which will be visited over the course of the motion event. However, these points do not need to be visited in any particular order, and there does not need to be a specific number of locations which are visited (as long as there is more than one).

Crucially, the felicity of HITPA'EL motion verbs with a path like that in (123) hinges on its indefinite marking. As such, an equivalent sentence with a definite-marked path is infelicitous:

- (124) #*hithalx-u* *l=kol ha=mkom-ot she='od lo*
 REFL.walk-IMP.PL to=all the=place-PL COMP=still NEG
xakar-tem
 investigate-2PL.M.PST

(intended) ‘Walk about to all the places that you have not yet looked into.

(constructed)

The distinction derives from the fact that indefinite nouns typically represent entities which have cumulative reference. In a formal sense, cumulative reference, defined as in Krifka (1998), makes it such that any two entities with the same property joined together will still retain that property:

$$(125) \quad \forall P[CUM(P) \leftrightarrow \forall x, y[P(x) \wedge P(y) \rightarrow P(x \oplus y)]]$$

‘A property P is cumulative iff for any two P -entities x and y , their join is also a P -entity.’ (Krifka 1998, p. 200, (3))

If an entity has cumulative reference, no matter how many parts it has, it is still the same type of entity. As such, there is no necessarily largest such entity which can serve to bound the motion, and therefore no largest such event of motion, satisfying antimaximality.⁶

Specifically, cumulative reference allows for a path which can be prospectively unboundedly large. Assume there are two locations, x and y , both of which satisfy the property of being a ‘place which one has not yet looked into.’ Their join, $x \oplus y$, generates the minimal path which would satisfy ‘places that you have not yet looked into’. Because its reference is cumulative, the join of $x \oplus y$ with another such location would also satisfy ‘places that you have not yet looked into’, as would the addition of another one, and so on. Thus, a goal with cumulative reference such as the one in (123) can produce paths which are unboundedly large, and as such, no largest path of that type can be found.

However, the number of locations described by a noun with cumulative reference cannot be necessarily infinite, and likely will be by, for example, real-world

⁶Relatedly, under the Movement Relation from Krifka (1998) or the Figure/Path Relation from Beavers (2012), cumulative reference of the goal expression ensures the event to have cumulative reference as well, ruling out telicity.

constraints on the number of locations which can be described by the dative-marked goal. For example, a noun like *bar-im b=Tel Aviv* ‘bars in Tel Aviv’ only has a finite number of entities which fall under its extension, namely, all of the bars in Tel Aviv and nothing else. Even in cases where the number of locations is tightly constrained, the motion is still able to continue indefinitely. To see this, consider (126).

- (126) (Context: The speaker is talking about their day out in a small little neighborhood. They went window shopping at all the stores in the neighborhood, but because it’s so small, they just kept going to the same 3 stores over and over.)

hithalax-ti *l=xanuy-ot b=a=fxuna* *kol ha=yom*
 REFL.walk-1SG.PST to=store-PL in=the=neighborhood all the=day

‘I walked about to all the stores in the neighborhood all day.’ (constructed)

Despite the fact that there are not a potentially unbounded number of discrete locations which could be visited over the course of the motion, the path represented by the cumulative dative-marked *xanuy-ot b=a=fxuna* is still licit. In this case, once the three possible locations are joined into the path of motion and visited over the course of the event, it is possible to join them as part of the path again and continue the motion by visiting those locations again. Roughly, if x, y, z all satisfy the property of being shops in the neighborhood, so does $x \oplus y \oplus z$ as well as $x \oplus y \oplus z \oplus x \oplus z \dots$ ⁷ The same logic therefore applies, as even once a final possible unique location has been visited, there is still a possible extension of the path where certain locations are revisited.

Naturally though, in the world of evaluation, the path which is traversed will have a final point which is reached *qua* its goal, and so the event will cease in that world. However, because another potential point could be added to the path, there is

⁷This is not a path in the strict sense of Krifka (1998), as there paths cannot loop.

another event of motion in that world which can occur along that path which is larger than the event in the evaluation world. Every addition of a point along the path would additionally create a continuation of the event of motion which would then be a larger event than the one in the evaluation world. Thus, every event which occurs along the path represented by this dative-marked goal with cumulative reference which in turn ceases cannot be maximal. It would not cease in the possible world which contains its one monotonically larger path and therefore a monotonically larger event. As such, antimaximality is satisfied, as no maximal event can be found, only events which cease along the path which is actually traversed but could continue along another potential one which continues from where the event of motion actually stops.

3.4.2.3 Periphrastic goals

Finally, in (127), a dative-marked goal which is a true semantic goal can appear with a HITPA'EL roaming motion verb when it is in a different clause:

- (127) *yaca-nu me=ha=malon, hithalax-nu b=a=sxuna*
 go.out-1PL.PST from=the=hotel REFL.walk-1PL.PST in=the=neighborhood
v=hega-nu l=bar l'yad ha=malon
 and=CAUS.arrive-1PL.PST to=bar next the=hotel
 ‘We left the hotel, walked around in the neighborhood, and made it to a bar
 next to the hotel.’ (constructed)

In this case, the arrival to the final point of the path *qua* its goal occurs outside of the event in which the roaming motion happens. Following Jerro (2016, 2023), a path contains three components: a source where the path begins, a route constituting all of the intermediary locations visited, and a goal where the path ends:

- (128) $\forall e[\exists p[\text{path}'(e, p)] \leftrightarrow \exists x, y, z[\text{source}'(e, x) \wedge \text{route}'(e, y) \wedge \text{goal}'(e, z)]]$

(Jerro 2023, p. 1450, (42))

In (127), the clause containing the HITPA'EL roaming motion verb only contains the

source and the route, with no necessary final goal, while the clause containing the verb of arrival contains the specific goal.

Because the HITPA'EL roaming motion verb is in an entirely separate clause, it represents motion on an entirely separate path which is able to continue unboundedly. As such, it can appear with the types of dative-marked goals which generate antimaximality, such as a cycle (cf. (123)):

- (129) *yaca-nu* *me=ha=malon*, *hithalax-nu* *l=xanuy-ot*
 go.out-1PL.PST from=the=hotel REFL.walk-1PL.PST to=store-PL
b=a=fxuna *v=hega-nu* *l=bar* *l'yad* *ha=malon*
 in=the=neighborhood and=CAUS.arrive-1PL.PST to=bar next the=hotel
 'We left the hotel, walked to stores in the neighborhood, and made it to a bar
 next to the hotel.' (constructed)

As such, the motion associated with the HITPA'EL roaming motion verb is still fundamentally unbounded, and only after that event has ceased will the final location be reached.

The event of roaming motion therefore has no commitments to where the event must end, because the necessary final endpoint is part of the event of arrival. The roaming motion will continue along the route of the path, but will always cease before reaching the final point expressed in the separate clause. As such, the point along the route where the roaming motion ceases will not be the necessary final point where the roaming motion must cease, and therefore the event which reaches that point will not be maximal. As a continuation along the route is always possible for any point where motion ceases along the route, the roaming motion event will not have any possible largest event. As such, this roaming motion satisfies antimaximality.

3.4.2.4 The ungrammaticality of datives, revisited

Here, I have surveyed the range of dative-marked goals which can appear with HITPA'EL roaming motion verbs and demonstrated that all of those types of goals are

ones which produce motion without a necessary final endpoint and therefore where additional motion is always possible. This property, which I termed antimaximality, characterizes these roaming motion verbs, where the predicate describing roaming motion does not have a maximal events in its denotation. This property can also explain why a dative-marked goal representing a true goal, as in (85) repeated here, is ungrammatical.

- (130) **hithalax-ti* *l=a=xanut*
REFL.walk-1SG.PST to=the=store
(intended) ‘I walked about/roamed to the store.’ (constructed)

The path over which the motion is occurring here has a clear maximal point: the store. After the store is reached, no additional continuation is possible which would also be part of ‘walking to the store’. Accordingly, it is infelicitous to assert (85) with a non-roaming motion verb while also asserting that the motion to the store subsequently continued, as in (131).

- (131) #*halax-ti* *l=a=xanut* *v=az* *himfax-ti* *la=lexet*
walk-1SG.PST to=the=store and=then CAUS.continue-1SG.PST INF-walk
l=a=xanut
to=the=store
(intended) ‘I walked to the store and then continued to walk to the store.’

(constructed)

In this case, the goal here, being a concrete location which bounds the path and therefore the event, generates a maximal reading where the walking has reached its final possible endpoint in the largest possible event. Subsequently, because the event is maximal, asserting a continuation of the motion event is not possible, as there can be no larger event constituting a potential continuation.

To contrast, the three goals which HITPA’EL roaming motion verbs can appear with each allow for an explicit continuation:

- (132) a. *hem hitrocec-u* *l=po* *u=l=fam* *v=az*
they REFL.run-3PL.PST to=here and=to=there and=then
himfix-u *l-hitrocec* *l=po* *u=l=fam*
CAUS.continue-3PL.PST INF-REFL.walk to=here and=to=there
‘They ran about back and forth and then kept on running about back and forth.’ (cf. (119))
- b. *hithalax-nu* *l=xanuy-ot* *b=a=fxuna* *v=az*
REFL.walk-1PL.PST to=store-PL in=the=neighborhood and=then
himfax-nu *l-hithalex* *l=xanuy-ot* *b=a=fxuna* (*’od*
CAUS.continue INF-REFL.walk to=store-PL in=the=neighborhood (more
kcat)
a.little)
‘We walked about to stores in the neighborhood, then walked about to stores in the neighborhood (a little longer).’ (cf. (123))
- c. *yaca-nu* *me=ha=malon*, *hithalax-nu*
go.out-1SG.PL.PST from=the=hotel REFL.walk-1PL.PST
b=a=fxuna, *himfax-nu* *l-hithalex* *’od* *kcat*,
in=the=neighborhood CAUS.continue INF-REFL.walk more a.little
v=hega-nu *l=bar* *l’yad* *ha=malon*
and=CAUS.arrive-1PL.PST to=bar next the=hotel
‘We left the hotel, walked around in the neighborhood, kept on walking about a little more, and made it to a bar next to the hotel.’ (cf. (127))
- (constructed)

Where in (131), a continuation of the event of motion is not possible because reaching the store occurs in the maximal event of that type, in all of the cases in (132), because the events can never be maximal, a continuation is possible. Thus, because the HITPA’EL roaming motion verb is antimaximal, it cannot appear with a dative-marked goal which represents a concrete location which fundamentally bounds the event, but it can appear with those that can, because they do not require the event to be maximal and allow for a potential continuation.

It is also important to note that antimaximality as it is construed here leaves open the possibility that infinitely many events are possible, because there is no largest

event to bound the size of the denotation of the antimaximal predicate. Likely what resolves this is establishing the modal base determining which potential continuations in which worlds are relevant. This would constrain the events which can serve as potential continuations to only those which are in those worlds relevant under the modal base.

One possibility is that the modal base for these verbs is unintentionality, where all other possible worlds which are accessible are those where the motion is unintentional. In this kind of analysis, maximality would be computed relative to all the worlds consistent with the agent not intending to continue moving, and as such will put an upper bound on the size of the roaming events. This kind of base would accord with the intuition that roaming does not involve an intention to reach the various points along a path which that motion occurs on. In this case, antimaximality would have to be redefined to be sensitive to that modal base of unintentionality, or antimaximality would have to scope over a predicate defined relative to that modal base. I leave the question as to what the relevant modal base is and how antimaximality would interact with it to future research. Instead, I now turn to the question of how the antimaximality of the HITPA'EL roaming motion verbs generates a contrast between the HITPA'EL verb forms and their correlated non-roaming PA'AL verb forms.

3.5 How HITPA'EL contrasts semantically with PA'AL

3.5.1 Affectedness

Up to this point, I have established that the HITPA'EL roaming motion verbs are semantically antimaximal, and that property can derive both their goal resistance and their distribution with dative-marked nouns. Specifically, HITPA'EL roaming motion verbs can appear with dative-marked nouns, and those nouns can serve as goals as long as those goals are consistent with the antimaximality requirement in that they do not represent a necessary final endpoint. This analysis in turn rejects a syntactic role for the HITPA'EL template in that the templatic membership in HITPA'EL cannot

on its own accurately determine the range of verbs which can appear in it just on the basis on its argument structures.

As such, HITPA'EL as a template deriving these roaming motion verbs is not contrasting with their PA'AL counterparts on the basis of their argument structure. The two templatic classes of motion verbs do not substantively differ in their argument structure, in that both are equally acceptable with dative-marked goals, and so the templates are not deriving a syntactic contrast between them. Instead, I will suggest that the HITPA'EL template is mediating the templatic alternation by marking a truth-conditional strength difference. Specifically, I will demonstrate that both the HITPA'EL roaming motion verbs and their non-roaming PA'AL counterparts represent the same type of change, namely non-quantized change, but the antimaximality condition on the HITPA'EL forms constitutes an additional entailment on top of the shared requirement for non-quantized change. The HITPA'EL motion verbs therefore have stronger truth conditions than their PA'AL counterparts, and in turn indicate a markedness difference between the two templates deriving this truth-conditional difference.

An initial hypothesis is that the antimaximality of the HITPA'EL roaming motion verbs indicates a more specific kind of change than the non-antimaximal change represented by the non-roaming PA'AL motion verbs. Because HITPA'EL is specified to be antimaximal, it would be more constrained in the kinds of changes which it is consistent with, and therefore would be more specific. In this context, specificity of change is a type of affectedness. Roughly, affectedness can notionally be understood as the amount by which an entity 'has something happen to it' in the course of an event. This understanding also allows for there to be greater or lesser levels of affectedness, where a verb expressing lower levels of affectedness requires less to have happened to an entity than verbs with higher levels of affectedness.

Formally, I am following the analysis of affectedness given in Beavers (2011), where different levels of affectedness are understood as different levels of specificity of

the type of change a theme undergoes on a scale, an ordered set of points measuring the degree to which a property holds of an entity (Kennedy 1997; Hay et al. 1999; Kennedy & McNally 2005; Wechsler 2005a; Kennedy & Levin 2008, among many others). Beavers identifies four different levels of affectedness for a predicate ϕ based on what kind of change it entails for a theme x . These different levels of change are distinguished based on several different diagnostics which taken together form an implicational relationship between each of the levels of affectedness. I review each of these in turn.

First, quantized change entails that the theme reaches a predicate-specific result state. Verbs entailing quantized change are telic and entail a change, specifically to that final result state, and are compatible with the *what happened to it* diagnosing force recipients:⁸

- (133) a. *yosi halax* *l=a=xanut* *tox fa'a*
 yosi walk[3SG.M.PST] to=the=store in hour
 'Yosi walked to the store in an hour.'
- b. *#yosi halax* *l=a=xanut* *aval lo higi'a*
 yosi walk[3SG.M.PST] to=the=store but NEG CAUS.arrive[3SG.M.PST]
l=a=xanut
 to=the=store
 (intended) 'Yosi walked to the store but did not make it to the store.'
- c. *ma fe=kara* *l=a=kos* *ze fe=Yosi*
 what COMP=happen[3SG.M.PST] to=the=cup that COMP=Yosi
favar *'ot-a*
 break[3SG.M.PST] ACC-3SG.F
 'What happened to the cup is that Yosi broke it.' (constructed)

Telicity here is diagnosed by the classic *in an hour* test from Dowty (1979), here applied for Hebrew as *tox fa'a*. Felicity with *tox fa'a*, as demonstrated in (133a),

⁸For independent reasons, the *what happened to it* test does not apply easily to motion verbs, and so where I would apply it to *halax*, I use a different verb representing a change of state.

indicates a telic predicate. Entailing a change is diagnosed through the classic negation test, where asserting the verb but negating the putatively entailed change generates an infelicitous sentence. Here, the change entailed by *halax l=a=xanut* ‘walk to the store’ is reaching the store, and the infelicity of (133b) demonstrates that it is indeed an entailment.

Second, non-quantized change entails that the theme reaches some result state, but not one lexically specified by the predicate (Hay et al. 1999). Such verbs are atelic, but still entail a change and pass the *what happened to it* test, as indicated for the verb *halax* lacking an explicit goal:

- (134) a. *yosi halax kol ha=yom*
 yosi walk[3SG.M.PST] all the=day
 ‘Yosi walked all day.’
- b. *#Einav halx-a b=a=sxuna aval lo zaz-a*
 Einav walk-3SG.F.PST in=the=neighborhood but NEG move-3SG.F.PST
 ‘Einav walked in the neighborhood but didn’t move.’
- c. *ma she=kara l=a=marak ze she=hu*
 what COMP=happen[3SG.M.PST] to=the=cup that COMP=Yosi
hitkarer
 REFL.cool[3SG.M.PST]
 ‘What happened to the cup is that it cooled down.’ (constructed)

Here, the change entailed is that motion happened, which is being negated in (134b) and in turn generates an infelicitous sentence. Atelicity is probed in (134a) using the *for an hour* test, also from Dowty (1979), here realized with *kol ha=yom* ‘all day’. Felicity with *kol ha=yom* ‘all day’ indicates atelicity.

Third, potential for change entails that the theme has some relationship to a scale, but not necessarily one wherein it changes on that scale. The class of verbs which entail potential for change are typically those which have force recipients for a theme (Croft 1991, 1998; Rappaport Hovav & Levin 2001), such that the theme receives some force from the agent. For these verbs, the theme does not necessarily need to

change, but because a change is possible, the predicates are still scalar (Tenny 1994). As such, change is not entailed for such verbs, but they do pass the *what happened to it* test, and they are atelic:

- (135) a. *yosi hika* *et ha=kir aval ha=kir lo*
 yosi CAUS.hit[3SG.M.PST] ACC the=wall but the=wall NEG
hifana
 REFL.change[3SG.M.PST]
 ‘Yosi hit the wall, but the wall didn’t change.’
- b. *ma fe=kara l=a=kir ze fe=yosi hika*
 what COMP=happen[3SG.M.PST] to=the=wall that COMP=YOSI hit
ot-o
 ACC-3SG.M
 (intended) ‘What happened to the wall is that Yosi hit it.’
- c. *yosi hika et ha=kir kol ha=yom*
 yosi CAUS.hit[3SG.M.PST] ACC the=wall all the=day
 ‘Yosi hit the wall the whole day.’ (constructed)

Here, it is possible to assert the verb *hika* ‘hit’ while also negating that any change happened to its theme *ha=kir* ‘the wall’ at all, and as such, *hika* does not entail a change. In addition, *hika* is fully felicitous in the context of *ma she=kara* ‘what happened’, and so its theme is a force recipient.

Finally, being unspecified for change entails that the theme is participating in an event where there is not necessarily a scale associated with it. Such verbs, here *caxak* ‘laugh’, do not entail any change at all and their themes are not a force recipient. As such, no change is entailed of these verbs, they do not pass the *what happened to it* test, and accordingly are atelic:

- (136) a. *Yosi caxak aval fum davar lo hifana*
 yosi laugh[3SG.M.PST] but nothing NEG REFL.change[3sg.m.pst]
 ‘Yosi laughed but nothing changed.’

- b. $\#ma$ $she=kara$ $l=yosi$ ze $she=hu$
 what COMP=happen[3SG.M.PST] to=yosi that COMP=he
caxak
 laugh[3SG.M.PST]
 (intended) ‘What happened to Yosi is he laughed.’
- c. *Yosi caxak* *kol ha=yom*
 Yosi laugh[3SG.M.PST] all the=day
 ‘Yosi laughed all day.’ (constructed)

Beavers (2011) formalizes these four levels of affectedness as in (137).

- (137) For a theme x , an event e , a predicate ϕ :
- a. Quantized change: $\phi \rightarrow \exists e \exists s [result'(x, s, \mathbf{g}_\phi, e)]$
 ‘ ϕ encodes quantized change if the theme x reaches a predicate-specific final result $\mathbf{g}_\phi$ on a scale s in an event e .’
 - b. Non-quantized change: $\phi \rightarrow \exists e \exists s \exists g [result'(x, s, g, e)]$
 ‘ ϕ encodes non-quantized change if the theme x reaches some final result g on a scale s in an event e .’
 - c. Potential for change: $\phi \rightarrow \exists e \exists s \exists \theta [\theta(x, s, e)]$
 ‘ ϕ has potential for change if the theme x has some relationship θ to a scale s in an event e .’
 - d. Unspecified for change: $\phi \rightarrow \exists e \exists \theta [\theta(x, e)]$
 ‘ ϕ is unspecified for change if the theme x has some (non-scalar) relationship to an event e .’ (Beavers 2011, p. 358, (60))

Each level of affectedness is monotonically stronger in truth conditions than the level below it. This corresponds to the fact that the diagnostics under consideration pick monotonically smaller subsets of predicates. Thus, a predicate which entails a higher level of affectedness also entails all of the levels of affectedness below it, generating an implicational hierarchy.

(138) **Affectedness Hierarchy:**

$$\forall x, \phi, e [\exists s[result'(x, s, \mathbf{g}_\phi, e)] \rightarrow \exists s \exists g[result'(x, s, g, e)] \rightarrow \exists s \exists \theta[\theta(x, s, e)] \rightarrow \exists \theta'[\theta'(x, e)]] \quad (\text{Beavers 2011, p. 359, (62)})$$

The hierarchical nature of affectedness ensures that any predicate which picks out one level of the hierarchy also picks out every level below it. That is, a predicate which entails, for example, quantized change will also entail non-quantized change, as entailing that a specific goal is reached also means entailing that some non-specific goal is met. This monotonic relationship between levels of affectedness where higher levels of affectedness are strictly more specific than lower levels will be crucial to understanding the relationship between roaming verbs and their non-roaming counterparts, as I do in the next section.

3.5.2 The specificity of HITPA'EL vs. PA'AL motion verbs

As already demonstrated in (134), the PA'AL motion verb *halax* ‘walk’ entails non-quantized change, as it passes the two sufficient diagnostics of atelicity and entailing change. Thus, if the antimaximality connected to HITPA'EL generates a more specific change than PA'AL due to the additional requirement antimaximality represents, then the HITPA'EL roaming motion verbs would pass the diagnostics for a higher level of change, namely quantized change. However, such is not the case. First, HITPA'EL roaming motion verbs do entail a change in that they require displacement. That is, HITPA'EL roaming motion verbs require motion to actually happen:

- (139) #*Einav hithalx-a* *b=a=fxuna* *aval lo*
 Einav REFL.walk-3SG.F.PST in=the=neighborhood but NEG
 zaz-a
 move-3SG.F.PST
 ‘Einav walked around the neighborhood but didn’t move.’ (constructed)

However, these verbs are not telic, but rather atelic, as demonstrated in (140).

- (140) *ani lo mesugel-et l-hithalex kol ha=yom,*
 I NEG capable-3SG.F INF-REFL.walk all the=day
hitlonen-a
 REFL.complain-3SG.F.PST
 ‘I can’t walk around all day, she complained.’ (corpus)

As such, both the HITPA’EL roaming motion verbs and their non-roaming counterparts in PA’AL entail the same level of change, namely non-quantized change. The antimaximality of HITPA’EL roaming motion verbs is not actually changing the specificity of the change entailed by the verb.

Even though the verbs in both templates have the same level of affectedness, the additional antimaximality requirement of the HITPA’EL roaming motion verbs do have a meaning which includes an entailment of a more specific type of change. Specifically, non-roaming PA’AL motion verbs are capable of describing both quantized and non-quantized change depending on the presence of a dative-marked goal, but antimaximality excludes quantized change from the HITPA’EL roaming motion verbs. Non-roaming PA’AL verbs display variable goal expression, typical of their membership in the class of manner of motion verbs (Levin 1993, sects. 1.4.1, 51.1), demonstrated in (141) with *halax* ‘walk’.

- (141) a. *Ron halax l=a=xanut*
 Ron walk[3SG.M.PST] to=the=store
 ‘Ron walked to the store.’
 b. *Ron halax*
 Ron walk[3SG.M.PST]
 ‘Ron walked.’ (constructed)

halax with a dative-marked goal does in fact entail quantized change, as demonstrated above in (133), where it is telic and entails a change. The dative marked goal *l=a=xanut* serves as the predicate-specific goal of the predicate represented by the VP *halax l=a=xanut*. Thus, *halax* is consistent with both quantized and non-quantized change, which is conditioned by the presence of a dative-marked goal. This

pattern follows directly from the implicational nature of levels of affectedness, where entailing quantized change implies entailing non-quantized change.

Relatedly, non-roaming PA'AL verbs are also able compatible with antimaximality, though do not entail it. Specifically, just as they can appear variably with dative marked goals, they can equally appear with the kinds of goals which are compatible with HITPA'EL roaming motion verbs:

- (142) *halax-nu l=mkom-ot she='od lo xakar-nu*
 walk-1PL.PST to=place-PL.F COMP=still no investigate-1PL.PST
 'We walked to places we had not yet looked into.' (constructed)

In (142), non-roaming PA'AL *halax* appears with a PP representing a path with cumulative reference. This sort of dative-marked goal generates antimaximality (cf. (123)), and as such, the whole VP *halax l=mkom-ot she='od lo xakar-nu* is an antimaximal predicate. Thus, *halax* is underspecified for antimaximality, in that it does not entail the lack of antimaximality. As such, it is possible to augment *halax* to derive an antimaximal predicate, as the goal in (142) does.

This variability is not present at all in the HITPA'EL roaming motion verbs. As has already been established, HITPA'EL has a much more limited range of dative-marked goals which it can appear with, and in particular it cannot appear with the kind of dative-marked goal representing a final endpoint such as *l=a=xanut* in (141a), as demonstrated in (85) repeated here.

- (143) **hithalax-ti l=a=xanut*
 REFL.walk-1SG.PST to=the=store
 (intended) 'I walked about/roamed to the store.' (constructed)

Instead, it is strictly constrained to exactly those dative-marked goals which are compatible with antimaximality. In addition, the HITPA'EL roaming motion verbs which appear with these goals are atelic, and therefore still entail non-quantized change:

- (144) *kol ha=yom hithalax-nu l=mkom-ot fe='od lo*
 all the=day REFL.walk-1PL.PST to=place-PL.F COMP=still no
xakar-nu
 investigate-1PL.PST
 ‘All day, we walked to places we had not yet looked into.’ (constructed)

Thus, despite the fact that both non-roaming PA’AL motion verbs and HITPA’EL roaming motion verbs both pattern as if they entail nonquantized change, the HITPA’EL roaming motion verbs specifically exclude interpretations of quantized change which are possible in the non-roaming PA’AL motion verbs. While their levels of affectedness are the same, in that they both entail non-quantized change, they still differ in how their change is specified, where one is additionally antimaximal. This paradigm thus complicates the notion of affectedness as pitched in (168), as properties such as antimaximality constitute additional specificational content which eschew the implicational relationship between quantized and non-quantized change.

In summary, both *halax* and *hithalex* entail non-quantized change, but *hithalex* selects for only those events of non-quantized change which generate antimaximality. *halax* is entirely underspecified: based on its dependents, it can express quantized change or be antimaximal. *hithalex* is explicitly specified to be antimaximal, and so excludes two types of change which are compatible with *halax*: non-antimaximal non-quantized change, and quantized change. I represent the relationship between these two predicates and their specification for change with the containment diagram in Figure 3.3. As depicted here, both *halax* and *hithalex* entail non-quantized change, but *hithalex* is additionally antimaximal, and therefore excludes other types of non-quantized change. *halax* is not specified further, and so can be either quantized or antimaximal, depending on the dative-marked goal it appears with.

3.5.3 Analysis

This relationship follows directly from the fact that HITPA’EL roaming motion verbs entail antimaximality, while their non-roaming PA’AL counterparts do not. This

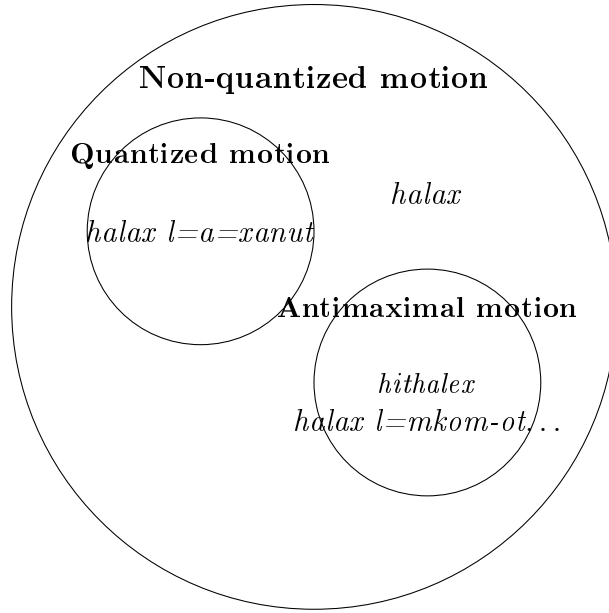


Figure 3.3: Change specifications of *halax* and *hithalex*

additional requirement in turn creates stronger truth conditions for the HITPA'EL roaming motion verbs than those of their non-roaming PA'AL counterparts. First, I define the denotation of the non-roaming PA'AL *halax* as in (145).

$$(145) \quad \llbracket halax \rrbracket = \lambda g \lambda y \lambda s \lambda e [walk'(e) \wedge result'(g, s, y, e)]$$

This denotation includes two parts. First, $walk'(e)$ is an event identifier ensuring that the event is in fact a walking event. Second, $result'(g, s, y, e)$ enforces the theme y reaching of a result state *qua* final point of motion g along the path s in the course of the event e . The result state is an open argument which can be saturated by a dative-marked goal to produce (141a) and quantized change, or can be left open and will be existentially bound at the end of the derivation, producing (141b) and non-quantized change.

Additionally, *halax* is fully consistent with antimaximality without actually entailing it. If *halax* takes a dative-marked goal which generates antimaximality, as in (142), the resulting predicate will be antimaximal. The antimaximality of *halax*

relies on its dependents, where the goal it combines with can determine whether it is antimaximal or not. Thus, though *halax* itself is not entailed to be antimaximal, it is fully consistent with an antimaximal interpretation with the right combination of goal, scale, and theme. Antimaximal change is a more specific kind of non-quantized change (cf. Figure 3.3), and so, if a verb entails non-quantized change, it must also be able to express antimaximal change as well. *halax* entails non-quantized change, but is not specified any further. The dependents of *halax* can specify its meaning more, and can specifically specify whether the change is antimaximal or quantized. As such, *halax* is compatible with the range of dative-marked goals, and its meaning follows from the specific kind of goal, scale, or theme it combines with.

To contrast, the roaming HITPA'EL motion verb *hithalex* also entails nonquantized change, but additionally has a lexical entailment requiring none of its events to be maximal. To capture this, the operator which defines antimaximality scopes over the denotation of the walking predicate in (145).

- (146) a. $\llbracket hithalex \rrbracket = \lambda g \lambda y \lambda s \lambda e [ANTIMAX(\llbracket halax \rrbracket(g, y, s, e))]$
 b. $\llbracket hithalex \rrbracket = \lambda g \lambda y \lambda s \lambda e [ANTIMAX(walk'(e) \wedge result'(g, s, y, e))]$

ANTIMAX modifies the denotation of *halax* to generate a predicate without any maximal events, following the definition in (118). Accordingly, none of the events which fall under the extension of *hithalex* will be maximal. As this contrasts with *halax*, *hithalex* is specified lexically to be antimaximal, and therefore can only appear with dative-marked goals which are compatible with antimaximality.

Therefore, because the denotation of *hithalex* is antimaximal, none of the walking events can be a maximal walking event. As such, *hithalex* can never describe quantized change *qua* entail a predicate-specific result state. In turn, it can never appear with a dative-marked goal which would make it maximal, thus constraining the range of dative-marked goals which it can appear with to those discussed in section 1.4, whose interpretations make it impossible to find a maximal event predicated over them as a true final point along their paths do not exist.

To contrast, *halax* is still compatible with quantized change because it is not explicitly specified for antimaximality, but rather entails non-quantized change, which includes antimaximality, but only for some contexts. It can appear with dative-marked goals which would make it antimaximal, but is not exclusively limited to do so. As such, it is able to accept dative-marked goals which are incompatible with antimaximality, such as those which represent a final endpoint and would make it represent quantized change.

Finally, I return to the question of what the role of the template is in deriving this alternation. Both HITPA'EL *hithalex* and PA'AL *halax* entail non-quantized change. However, *hithalex* additionally entails that it is antimaximal, which in turn provides an entailment restricting quantized change. To contrast, *halax* is only consistent with antimaximality, and is additionally able to represent quantized change, depending on its choice of dependents. Therefore, the only relevant entailment associated with *halax* is that of its non-quantized change, which makes *halax* compatible with either quantized change or antimaximality. However, neither are entailed, in that there are contexts which produce neither quantized change nor antimaximality. Therefore, the set of entailments associated with *hithalex* is therefore stronger than the set of entailments associated with *halax* because it is larger, represented in (147).⁹

$$(147) \quad \left\{ \begin{array}{c} \underline{\textit{halax}} \\ \text{non-quantized change} \end{array} \right\} <_S \left\{ \begin{array}{c} \underline{\textit{hithalex}} \\ \text{non-quantized change} \\ \text{antimaximal} \\ \neg\text{quantized change} \end{array} \right\}$$

Therefore, the denotation of *hithalex* has stronger truth conditions than *halax*, as the set of entailments associated with *hithalex* are the set of entailments associated with *halax* are a subset of those associated with *hithalex*. Additionally, this strength difference follows the markedness difference between the templates of the two verbs,

⁹Even though antimaximality entails lacking quantized change, I represent them both here for completeness.

according to the VCMP. Namely the stronger *hithalex* is in the more marked HITPA'EL template, while the weaker *halax* is in the weaker PA'AL template.

3.6 Conclusion

In this chapter, I have argued that the templatic alternation which derives HITPA'EL roaming motion verbs is a templatic alternation which affects only the semantics of the verb, not its syntax or argument structure. Specifically, I have shown that the goal resistance of HITPA'EL roaming motion verbs is not an outgrowth of its intransitivity as enforced by the HITPA'EL template, but rather from their anti-maximality limiting the range of dative-marked goals they can take. The HITPA'EL template in this case is not performing an argument structural function by changing the transitivity of the verb, but rather by strengthening the truth conditions of the verb it forms relative to its PA'AL correlate. In the case of the roaming motion verbs which it forms, the strengthening takes the form of the addition of an antimaximality entailment.

In addition, I have indicated that the strengthening effect the HITPA'EL template is displaying is due to the markedness difference between the two templates. These results give a first indication as to what the semantic function of the templates is. They exist as a system of contrast in markedness, where the templates fall on a markedness hierarchy with one another, where more marked templates generate stronger meanings relative to their less marked counterparts. Such is the case with HITPA'EL here contrasting with PA'AL, where the more marked HITPA'EL template has a stronger meaning than its less marked PA'AL counterpart. The rest of this dissertation will investigate these various markedness relationships across the various templates in contrast with one another.

Finally, this chapter has addressed an alternation with a strictly semantic effect, and developed the notion of strengthening on the basis of the semantic effect demonstrated here. However, there are other templatic alternations which do have

clear argument structural effects which must then interact with the semantic strengthening proposed here. For example, the other functions of HITPA'EL as demonstrated in (99) - (101), repeated here, also satisfy the VCMP.

(99) Inchoative:

- a. *Ron bifel* *et ha=marak b=a=femef*
 Ron INTNS.cook[3SG.PST.M] ACC the=soup in=the=sun
 'Ron cooked the soup in the sun.'
- b. *ha=marak hitbafel* *b=a=femef*
 the=soup REFL.cook[3SG.PST.M] in=the=sun
 'The soup cooked in the sun.'

(100) Reflexive:

- a. *Yicxak 'iper* *et Tomi*
 Tomi INTNS.make.up[3SG.PST.M] ACC face=PL.3POSS
 'Tomi made his face up.'
- b. *Tomi hit'aper*
 Tomi REFL.make.up[3SG.PST.M]
 'Tomi put makeup on (himself).'

(101) Reciprocal:

- a. *Yosi xibek* *et jager*
 Yosi INTNS.hug[3SG.M.PST] ACC Jagger
 'Yosi hugged Jagger.'
- b. *Yosi v=jager hitxabk-u*
 Yosi and=Jagger REFL.hug-3PL.PST
 'Yosi and Jagger hugged. (Kastner 2017, p. 3, (4)-(6))

In each of these cases, the difference between the PI'EL and HITPA'EL forms is argument structural, but each of these argument structural differences also correspond with an additional entailment for the HITPA'EL form which the PA'AL form also has. The reflexive HITPA'EL verbs have an entailment of reflexivity which their PA'AL forms do not have, and the reciprocal HITPA'EL verb has an entailment of reciprocity

which the PA'AL forms do not have. The relationship between a PI'EL causative and a HITPA'EL inchoative is less clear, but I will argue in chapter 6 that the latter has reflexive semantics, and therefore satisfies the VCMP through that additional reflexive entailment. In the next chapter, I will investigate more systematically how the VCMP can derive argument structural effects through an investigation of the templatic realization of the locative alternation.

Chapter 4: HIF'IL locative verbs and their argument structural frames

4.1 Introduction

In the previous chapter, I established that Hebrew templates are able to contrast with just a specific semantic effect, rather than an argument structural one. In particular, HITPA'EL roaming motion verbs contrast with their related PA'AL non-roaming manner of motion verbs through the addition of an entailment of antimaximality to the former, rather than a restriction of dative marking. This in turn follows from a correlation between markedness and strength, such that the additional entailment of antimaximality makes the verb derived in the more marked HITPA'EL template stronger than the verb derived in the less marked PA'AL template. In turn, the argument structural facts of the templatic contrast fall out; the goal resistance of HITPA'EL roaming motion verbs is actually an incompatibility with dative marked goals which indicate a final endpoint.

Here, I turn to a templatic contrast with a clear argument structural effect, and derive the contrast from differences in entailments between the two verbs. Specifically, I am going to focus on the canonical transitive and causative template HIF'IL and indicate that formation of a verb in HIF'IL requires causal entailments, but not necessarily the addition of a causer.

In general, HIF'IL creates transitive and causative verbs, and participates in a variety of causativization alternations across verb classes:

(148) Causative-inchoative:

- a. *ha=ya'ar ba'ar*
the=forest burn[3SG.M.PST]
'The forest burned.'

PA'AL

- b. *Ya'el hiv'ir-a et ha=ya'ar*
 Yael CAUS.burn-3SG.F.PST ACC the=forest
 'Yael burned the forest.' HIF'IL (constructed)

(149) Synthetic causative:

- a. *ha=kelev rac*
 the=dog run[3SG.M.PST]
 'The dog ran.' PA'AL
- b. *'afer heric et ha=kelev*
 Asher CAUS.run[3SG.M.PST] ACC the=dog
 'Asher made the dog run.' HIF'IL (constructed)

In (148), a HIF'IL verb is the causative alternant of a causative-inchoative pair for the change of state verb *ba'ar* 'burn'. Relatedly, in (149), the HIF'IL verb adds a causer to a verb representing an activity and entailments of the causer causing the activity. These two functions correspond to the general function of HIF'IL as representing a process of causativization (Horvath & Siloni 2011b; Laks 2013), specifically deriving a causative from a non-causative verb through the addition of an additional participant.¹

However, HIF'IL does not always contrast with a verb in another template to derive a causative version. Specifically, for some roots, HIF'IL can form locative verbs, such as *he'emis* 'load' in (150).

- (150) a. *Noa he'emis-a et ha=saki-ot al ha=masa'it*
 Noa CAUS.load-3SG.PST.F ACC the=bag-PL.F on the=truck
 'Noa loaded the bags onto the truck.'
- b. *Noa he'emis-a et ha=masa'it b=a=saki-ot*
 Noa CAUS.load-3SG.PST.F ACC the=truck in=the=bag-PL.F
 'Noa loaded the truck with the bags.' (constructed)

In general, locative verbs take two internal arguments, a locatum and a location, and involve movement of the locatum relative to the location. In (150), the location is

¹I will discuss this function more in the next chapter when I include it in my analysis of the causative-inchoative alternation.

the truck, the locatum is the bags, and the movement involves the bags entering the truck. Locative verbs can undergo an alternation which reorders the locatum and the location arguments, switching which is marked as a direct object and which is marked as an oblique (see, among many others, Hall 1965; Rappaport & Levin 1988; Tenny 1992, 1994; Levin & Rappaport Hovav 1995; Mateu 2000; Koenig & Davis 2006; Beavers 2006, 2010; Erteschik-Shir & Rapoport 2010). Throughout this chapter, I will refer to the argument marking pattern in (150a), where the locatum is marked as the direct object, as the ‘locatum object frame’, while I will refer to the pattern in (150b), where the location is marked as the direct object, as the ‘location object frame’.

In (150), the non-alternating verb *he’emis* is able to appear with both argument marking patterns, but in Hebrew, not all locative verbs can appear in both frames (see Segal 2020, p. 3, List 1-5, for a full list of which verbs can appear in which frames). For example, PA’AL *taval* ‘dip’ cannot, and can only appear in the location object frame:

- (151) a. *Noa tavl-a et ha=xalah b=a=marak*
 Noa dip-3SG.F.PST ACC the=challah in=the=soup
 ‘Noa dipped the challah into the soup.’ (constructed)
- b. ~~#~~ *Noa tavl-a et ha=marak b=a=xalah*
 Noa dip-3SG.F.PST ACC the=soup in=the=challah

Conversely, PI’EL *tibel* ‘season’ can only appear in the locatum object frame, and cannot appear in the location object frame:

- (152) a. *Noa tibl-a et ha=salat b=femen zayit*
 Noa INTNS.season-3SG.F.PST ACC the=salad in=oil olive
- b. ~~#~~ *Noa tibl-a et femen ha=zayit b=a=salat*
 Noa INTNS.season-3SG.F.PST ACC oil the=olive in=the=salad
 ‘Noa seasoned the salad with olive oil.’ (constructed)

The question then becomes under what circumstances a verb can appear in a given frame? Segal (2020) derives the felicity of a locative verb in an argument structural

frame through an interaction of the entailments of the verb and the content of the preposition. In general, a verb is licit in a frame when its entailments are fully compatible with the entailments of the preposition.

However, though *he'emis* in (150) can appear in both frames, other HIF'IL locative verbs are restricted to only one. Specifically, HIF'IL *hit'in* 'load' has a strong preference for the location object frame, with the locatum object frame dispreferred:

- (153) a. *Noa hit'in-a et ha=masa'it b=a=saki-ot*
 Noa CAUS.load-3SG.PST.F ACC the=truck in=the=bag-PL.F
 'Noa loaded the truck with the bags.'
- b. ?*Noa hit'in-a et ha=saki-ot al ha=masa'it*
 Noa CAUS.load-3SG.PST.F ACC the=bag-PL.F on the=truck
 'Noa loaded the bags onto the truck.' (constructed)

Unlike *he'emis* in the same HIF'IL template, *hit'in* has a correlate in the PA'AL template, *ta'an* 'load', where a PA'AL equivalent does not exist for *he'emis* (e.g. **'amas*). Conversely to *hit'in*, *ta'an* prefers the locatum object frame, with the location object frame dispreferred:

- (154) a. ?*Noa ta'an-a et ha=masa'it b=a=saki-ot*
 Noa load-3SG.PST.F ACC the=truck in=the=bag-PL.F
 'Noa loaded the truck with the bags.'
- b. *Noa ta'an-a et ha=saki-ot al ha=masa'it*
 Noa load-3SG.PST.F ACC the=bag-PL.F on the=truck
 'Noa loaded the bags onto the truck.' (constructed)

Thus, the contrast between the PA'AL *ta'an* and the HIF'IL *hit'in* is not one where the HIF'IL verb has an additional argument which has a causer role, but instead realizes an internal argument alternation where the marking of two internal argument alternations change (cf. Herslund 1995, for Danish).

In this chapter, I will demonstrate that for these alternating loading verbs, membership in the HIF'IL template fixes the argument-structural frame of the verb,

but only when it contrasts with another verb in PA'AL. Specifically, though data like (154) might indicate that PA'AL *ta'an* can only appear in the location object frame, a corpus analysis will demonstrate that it is able to appear in both the locatum object and location object frames. To contrast, that same corpus analysis will demonstrate that HIF'IL *hit'in* can only appear in the location object frame. However, *he'emis* can appear in both frames like *ta'an*, despite being in the more marked HIF'IL template, because it does not have a related verb in a weaker template. This felicity pattern is consistent with the VCMP, as a loading verb in the more marked template has a stronger meaning by entailing holistic change for one of its arguments, where the loading verb in the weaker template has no such entailments. These entailments in turn fix the options for the argument structure of each of these verbs, following a principle like the Morphosyntactic Alignment Principle (Beavers 2010). In addition, I explore data suggesting that in the location object frame, the locatum oblique is instrumental, in the sense that it is a causal participant in an intermediary subevent. I will use these data to explain variation in the argument-structural frames which HIF'IL loading verbs can appear in, specifically by analyzing HIF'IL as an instance of the causative-instrumental syncretism (Jerro 2017).

4.2 The semantics of the Hebrew locative alternation

In Hebrew, Segal (2020) identifies three different argument structural frames which locative verbs can appear in, depending on the syntactic position of the location and locatum. First, the location object frame marks the location as an object and the locatum as an oblique; this is the frame from (150b). Second, the locatum object frame marks the locatum as an object and the location as an oblique, as demonstrated in (150a). Finally, it is possible for the locatum to be in the subject position, and the location to be marked as oblique, constituting the locatum subject frame (for Segal, the State frame):

- (155) *boc lixlex* *et ha=safsal*
 mud INTNS.dirty[3SG.M.PST] ACC the=bench
 ‘Mud dirtied the bench.’ (Segal 2020, p. 2, (3c))

As shown in (151) and (152), not all of these argument structural frames are available to all locative verbs. The question then becomes what the conditions are that allow a verb to enter an argument structural frame. I review two main approaches to addressing this question, a compatibility approach from Segal & Landau (2012) and Segal (2020), and the lexical entailment approach from Beavers (2010), though I will ultimately affiliate myself with the latter.

Both Segal & Landau (2012) and Segal (2020) take a compatibility approach to this question, where a locative verb is able to appear in an argument structural frame if its entailments are compatible with the entailments of the oblique which it appears with. In general, locative verbs entail two changes along two scales: a change of location for the locatum and a change of state for the location.² Additionally, in both argument structural frames, both changes are entailed, and so cannot be denied, regardless of the argument structural frame. I demonstrate this in (156) through the infelicity of asserting that the changes happened while equally asserting the locative verb.

- (156) a. *#Noa he'emis-a* *et ha=saki-ot al ha=masa'it aval*
 Noa CAUS.load-3SG.PST.F ACC the=bag-PL.F on the=truck but
ha=saki-ot lo zaz-u
 the=bag-PL.F NEG move-3PL.PST
 ‘Noa loaded the bags onto the truck but the bags didn’t move.’

²Both Segal & Landau (2012) and Segal (2020) assume a restriction on a single predicate entailing two changes, their Unique Scalar Change Constraint (cf. Goldberg 1991; Tenny 1994; Tortora 1998) in order to permit the preposition to introduce changes which the verb does not entail. I do not assume such constraint here, following e.g. Beavers (2010); Beavers & Koontz-Garboden (2017), but the compatibility approach does not require such a constraint, as it is possible to exclude a verb from a frame based on conflicts between the meaning of the verb and the meaning of the preposition.

b.# *Noa he'emis-a* *et ha=saki-ot al ha=masa'it aval*
 Noa CAUS.load-3SG.PST.F ACC the=bag-PL.F on the=truck but
ha=masa'it lo hitmal'-a (b'xlal)
 the=truck NEG fill.up-3SG.F.PST (at all)
 'Noa loaded the bags onto the truck but the truck didn't fill up (at all).'

(constructed)

Similarly, (157) shows how neither change can be denied in the location object frame.

(157) a.# *Noa he'emis-a* *et ha=masa'it b=a=saki-ot aval*
 Noa CAUS.load-3SG.PST.F ACC the=truck in=bag-PL.F but
ha=saki-ot lo zaz-u
 the=bag-PL.F NEG move-3PL.PST
 'Noa loaded the truck with the bags but the bags didn't move.'

b.# *Noa he'emis-a* *et ha=masa'it b=a=saki-ot aval*
 Noa CAUS.load-3SG.PST.F ACC the=truck in=bag-PL.F but
ha=masa'it lo hitmal'-a (b'xlal)
 the=truck NEG REFL.fill.up-3SG.F.PST (at all)
 'Noa loaded the truck with the bags but the truck didn't fill up (at all).'

(constructed)

Thus, in general, for a verb to enter a frame, the preposition marking the oblique must have a meaning which is compatible with the change the oblique is a participant in (see e.g. Beavers 2009).

Segal & Landau (2012) focused on verbs of detaching, which, conversely to locative verbs, involve motion of a locatum away from a location:

(158) *Dan nigeu* *et ha='avak me=ha=madaf*
 Dan INTNS.wipe[3SG.M.PST] ACC the=dust from=the=shelf
 'Dan wiped the dust from the shelf.' (Segal & Landau 2012, p. 226, (7c))

Just as in (150), verbs of detaching can also undergo an argument alternation which changes which of the arguments is marked as an object and which is marked as an oblique:

- (159) *Dan nigev* *et ha=madaf me=ha='avak*
 Dan INTNS.wipe[3SG.M.PST] ACC the=shelf from=the=dust
 ‘Dan wiped the shelf of the dust.’ (Segal & Landau 2012, p. 227, (8c))

While in both argument structural frames the preposition is *me=*, Segal & Landau demonstrate that *me=* in the locatum object frame in (158) is distinct from *me=* in the location object frame. Namely, when *me=* marks a location, it serves as a full preposition with locative semantics, while when *me=* marks a locatum, it is semantically inert, marking the participant in a result state predication headed by a default preposition.

In particular, *me=* has a free alternant *min*, which is typically present with verbs of motion to represent a source:

- (160) *Dani yaca' min ha=xeder*
 Dani go.out[3SG.M.PST] from the=room
 ‘Dani went out from the room.’ (Segal & Landau 2012, p. 228, (11b))

When *me=* appears with a verb of detaching, this alternation between free and clitic forms is only possible when marking a location oblique, and is ungrammatical when marking a locatum oblique:

- (161) a. *Dani maxak et ha=nusax-ot min ha=lu'ax*
 Dani erase[3SG.M.PST] ACC the=equation-PL from the=board
 b. **Dani maxak et ha=lu'ax min ha=nusax-ot*
 Dani erase[3SG.M.PST] ACC the=board from the=equation-PL
 ‘Dani erased the equations from the board.’

(Segal & Landau 2012, p. 228, (13))

Therefore, they conclude that *me=* in the locatum object frame is a true locative preposition introducing change of location semantics independently of any other element in the argument structure frame. Specifically, it does so because it is the only

without the spatial noun, where the only requirement is for the subject to be at the library, not necessarily inside:

- (164) *ani a-xakeh l-xa b=a=sifriya*
 I 1SG.FUT-INTNS.wait to-2SG.M in=the=library
 ‘I’ll wait for you at the library.’ (Segal 2020, p. 75, (126a))

The spatial noun has been taken as a realization of a typically silent Place head representing a region of the location, which compounds with the location itself to specify where relative to the location the locatum moves (for details on the specific syntactic structure see e.g. Svenonius 2003; Botwinik-Rotem 2008; Botwinik-Rotem & Terzi 2008; Kracht 2008).

Accordingly, the spatial prepositional complex in (163) can only appear with a locative verb when it marks a location, i.e. in the locatum object frame, and is ungrammatical when it marks a locatum, i.e. in the location object frame:

- (165) a. *hu mile mayim b=tox ha=’ambatiya*
 he INTNS.fill[3SG.M.PST] water in=inside the=bathtub
 (lit.) ‘He filled water into the tub.’
 (based off of Segal 2020, p. 79, (134))

- b. **ha=po’al-im he’emis-u et ha=masa’it b=tox argaz-e*
 the=worker-PL CAUS.load-3PL.PST ACC the=truck in=inside the
ha=per-ot
 box-POSS.PL the=fruit-PL
 (intended) ‘The workers loaded the truck inside the fruit crates.’
 (based off of Segal 2020, p. 81, (139a))

Therefore, under this analysis, only the oblique in the locatum object frame is compatible with spatial semantics such as change of location, paralleling the oblique marking a location in the case of the verbs of detaching. Similarly, the locatum oblique is only

compatible with a change of state, specifically a change in possession where the location comes to have control over the locatum (see Segal 2020, sect. 4.5.1 for her specific formal implementation in terms of ‘central coincidence relations’.)

The specific marking of the location or the locatum as an oblique also has a separate effect of determining the specificity of the two changes associated with each of the participants. Specifically, the locative alternation demonstrates the ‘holistic affectedness effect’ (Anderson 1971), where an alternating argument is entailed to be holistically affected when it is marked as a direct object, in that all of it undergoes a change, while when it is marked as an oblique, only part of it must undergo a change:

- (166) a. John smeared paint on the wall, but most of the wall didn’t get any paint on it.
 b. John smeared the wall with paint, #but most of the wall didn’t get any paint on it. (Anderson 1971, p. 389, (8))

Thus, the difference in argument marking correlates with the semantics, where differing entailments associated with the alternating argument leads to different marking. This is the claim of the Morphosyntactic Alignment Principle discussed in chapter 2 (Beavers 2010):

- (167) **Morphosyntactic Alignment Principle (MAP):** When participant x may be realized as either a direct or oblique argument of verb V , it bears L-thematic role R as a direct argument and L-thematic role $Q \subseteq_M R$ as an oblique. (Beavers 2010, p. 848, (69))

Specifically, the MAP requires oblique marked arguments to have equal or weaker entailments associated with them than direct marked correlates.

Thus, for the locative alternation, the difference in which marking strategy is utilized for which argument reflects a difference in specificity for the change associated with that argument. For Beavers, the contrast is between quantized and non-quantized

change, with quantized change entailed of a participant when it is marked as a direct object, while non-quantized change is entailed when it is marked as an oblique. I repeat the definition of quantized and non-quantized change from Beavers (2011) below:

- (168) a. Quantized change: $\phi \rightarrow \exists e \exists s [\text{result}'(x, s, \mathbf{g}_\phi, e)]$
 ‘ ϕ encodes quantized change if the theme x reaches a predicate-specific final result $\mathbf{g}_\phi$ on a scale s in an event e .’
 b. Non-quantized change: $\phi \rightarrow \exists e \exists s \exists g [\text{result}'(x, s, g, e)]$
 ‘ ϕ encodes non-quantized change if the theme x reaches some final result g on a scale s in an event e . (Beavers 2011, p. 358, (60a-b))’

Having demonstrated in (156) and (157) that locative verbs entail two changes, change of location for the locatum and change of state for the location, which is thus minimally non-quantized change, the only difference between quantized and non-quantized change is in telicity.

Specifically, predicates of change entailing quantized change are telic, while predicates of change entailing non-quantized change are atelic. Thus, the change associated with the direct object can give rise to telicity, while the change associated with the oblique does not. Beavers (2010) probes this distinction through bare plurals, which forces atelicity for the participant represented by the bare plural, but only for predicates of quantized change (cf. Beavers 2011):

- (169) a. *Noa he'emis-a et ha=masa'i-ot b=a=saki-ot tox*
 Noa CAUS.load-3SG.F.PST ACC the=truck-PL in=the=bag-PL inside
 fa'a
 hour
 ‘Noa loaded the trucks with the bags in an hour.’
 b. *Noa he'emis-a masa'i-ot b=a=saki-ot kol ha=yom*
 Noa CAUS.load-3SG.F.PST truck-PL in=the=bag-PL all the=day
 ‘Noa loaded trucks with the bags all day.’ (constructed)

In (169a), the locatum object is definite, and so the whole sentence is telic, as demonstrated by felicity with *tox fa'a* ‘in an hour’. In turn, the change associated with that object is quantized. To contrast, in (169b), the locatum object is a bare plural, the sentence is atelic, as demonstrated by felicity with *kol ha=yom* ‘all day’. Such an effect does not hold for the expression of the oblique, where the oblique being marked as a bare plural does not change the telicity of the sentence:

- (170) a. *Noa he'emis-a et ha=masa'i-ot b=a=saki-ot tox*
 Noa CAUS.load-3SG.F.PST ACC the=truck-PL in=the=bag-PL inside
fa'a
 hour
 ‘Noa loaded the trucks with the bags in an hour.’
- b. *Noa he'emis-a et ha=masa'i-ot b=saki-ot tox fa'a*
 Noa CAUS.load-3SG.F.PST ACC the=truck-PL in=bag-PL inside hour
 ‘Noa loaded the trucks with bags in an hour.’ (constructed)

In both sentences in (170), the sentence is telic, as shown through felicity with *tox fa'a*, despite the difference in definiteness in the location oblique. Therefore, the change associated with the direct object is quantized, while the change associated with the oblique is non-quantized.

Accordingly, the implicational relationship between quantized and non-quantized change satisfies the requirement of minimal contrast which the MAP requires. Namely, the MAP requires the thematic role *qua* set of lexical entailments associated with the participant when it is marked as an object to be equal to or smaller than its thematic role when marked as an oblique. This requirement is captured through minimal context in (171).

- (171) **Minimal Contrast:** Q is minimally weaker than R ($Q \subseteq_M R$) on a hierarchy of L-thematic roles iff $Q = R$ or $Q \subset R$ and there is no role P on the hierarchy such that $Q \subset P \subset R$. (Beavers 2010, p. 848, (68))

The non-quantized change associated with the direct object is minimally weaker than the quantized change associated with the oblique in the other variant. Namely, quantized change entails non-quantized change, such that change to predicate-specific result state is contained within the notion of change to any result state, regardless of which it is. This relationship is part of the affectedness hierarchy, the implicational hierarchy between the four levels of affectedness (see the previous chapter for a discussion):

(172) **Affectedness Hierarchy:**

$$\forall x, \phi, e [\exists s [\text{result}'(x, s, \mathbf{g}_\phi, e)] \rightarrow \exists s \exists g [\text{result}'(x, s, g, e)] \rightarrow \exists s \exists \theta [\theta(x, s, e)] \rightarrow \exists \theta' [\theta'(x, e)]] \quad (\text{Beavers 2011, p. 359, (62)})$$

Thus, the difference in argument marking derives directly from the semantics, where differing entailments associated with the alternating argument leads to different marking.

This type of analysis gives the denotations in (173) for *he'emis* ‘load’ from (150) in its two argument structural frames (cf. Beavers 2010, p. 839, (49)):

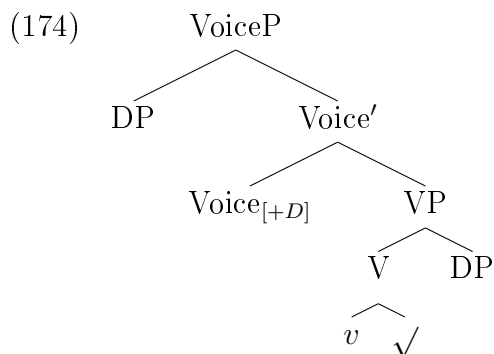
$$\begin{aligned} (173) \quad \text{a. } \llbracket \text{he'emis et ha=saki-ot al ha=masa'it} \rrbracket &= \lambda e \exists s_1 \exists s_2 [\text{load}'(\mathbf{b}, \mathbf{t}, e) \wedge \\ &\quad \exists g [\text{result}'(\mathbf{t}, s_1, g, e)] \wedge \text{result}'(\mathbf{b}, s_2, \mathbf{g}_{\text{ON}(\mathbf{t})}, e)] \\ \text{b. } \llbracket \text{he'emis et ha=masa'it b=a=saki'ot} \rrbracket &= \lambda e \exists s_1 \exists s_2 [\text{load}'(\mathbf{b}, \mathbf{t}, e) \wedge \\ &\quad \exists g [\text{result}'(\mathbf{b}, s_1, g, e)] \wedge \text{result}'(\mathbf{t}, s_2, \mathbf{g}_{\text{FULL}}, e)] \end{aligned}$$

Each of these denotations follow the argument structural frame, where the participant marked as an object undergoes quantized state to a specified goal, full of bags for the truck and all on the truck for the bags. Thus, the entailments of holistic change, here represented as quantized change, follow the argument which is marked as an object. In the next section, I will assess whether the HIF'IL template can also determine the argument structure of a locative verb, and in turn provide an analysis where HIF'IL is providing entailments of holistic change, following the VCMP.

4.3 The argument structural frames of HIF'IL

4.3.1 Three HIF'IL locative verbs

Recall from the first section that the canonical function of the HIF'IL template is to form causative, transitive verbs. This function is demonstrated in (148) and (149), where HIF'IL verbs contrast with non-HIF'IL alternants to generate a version with an additional causer and entailments of causation. To that end, Kastner (2020) captures the strong causativity and transitivity of the HIF'IL template through a specifier requirement for the Voice head which is realized as the HIF'IL template, Voice_[+D]. The [+D] feature on the Voice head is a EPP-feature which requires a DP to Merge into the specifier of Voice. This feature ensures that the verb will have an underlying subject which through the denotation of the Voice head will serve as the causer of the event introduced by the root. This structure is given in (174).



(Kastner 2020, p. 132, (12))

Under this structure, all HIF'IL locative verbs should be causatives, as specified by the Voice head. Additionally, as HIF'IL locative verbs contrast with locative verbs in other templates, it should equally contrast with another locative verb in a different template. That is, where the HIF'IL verb is obligatorily causative, the other verb need not be.

Given the function of HIF'IL, consider the locative root $\sqrt{\text{XNH}}$, related to parking (e.g. of vehicles). In PA'AL, that root forms *xanah* 'park', shown in (175).

- (175) *ha=rɛxɛv xanah* *b=a=midraxa*
the=car park[3SG.M.PST] in=the=sidewalk
‘The car was parked on the sidewalk.’ (constructed)

xanah is in the locatum subject frame (cf. (155)), and as such represents a state, here of the car being parked on the sidewalk. As such, it is possible to assert (175) but deny that the parking was caused (if, for example, it was built on the sidewalk and never moved), demonstrated in (176).

- (176) *ha=rɛxɛv xanah* *b=a=midraxa* *aval af 'exad lo*
the=car park[3SG.M.PST] in=the=sidewalk but no.one NEG
hexna *'oto* *fam*
CAUS.park[3SG.M.PST] ACC-3SG.M there
‘The car had been parked on the sidewalk but no one parked it there.’
(constructed)

As a PA'AL verb, Kastner (2020) casts it as a realization of unspecified Voice, and it has no requirements on its external argument.

Accordingly, when $\sqrt{\text{XNH}}$ generates a verb in HIF'IL, it creates *hexnah* ‘cause to park’, a causative alternant of *xanah*, which cannot appear as a causative:

- (177) a. *Yosi hexnah* *et ha=rɛxɛv b=a=midraxa*
Yosi CAUS.park[3SG.M.PST] ACC the=car in=the=sidewalk
‘Yosi parked the car on the sidewalk.’ (constructed)
- b. **Yosi xanah* *et ha=rɛxɛv b=a=midraxa*
Yosi park[3SG.M.PST] ACC the=car in=the=sidewalk

hexnah is the causativized version of *xanah*: an external causer, Yosi, is added relative to the argument structure of *xanah*, which has no causer. The new external causer subsequently causes the the event of parking to happen, such that Yosi caused the car to be parked on the sidewalk. This type of contrast is exactly that anticipated under the analysis of HIF'IL from (174), such that the HIF'IL alternant is specified to be a causative where the PA'AL alternant need not be.

However, HIF'IL does not always have the effect of causativizing a non-HIF'IL locative verb, as it does with *xanah*. Specifically, with the root $\sqrt{T'N}$ related to loading, both the PA'AL and HIF'IL verbs appear in full causative, transitive argument structural frames. This is demonstrated in (153) and (154), repeated here.

- (178) *Noa ta'an-a et ha=saki-ot al ha=masa'it*
 Noa load-3SG.PST.F ACC the=bag-PL.F on the=truck
 'Noa loaded the bags onto the truck.'
- (179) *Noa hit'in-a et ha=masa'it b=a=saki-ot*
 Noa CAUS.load-3SG.PST.F ACC the=truck in=the=bag-PL.F
 'Noa loaded the truck with the bags.' (constructed)

In contrast to $\sqrt{XNH}$, the HIF'IL form of $\sqrt{T'N}$, *hit'in*, is not the PA'AL verb *ta'an* with an additional causer, but rather realizes a different argument structural frame compared to its PA'AL correlate. Additionally, while *hit'in* can appear in the location object frame, *hexnah* must appear in the locatum object frame, and is ungrammatical otherwise:

- (180) **Yosi hexnah et ha=midraxa b=a=rexev*
 Yosi CAUS.park[3SG.M.PST] ACC the=sidewalk on=the=car
 (intended) 'Yosi parked the street on the car.' (constructed)

$\sqrt{T'N}$ contrasts further with $\sqrt{MS}$, also related to loading. $\sqrt{T'N}$ also forms a verb in HIF'IL which can appear in both argument structural frames, demonstrated in (150) and repeated here.

- (181) a. *Noa he'emis-a et ha=saki-ot al ha=masa'it*
 Noa CAUS.load-3SG.PST.F ACC the=bag-PL.F on the=truck
 'Noa loaded the bags onto the truck.'
- b. *Noa he'emis-a et ha=masa'it b=a=saki-ot*
 Noa CAUS.load-3SG.PST.F ACC the=truck in=bag-PL.F
 'Noa loaded the truck with the bags.' (constructed)

Thus, the HIF'IL template does not have a uniform effect across locative verbs when compared to their PA'AL correlates, and raises the question of whether there is a uniform effect that can be attributed to a verb's membership in the HIF'IL template. Here, I will assess the extent to which HIF'IL locative verbs prefer one argument structural frame over the other, starting with the contrast between *hit'in* and *he'emis*.

4.3.2 Verbal markedness and argument frames

I start by comparing the different argument structural preferences for *hit'in* and *he'emis*, both meaning 'load'. Both of these verbs are formed off of roots that are related to loading, and consequently exist in the same semantic space as one another. As such, a comparison of the argument structural preferences of these two verbs would control for idiosyncratic differences between the two verbs. In turn, it will isolate the grammatical or structural differences which would influence argument structural preference for one frame over another.

Towards that end, the main difference between *hit'in* and *he'emis* is whether it has an alternant in another template or not. *hit'in* shares a root with the PA'AL verb *ta'an*, but no such verb exists for *he'emis*. In fact, *he'emis* is the only verb formed of its root $\sqrt{\text{'MS}}$.⁴ As such, there are no verbs which *he'emis* can contrast with. Under the VCMP, differences in truth-conditional strength hinge on a difference in markedness, understood as formal complexity, with more marked forms having stronger truth conditions than less marked forms. Conversely, when no markedness relationships exist for a form, no truth-conditional strength difference is necessitated. Such should be the case for *he'emis*, where a lack of contrast with any other templatic forms would indicate that *he'emis* has no requirement to be stronger or weaker than another verb form.

⁴ $\sqrt{\text{'MS}}$ does however form a deverbal adjective *'amus* 'loaded/burdened'. This kind of deverbal adjective is typically derived from a correlated PA'AL verb (Meltzer-Asscher 2011; Doron 2013), and so might indicate a present but unrealized PA'AL verb **'amas*. However, in the absence of any evidence of such an underlying form, I treat it as if one does not exist.

As these three verbs under consideration are verbs of loading, they all mainly vary in the holistic change entailed of either of its two arguments. Specifically, as summarized in the previous section, the choice of argument marking determines which argument undergoes holistic change. As such, syntactic compatibility with either the location object frame or the locatum object frame indicates semantic compatibility with holistic change of state or holistic change of location respectively. Accordingly, if a verb specifies for one of these holistic changes, its range of argument structural frames are similarly limited, namely to the one which is compatible with the holistic change which the verb entails. I will therefore be using compatibility with location object or locatum object frames as a stand-in for which types of change are entailed by the loading verb. Where a verb is compatible with the location object frame, it will entail holistic change of state, and where the verb appears with the locatum object frame, it will entail holistic change of location.

Therefore, for the Hebrew verbs considered here, the VCMP predicts a locative verb with a contrasting correlate in a less marked template should have a significant preference for one of the two argument structural frames, where its less marked correlate should not. Similarly, a locative verb without a contrasting correlate should have no such preference. Such is the difference between *hit'in* and *he'emis*, where the more marked HIFIL *hit'in* contrasts with the less marked PA'AL *ta'an* and therefore should prefer one of the two argument structural frames, where *ta'an* need not. Similarly, *he'emis*, being the only verb of its root, should also have no necessary preference for one argument structural frame over another, despite being in the more marked HIF'IL template.

To that end, I assessed the collocative frequency of *ta'an*, *hit'in*, and *he'emis* with objects representing either locata or locations. For each of the three verbs, I extracted from heTenTen 2014 all of the nouns marked as its direct object, as determined by collocation with *et* directly adjacent to the verb, and number of tokens of each of those nouns functioning as direct object. From this list, I selected all of the nouns which appeared as a direct object for all three verbs, and excluded all of the

nouns which only appeared with just one or two of the verbs. In general, restricting the dataset in this way had the effect of helping ensure that all three verbs were operating in the same semantic space, such that all three feasibly had the interpretation of ‘load’ in the sense of moving a physical object onto a physical location. Specifically, *ta’an* is polysemous between a loading reading and ‘claim’, demonstrated in (182).

- (182) *hi ta’an-a fe=hi ha=nesixa fel ha=bayit*
 she claim-3SG.F.PST COMP=she the=princess POSS the=house
 ‘She claimed that she’s the princess of the house.’ (corpus)

he’emis has no such readings, and so nouns related to claiming, such as *te’ana* ‘claim’ were excluded. Similarly, *hit’in* is the primary loading verb used in computer-related contexts, as well as ‘charge’ in the sense of charging a battery.

- (183) a. *Iris hit’in-a et ha=kovec ’al disk on ki*
 Iris CAUS.load-3SG.F.PST ACC the-file on thumbdrive
 ‘Iris loaded the file onto the thumbdrive.’
 b. *Iris hit’in-a et ha=pelefon shel-a*
 Iris CAUS.load-3SG.F.PST ACC the=cell.phone POSS-3SG.F
 ‘Iris charged her cellphone.’ (constructed)

These technical uses of *hit’in* might represent a metaphorical type of loading. However, in order to be conservative about the readings permitted in this study, such nouns where only *hit’in* had them as the direct object were excluded. This avoided the possibility that this sense of *hit’in* might lead to different argument structural properties.⁵

As such, by limiting the range of objects under consideration to those which occur with all three nouns, I ensured that the three verbs shared the same sense, and thus the only differences between their distributions would be conditioned by different

⁵Three such nouns remained in the sample after this exclusion: *maxfev* ‘computer’, *disk* ‘disk’, and *daf* ‘page’.

Noun	Translation	Classification
<i>even</i>	rock	Locatum
<i>if</i>	person	Locatum
<i>guf</i>	body	Locatum
<i>davar</i>	thing	Locatum
<i>xavila</i>	package	Locatum
<i>xefec</i>	thing	Locatum
<i>mocar</i>	product	Locatum
<i>kli</i>	tool	Locatum
<i>mat'en</i>	charger	Locatum
<i>sxora</i>	cargo	Locatum
<i>parit</i>	item	Locatum
<i>oto</i>	car	Location
<i>atar</i>	site	Location
<i>disk</i>	disk	Location
<i>daf</i>	page	Location
<i>xayim</i>	life	Location
<i>xalal</i>	space	Location
<i>maxfev</i>	computer	Location
<i>mixonit</i>	car	Location
<i>maxfir</i>	appliance	Location
<i>macber</i>	pile/collection	Location
<i>nefek</i>	gun	Location
<i>sfina</i>	boat	Location
<i>rexev</i>	car	Location
<i>ta</i>	cell	Location
<i>tarmil</i>	backpack	Location

Table 4.1: Semantic classification of locative verb direct objects

grammatical or structural properties. For each of these nouns, I coded them either as a location or a locatum based on a qualitative assessment of their use in various sentences throughout heTenTen 2014 where they are the object of one of the three loading verbs. In general, a noun was coded as a location if notionally it represented a location, area, or space, such as *mixonit* ‘car’. To contrast, nouns coded as a locatum notionally were physical objects which co-occurred with a clear location, such as *xavila* ‘package’. Where there was no clear categorization for the noun, in that there were

contradictory examples in corpus where the noun appeared as both a locatum or a location, I additionally excluded it from the dataset, so as to be conservative. This mostly excluded nouns which would have led to a metaphorical type of loading, as well as numerals which could have modified either a location or a locatum. After these exclusions, there were 27 total nouns, 16 locations and 11 locata, shown in Table 4.1.

For the frequency of each of these nouns as the direct object of the three verbs, I calculated the collocation of each type of noun with each verb as a percentage of the total number of tokens with each verb collocating with a direct object. These percentages thus approximate a percentage occurrence for *ta'an*, *hit'in*, and *he'emis* in either the location object frame or the locatum object frame.⁶ I display these percentages in Figure 4.1. As anticipated, *hit'in*, as the HIF'IL verb has a large preference for one argument frame over another, namely the location object frame over the locatum object frame. *ta'an* has a similar preference for the location object frame over the locatum object, but to a much smaller margin, while *he'emis* only slightly prefers the locatum object frame over the location object frame, but to a marginal degree.

To assess the validity of these differences, I fit a quasibinomial logistic regression to the data, estimating the effect of loading verb and direct object category on the frequency of collocation with a location or a locatum. The dependent variable was percentage collocation, and the independent variables were loading verb and direct object category. From that regression, I calculated the estimated marginal means of each group and compared the mean percentage of collocation with a location vs. a locatum by verb. Across the three verbs, only the difference within *hit'in* was significant (log odds = 2.086, $p = 0.047$), with *hit'in* preferring location objects more than locatum objects (as indicated by the positive effect size). The other two differences were not significant.

⁶The total percentages will not add up to 100%, as the amounts are computed as a percentage of the total number of verbs with a direct object, and so those direct objects which are not part of the sample in Table 5.1 are not represented.

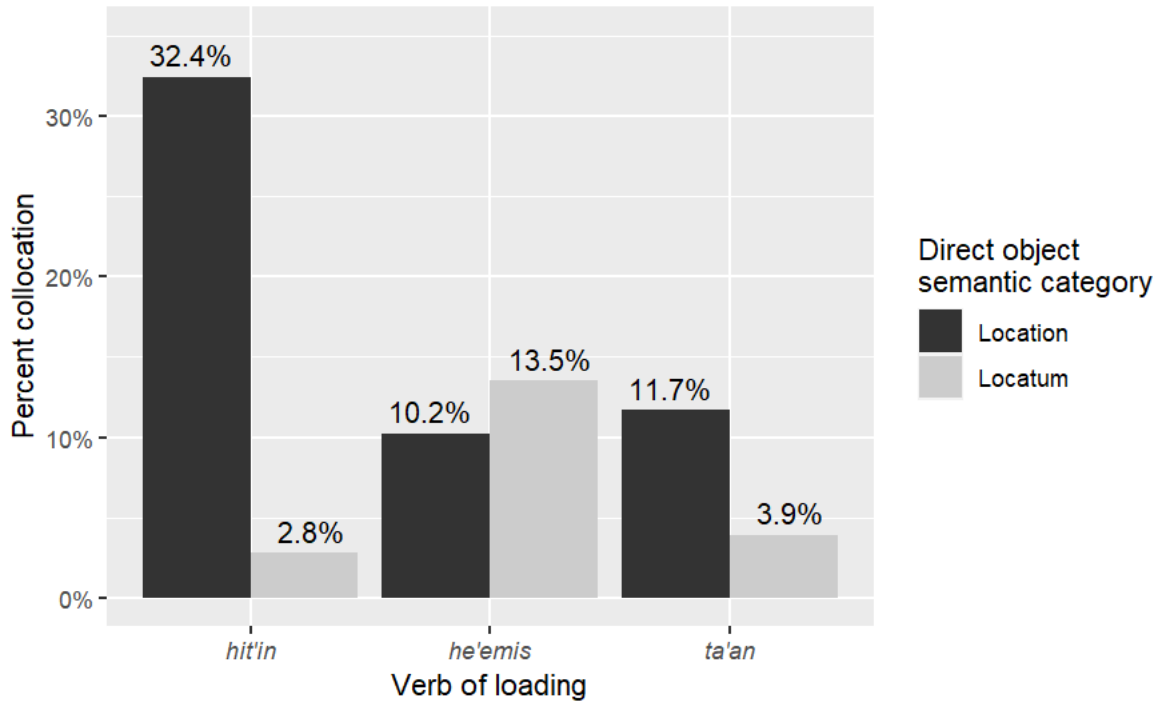


Figure 4.1: Percentage of location vs. locatum direct objects by load verb

4.3.3 Analysis

Among the three verbs, only *hit'in* has a significant preference for one argument structural frame over the other, namely the location object frame over the locatum object frame. The location object frame, being the frame which marks the location as a direct object and the locatum as an oblique, represents holistic change of state, namely of the locatum. Therefore, *hit'in*'s significant preference for the location object frame indicates a lexical specification for holistic change of state.⁷ Both *ta'an* and *he'emis* have no such preference, despite data like (154). *ta'an* and *he'emis* therefore have no such entailments, and are underspecified for holistic change, which is thus to be determined by the argument structural frame which it ultimately appears

⁷*hit'in* does have a 2.8% collocation with locatum objects, which is significantly higher than 0 following a one-sided t-test ($t = 7.9018$, $p = 0.00001312$). I return to the implications this might have in chapter 6.

in. I represent these preferences in the following table.

Verb	Root	Marked?	Holistic change of state	Holistic change of location
<i>ta'an</i>	$\sqrt{T'N}$	Less marked	Compatible	Compatible
<i>hit'in</i>	$\sqrt{T'N}$	More marked	Entailed	Not possible
<i>he'emis</i>	$\sqrt{MS}$	No markedness	Compatible	Compatible

Table 4.2: Holistic change specifications of Hebrew loading verbs

Because *ta'an* is underspecified, I represent its meaning as entailing two changes, but without specifying which change is quantized and which change is non-quantized. I do so through an underspecification analysis, where the variable argument structure frames derive from *ta'an* specifying only for non-quantized change in both cases. Under this analysis, *ta'an* as a lexical item entails only non-quantized change of state and change of location, as in (184).

$$(184) \quad \llbracket ta'an \rrbracket = \lambda y \lambda x \lambda e \exists s_1 \exists s_2 [load'(x, y, e) \wedge \exists g_1 [result'(x, s_1, g, e)] \wedge \exists g_2 [result'(y, s_2, g, e)]]$$

These changes are further specified when the direct object is introduced and saturates the first open argument position, thus fixing it as the participant undergoing holistic change, following the MAP.⁸ After specification, *ta'an* with a direct object would have two possible meanings, given in (185).

$$(185) \quad \begin{aligned} \text{a. } \llbracket ta'an \text{ et } ha=sakiot \rrbracket &= \lambda x \lambda e \exists s_1 \exists s_2 [load'(x, y, e) \wedge \exists g [result'(x, s_1, g, e)] \wedge \\ &\quad result'(\mathbf{b}, s_1, \mathbf{g}_{AT(\mathbf{x})}, e)] \\ \text{b. } \llbracket ta'an \text{ et } ha=masa'it \rrbracket &= \lambda x \lambda e \exists s_1 \exists s_2 [load'(x, y, e) \wedge \exists g [result'(x, s_1, g, e)] \wedge \\ &\quad result'(\mathbf{t}, s_1, \mathbf{g}_{FULL}, e)] \end{aligned}$$

⁸This analysis treats the MAP here as a filter on verb meaning which must be satisfied in order to create a licit expression, a constraint much like those in frameworks like HPSG (Pollard & Sag 1994). As a filter on meaning, I presume there is a process of coercion which performs this specification in order to satisfy the MAP, though I am agnostic as to what that process is.

Basically, *ta'an*, due to its low markedness relative to its HIF'IL correlate, specifies neither change to be holistic, and as such entails both changes only to be non-quantized.

On the other hand, membership in HIF'IL restricts the possible denotations for *hit'in* to just one of the two possible options for *ta'an*, namely the one which entails quantized change of state, linked to the location object frame.

$$(186) \quad \llbracket hit'in \rrbracket = \lambda y \lambda x \lambda e \exists s_1 \exists s_2 [load'(x, y, e) \wedge \exists g [result'(x, s_1, g, e)] \wedge result'(y, s_1, \mathbf{g}_{FULL}, e)]$$

This pattern follows specifically from the VCMP necessitating more marked forms to have stronger truth conditions than less marked forms. The only loading verb which entails one of the two possible changes to be holistic is *hit'in*. Additionally, *hit'in* is the only loading verb which contrasts with a verb in a less marked template, *ta'an*. As such, the VCMP necessitates for *hit'in* to have stronger truth conditions than *ta'an*, and this condition is satisfied by the holistic change of state entailments which *hit'in* has and *ta'an* lacks. Specifically, because the denotation of *ta'an* is underspecified for which change is holistic, the minimal set of entailments which constitute its truth conditions are non-quantized change for both change of location and change of state. However, *hit'in* has been limited just to a denotation which entails quantized change of state for the location. Thus, because quantized change is stronger than non-quantized change, the entailment of quantized change associated with *hit'in* makes its truth conditions stronger than those of *ta'an*, schematized in (187).

$$(187) \quad \left\{ \begin{array}{c} \underline{ta'an} \\ \text{Non-quantized change of location} \\ \text{Non-quantized change of state} \end{array} \right\} <_S \left\{ \begin{array}{c} \underline{hit'in} \\ \text{Non-quantized change of location} \\ \text{Quantized change of state} \end{array} \right\}$$

Similarly, even though *he'emis* is also in the more marked HIF'IL template, it does not have another verb to contrast with. As such, there is no markedness asymmetry for *he'emis* which the VCMP can operate over, and no requirement for it to have stronger truth conditions, here realized as an entailment of one holistic

change. As such, its denotation is identical to *ta'an*, underspecified for which change is holistic and so entailing only non-quantized change for both changes:

$$(188) \quad \llbracket he'emis \rrbracket = \lambda y \lambda x \lambda e \exists s_1 \exists s_2 [load'(x, y, e) \wedge \exists g_1 [result'(x, s_1, g, e)] \wedge \exists g_2 [result'(y, s_2, g, e)]]$$

Thus, because *he'emis* is underspecified as *ta'an* is, the direct object is able to specify which change is holistic and which change is not, following the MAP. This kind of meaning for *he'emis* is possible under the VCMP despite it being in the more marked HIF'IL template, because there are no other verbs in its root which contrast with it. As such, there is no requirement for it to have a stronger meaning, as *hit'in* does, despite being in the same template.

What remains is an understanding of the difference between *hit'in* and *hexnah* 'park', such that the former corresponds to the locative alternation while the latter corresponds to an instance of causativization. In the next section, I will assess this difference, focusing on the fact that *hit'in* prefers the location object frame, with a locatum oblique, while *hexnah* requires the locatum object frame, with a location oblique. Specifically, based on the syncretism between a locatum oblique and an instrumental oblique demonstrated in (189) and (190), I will suggest that in both cases the HIF'IL verb has causal entailments which the PA'AL verb does not, with the only difference being whether the causer is initial, generating a causative in the case of *hexnah*, or medial, generating an instrument in the case of *hit'in*.

4.4 A locative-instrumental syncretism

4.4.1 What is an instrument

In the location object frame, when the locatum is marked as an oblique, *b=* is able to vary with *'im* 'with':

- (189) a. *ha=po'al-im he'emis-u et ha=masa'it b=argaz-e*
the=worker-PL CAUS.load-3PL.PST ACC the=truck in=box-POSS.PL
ha=per-ot
the=fruit-PL
- b. *ha=po'al-im he'emis-u et ha=masa'it 'im argaz-e*
the=worker-PL CAUS.load-3PL.PST ACC the=truck with box-POSS.PL
ha=per-ot
the=fruit-PL
'The workers loaded the truck with the fruit crates.'

(Segal 2020, p. 82, (141a))

These prepositions are identical to the prepositions which can mark an instrument:

- (190) a. *Noa ta'an-a et ha=masa'it b=/'im ha=saki-ot*
Noa load-3SG.PST.F ACC the=truck in=/with the=bag-PL.F
'Noa loaded the truck with the bags.'
- b. *Yaniv gazar et ha=niyar b=/'im misparayim*
Yaniv cut[3SG.M.PST] ACC the=paper in=/with scissors
'Yaniv cut the paper with scissors.' (constructed)

On a very basic level, this syncretism between instrumental and locatum prepositions might indicate that the two are holding the same role. One possibility for that syncretism is that the locatum is an instrument. Here, I will overview the concept of an instrument, settling on the idea that an instrument is a participant in an intermediary event, and give some evidence that this syncretism holds and that the locatum oblique does in fact serve as an instrument.

In general, an instrument is an intermediary causer, which is acted on by an agent to cause a change of state. Koenig et al. (2008) follows Croft (1991) and Talmy (2000) in defining an event involving an instrument as 'an agent of some sort acts on the instrument and the result of this action directly or indirectly causes a change of state in a patient or theme' (p. 176). Such an event is that in (190b) in Hebrew. Here, *misparayim* 'scissors' is the instrument, introduced by the preposition *b=* 'in',

and is a means by which the agent *Yaniv* caused the paper to be cut. An instrument is therefore both a causer, in that it causes the change of state, and a type of force recipient, in that an agent also acts on it. In (190b), the agent Yaniv acted on the scissors, and as a result of being acted on, the scissors cut the paper.

As early as Fillmore (1968), the notion of instrument has been taken to be an instantiation of a thematic role, much like agent and theme. Reinhart (2002), in her feature based analysis of thematic roles, specifies it as $\langle +c, -m \rangle$, that is, specified for causation but with an explicit lack of mental state. As such, it shares the feature of causation ($\langle +c \rangle$) with both agents and causers, indicating a generalization over the three; Van Valin & Wilkins (1996) in fact subsumes them under a single thematic role of ‘effector’ (see also Ramchand 2008). Because of its similarities to agents and causers, instruments share some grammatical features with both, such as the instrumental subject alternation, where an instrument can serve as the subject in place of an agent or causer (e.g. Schlesinger 1989; DeLancey 1991; Rappaport Hovav & Levin 1992; Kamp & Rossdeutscher 1994; Alexiadou et al. 2006; Alexiadou & Schafer 2006; Schäfer 2012):

- (191) a. John broke the window with a rock.
 b. The rock broke the window.

In (191), the instrument *a rock*, in its canonical position in (191a), is additionally able to serve as the subject in (191b). This alternation derives from the fact that instruments, like agents or causes canonically when they are subjects, is a type of causer and is therefore able to serve a similar causal function to that of the subject.

If a locatum oblique is in fact an instrument, in the sense of saturating a thematic role and serving the same grammatical function, then they should pattern the same with respect to their distributional tests. Segal (2020, sect. 4.5.3.1) applies a battery of tests from Iwata (2008) to both instrument obliques and locatum obliques to demonstrate that the former is an adjunct and the latter is an argument (Halevy

2009; Doron & Dubnov 2017), and thus the two are not identical. Importantly, these tests are at their core argumenthood tests. As such, what they diagnose is not whether the locatum semantically is an instrument, in that it is an intermediary causer, but rather whether it is syntactically an adjunct as a canonical instrument oblique is. Ultimately, the five tests she applies all converge on the instrument oblique patterning like an adjunct, while the locatum oblique is an argument, and as such, the locatum oblique is not an instrument in this sense of instrument.

First, a locatum oblique and an instrument oblique can co-occur in the same sentence, indicating that the two do not have the same syntactic function:

- (192) *Dani cipa et ha='uga b=fokolat 'im marit*
 Dani INTNS.cover[3SG.M.PST] ACC the=cake in=chocolate with spatula
 ‘Dani covered the cake with chocolate with a spatula.’

(Segal 2020, p. 128, (222d))

Second, when a locatum oblique and an instrumental oblique occur in the same sentence, they cannot be scrambled. Specifically, the locatum oblique needs to be closer to the verb than the instrument oblique, and generates an infelicitous sentence if the order is reversed:

- (193) *#hu cipa et ha='uga 'im marit b=fokolat*
 he INTNS.cover[3SG.M.PST] ACC the=cake with spatula in=chocolate
 (intended) ‘He covered the cake with a spatula with chocolate.’

(Segal 2020, p. 129, (224b))

Third, it is impossible to insert an adverb between the direct object and a locatum oblique, while it is possible to do so between the direct object and an instrument oblique, indicating a tighter relationship to the verb for the locatum oblique than for the instrument oblique:

- (194) a. *hila he'emis-a et ha=masa'it b'mhirut b=argaz-im*
 Hila CAUS.load-3SG.F.PST ACC the=truck quickly in=box-PL
 (intended) 'Hila loaded the truck with boxes quickly.'
- b. *hila he'emis-a et ha=masa'it b'mhirut 'im malgeza*
 Hila CAUS.load-3SG.F.PST ACC the=truck quickly with forklift
 'Hila loaded the truck quickly with the forklift.'
- (Segal 2020, p. 130, (225c-d))

Fourth, it is possible to answer a *how*/'*ex* question with an instrument oblique, but not a locatum oblique:

- (195) '*ex hila he'emis-a et ha=masa'it*
 how Hila CAUS.load-3SG.F.PST ACC the=truck
 'How did Hila load the truck?'
- a. **b='im argaz-im*
 in=/with box-PL
 (intended) 'With boxes.'
- b. *b='im malgeza*
 in=/with forklift
 'With a forklift.'
- (Segal 2020, p. 130, (226c))

Finally, an instrument oblique can be stranded under *do so*/'*asa et ze* replacement, but not a locatum oblique:

- (196) a. *Itay cipa et ha='uga b=krem fokolat*
 Itay INTNS.cover[3SG.M.PST] ACC the=cake in=cream chocolate
**v=Hila as-ta et ze b=krem vanil*
 and=Hila do-3SG.F.PST ACC that in=cream vanilla
 (intended) Itay covered the cake with chocolate frosting and Hila did so
 with chocolate cream.'
- b. *Itay cipa et ha='uga 'im marit v=Hila*
 Itay INTNS.cover[3SG.M.PST] ACC the=cake with spatula and=Hila
as-ta et ze 'im sakin
 do-3SG.F.PST ACC that with knife

‘Itay covered the cake with a spatula and Hila did so with a knife.

(Segal 2020, p. 131, (227c-d))

Thus, Segal concludes that the locatum oblique is not an instrument identical to an instrumental oblique, as the former is an argument and the latter is an adjunct (cf. Rissman 2010, Rissman & Rawlins 2017, sect. 4.1).

The results from Segal (2020) stand in contrast to the approach to the locatum oblique from Koenig et al. (2003, 2008). Specifically, the diagnostics from Iwata (2008) she applies to the Hebrew data only diagnose the *syntactic* status of the two prepositional phrases, and therefore conclude that an instrument oblique is not the same syntactic object as a locatum oblique, and so do not hold the same thematic role (cf. Rissman 2010). However, they fail to diagnose whether the locatum oblique holds any other properties in common with an instrument, such as whether it has causal force. This notion of instrument which I am targeting here is specifically a semantic one, intending to characterize it not in terms of shared distributional properties, but rather similarities in their meaning. I turn to this notion of instrument now, and motivate a distinction between instruments introduced periphrastically, appearing as an object of verbs like English *use* or Hebrew *hiftemef*, and those introduced obliquely like with *b=* or *'im*. In both cases, an instrument is a non-initial causer in the causal chain, but periphrastic instrument are more peripheral to the causal chain, but more under the control of the agent, while oblique instruments are more central to the causal chain.

From the perspective of its meaning, the notion of instrument is a very broad one, and covers a vast range of participants of a variety of different types for a given event. Koenig et al. (2008, sect. 2) identifies 14 subtypes of obligatory instruments, and 12 subtypes of optional instruments, each of which differ in the relation they have to the event which they are a participant in. Under their classification, locative verbs in English are a part of the *fill*-type class, which entails the locatum as an instrument

causing a change in configuration:⁹

(197) *fill*-type verbs, schematically:

$$\begin{array}{l} \mathbf{cause}(s_1, s_2) \quad \wedge \quad \mathbf{ACT}(s_1, \mathbf{A}, \mathbf{I}) \quad \wedge \quad \mathbf{in}(s_2, \mathbf{I}, \mathbf{P}) \quad \wedge \quad \mathbf{cause}(s_2, s_3) \quad \wedge \\ \mathbf{CHANGE-OF-CONFIGURATION}(s_3, \mathbf{P}) \end{array} \quad (\text{Koenig et al. 2008, p. 190, (12)})$$

In this representation, the agent **A** acts on the instrument **I**, which causes the instrument to be in the patient *P*, which in turn causes the patient *P* to undergo a change in configuration. Here, the patient **P** is the location, which undergoes a change in configuration by virtue of the instrument **I** being in **P**. This analysis both positions the locatum as entailed by the verb, and attributes causal force to the locatum, by virtue of it causing the change in configuration.¹⁰

However, Koenig et al.’s classification of instrument types vary wildly in terms of the kinds of entailments which define them. For example, *cut*-type verbs, as in (190b), require their instrument to have direct contact with the theme and cause the incision by doing some cutting, while *eat*-type verbs, demonstrated in (198), only require the instrument to assist in causing the eating.

(198) Jill drank her soda with a straw. (Koenig et al. 2008, p. 198, (38b))

Here, the instrument *a straw* is not actually doing any drinking, but rather assists the agent *Jill* in performing the drinking. This type of instrument stands in direct contrast to the *cut*-type verb in (190b), where the instrument actually causes the cutting to happen. Unlike the instrument of *fill*-type verbs, those of *eat*-type verbs do not apply any force in the event, nor cause any change, but instead have an assisting role in the event.

⁹Koenig et al. (2008) operates within a version of situation semantics which I do not use here. Nonetheless, I reproduce their representation faithfully.

¹⁰Tangentially, Koenig et al.’s *fill*-type verbs explicitly entail that the locatum is an instrument, and thus it would be unsurprising that the locatum *qua* an instrument would pattern more like an argument than an oblique as demonstrated in (192)-(196), e.g. under a condition on semantic argument corresponding to a syntactic argument, such as the Semantic Obligatoriness Condition (Koenig et al. 2003).

Because of the wide range of properties which can be associated with an instrument, it therefore has been taken not as a semantic role with a fixed meaning associated with it (cf. role based approaches like Fillmore 1968; Iwata 2008), but rather a related cluster of properties which can be instantiated differently across contexts. In this way, an instrument is best construed as a prototype centered around a set of properties which a given participant can be a better or worse instantiation of (Schlesinger 1989; Dowty 1991; Primus 2002). Rissman et al. (2022) characterized the prototype of an instrument as a ‘direct extension of an intentional agent’ (p. 27), in that the most prototypical instruments need not actually cause some sort of change, but rather just need to be a means by which an agent affects the change itself. Specifically, they demonstrated that tools, body parts, and means of contact pattern together as most licit in instrumental contexts, with categories like means of transport, accidental agents, and, importantly here, locatums, as more marginal. Overall, they demonstrated gradient differences across 10 different categories of potential instruments, such that groups of them pattern closely to the prototypical instrument while others share some properties but are more marginal. As such, while a locatum may not be as canonical an instrument as something like a tool, it still might share enough properties to function as one.

Consequently, because there are so many different types of instruments which approximate the prototype, there can be several different means of expressing an instrument, which can be similar to one another, but not necessarily identical. For example, in addition to the oblique *b=* in (190b), it is possible to introduce an instrument periphrastically, in Hebrew with the verb *hifstamef*:¹¹

- (199) *Yaniv hifstamef* *b=a=misparayim li-gzor et ha=niyar*
 Yaniv REFL.use[3SG.M.PST] in=the=scissors INF-cut ACC the=paper

¹¹Note that the instrument *misparayim* is still marked with *b=*, the same oblique which marks an oblique instrument. I take this to be a case of a prepositionally-marked object and not a true instrumental PP, as note that the preposition cannot vary with the other instrumental preposition *'im* (**hifstamef 'im ha=misparayim*) and thus maintain a distinction between the two structures.

‘Yaniv used the scissors to cut the paper. (constructed)

Rissman & Rawlins (2017) identifies for English both the oblique structure in (190b) and a periphrastic structure like (199) as introducing an instrument, but differ in what kind of instrument they introduce. Namely, while both instruments share the property of being a causer in an intermediary event, the oblique, *with* in English and *b=* in Hebrew, specifically require the instrument to have a tight causal relationship to the overall event. To contrast, the periphrastic, *use* or *hifstamef*, only requires intentionality, in that the agent must intend to participate in the event by means of the instrument.

Specifically, they demonstrate first that oblique instrumentals are infelicitous when in a context where the agent does not intentionally perform an action:

- (200) a. #While sleeping, Jim used his sleeping bag to mop the floor.
 b. While sleeping, Jim mopped the floor with his sleeping bag.

(Rissman & Rawlins 2017, p. 513, (14a), (15a))

In (200), the agent Jim is performing the mopping while sleeping, presumably by moving back and forth in his sleep. This obviates any intentionality, as Jim did not intend to mop the floor, and only did so incidentally in his sleep. Accordingly, *use* is not felicitous, while *with* is. Similar facts obtain in Hebrew for *hifstamef*, as contrasted with *b=*:

- (201) a. # 'or *hifstamef* *b=sak* *ha=fena fel-o*
 Or REFL.use[3SG.M.PST] in=bag[POSS] the=sleep POSS-3SG.M
l-nakot et ha=ricpa kfe=hu yafan
 INF-clean ACC the=floor while=he sleep[3SG.M.PST]
 (intended) ‘Or used his sleeping bag to clean the floor while he was sleeping.’

- b. *'or nikah* *et ha=ricpa b=sak ha=fena*
 Or INTNS.clean[3SG.M.PST] ACC the=floor in=bag[POSS] the=sleep
fel-o kfe=hu yafan
 POSS-3SG.M while=he sleep[3SG.M.PST]
 ‘Or cleaned the floor with his sleeping bag while he was sleeping.’
 (constructed)

To contrast, they show that *with* has no intentionality requirements, but rather requires that the instrument be directly involved in the event as an extension of the agent’s force. Namely, *with* instrumentals greatly disprefer peripheral instruments which might hold a helping or a preconditioning role, and instead prefer instruments which participate in the event themselves:

(202) (where Nigel is chopping the lettuce on his desk)

- a. Nigel used his desk to chop the lettuce.
 b.#Nigel chopped the lettuce with his desk.

(Rissman & Rawlins 2017, p. 530, (56))

Here, the desk is holding a helping role in the event: it is not directly acting on the lettuce, but rather helping the agent Nigel to act on the lettuce, by being the surface which he chops it on. Accordingly, the desk is only peripherally involved in the event, in that it is not actually participating in the main activity of the event. Similar facts also hold for the Hebrew oblique instrumental *b=*:

- (203) a. *Yaniv hiftamef b=a=delpek la-xtox et ha=xasah*
 Yaniv REFL.use[3SG.M.PST] in=the=desk INF-cut ACC the=lettuce
 ‘Yaniv used the desk to cut the lettuce.’
 b.#*Yaniv xatax et ha=xasah b=a=delpek*
 Yaniv cut[3SG.M.PST] ACC the=lettuce in=the=desk
 (intended) ‘Yaniv cut the lettuce with the desk.’ (constructed)

Thus, while both oblique and instrumental constructions introduce instruments, the kind of instrument they introduce are different, according with the notion of instrument being a related cluster of properties, rather than a single category.

To that end, Rissman & Rawlins (2017) give an analysis of instruments which unify them solely as being a participant in a subevent which follows the initiation of the larger event by a separate agent, which they call an ‘instrumental subevent’. The individual instantiations of an instrument have their own requirements for what must be true about that instrument in that instrumental subevent, but in both cases, the instrument is a participant in a separate subevent which follows the initial event. For *use*, the instrument is specified to be within an event compatible with the agent’s intentions:

$$(204) \quad \llbracket use \rrbracket^w = \lambda x \lambda P \lambda e' [\text{ACTED-ON}(e', x, w) \wedge \exists e [e' \subset e \wedge P(w, e) \wedge \text{Ag}(e) = \text{Ag}(e') \wedge \forall w' \in \{\text{worlds compatible with Ag}(e)\text{'s intentions in } w\} [\text{ACTED-ON}(e', x, w') \wedge P(w', e) \wedge e' \subset e]]] \quad (\text{Rissman \& Rawlins 2017, p. 521, (31)})$$

In (204), the agent acts on the instrument x in a subevent e' of event e where the agent performs the action specified by P , and both events e and e' are true in all worlds compatible with the agent’s intentions. Thus, *use* is only licit when the agent intends to use the instrument and when the agent intends to perform the action in which they use the instrument.

To contrast, *with* specifically requires the instrument to be a core participant, and so specifies for the instrumental subevent to be a part of the ‘force-dynamic core’ of the event:

$$(205) \quad \llbracket with \rrbracket = \lambda x \lambda P \lambda e : \exists y [\text{Ag}(e, y)] [P(e) \wedge \exists e' [e' \subset \text{FORCE}(e) \wedge \text{ACTED-ON}(e', x) \wedge \text{Ag}(e') = \text{Ag}(e)]] \quad (\text{Rissman \& Rawlins 2017, p. 531, (60)})$$

Here, the predicate **FORCE** returns the force-dynamic core of the event e , and *with* specifies that the instrumental subevent e' is a part of that force-dynamic core. In

that instrumental subevent, the agent *y* acts on the instrument. Thus, *with* is only licit when the instrumental subevent is a core part of the event, and therefore the instrument is a core participant. Given that the instrumental subevent is specified to be part of the force-dynamic core of the event, peripheral instruments such as helping or preconditioning instruments are not permitted.

Thus, while *use* and *with* differ in what they require of their instrument, they both share the property of the instrument being a participant in an intermediary event. This notion of instrument is fundamentally different than the one which Segal (2020) probed for in the locatum oblique, as her syntactic tests only assessed whether the locatum has the same syntactic function as instruments marked with the instrumental oblique *b=*. Those tests specifically did not bear on whether a locatum is an instrument semantically, in that it satisfies some prototype of what an instrument is, potentially as established in Rissman et al. (2022). Due to the syncretism between the preposition marking an oblique locatum and an oblique instrument, one might anticipate that both would pattern like *with* and have the meaning shown in (205). Namely, a locatum oblique might be a separate participant in its own instrumental subevent where it causes some portion of the change, rather than marking a result state, as argued by Segal & Landau (2012) and Segal (2020).

4.4.2 Locata in instrumental contexts

Here, I survey data assessing the behavior of the locatum oblique on tests which characterize the instrument oblique as an intermediary causer. I start from the position that if the locatum oblique, marked with *b=* or *'im*, is an instrument, it would have the same meaning associated with it, namely that of being an intermediary causer along the causal chain. Based on the syncretism between the instrumental oblique and the locatum oblique, where both can be marked with *b=* or *'im*, it would be anticipated that the locatum could pattern like *with* from Rissman & Rawlins (2017). Namely, it should pattern as if it is a core participant serving as an intermediary causer, and therefore pass all of the same diagnostics for core participant-hood

that oblique-marked instruments do. I review three diagnostics in turn: ability to be pseudo-clefted, presence in a causal paraphrase, and preservation as a locatum in other instrumental contexts.

First, Rissman & Rawlins (2017) demonstrate that instrumental obliques can be embedded under a pseudo-cleft, there demonstrated in English with *what X did is Y*:

- (206) a. What Sarah did with the stick was poke the fire.
 b. What Sarah did with the chopsticks is eat the sushi.

(Rissman & Rawlins 2017, p. 526, (45c-d))

Pseudo-clefting as a diagnostic typically evinces a syntactic difference between an argument and an adjunct, shown with an adverb in (207).

- (207) What Sarah did quickly is eat the sushi.

Thus, passing this diagnostic in (206) indicates that the instrument oblique is an adjunct. However, just as an instrument oblique can be embedded under a pseudo-cleft, so too can a locatum oblique:

- (208) *ma fe=hila as-ta 'im ha=saki-ot hu l-male' et*
 what COMP=hila do-3SG.F.PST with the=bags COP INF-CAUS.load ACC
ha=masa'it
 the=truck
 'What Hila did with the bags is load the truck.' (constructed)

Despite the fact that other argumenthood diagnostics pick out the locatum oblique as an argument, the locatum oblique does pass the pseudo-clefting test, which diagnoses an adjunct. Both semantically optional and semantically obligatory instruments, like a locatum is under Koenig et al.'s (2008) typology (cf. (197)), pass this test, indicating a disconnect between semantic obligatoriness and argumenthood (see e.g. Rissman

et al. 2015). Thus, despite the fact that the locatum oblique is a syntactic argument, it still patterns as an instrumental oblique, indicating that the two overlap such that they can serve the same function, namely that of being an instrument.

Importantly, in (208), the oblique is still able to function as a locatum, despite being clefted out of its canonical position. Namely, in the sentence in (209), a resumptive pronoun indexed with the locatum is in the canonical position for the locatum.

- (209) *ma fe=hila as-ta 'im ha=saki-ot hu l-male' et*
 what COMP=hila do-3SG.F.PST with the=bags he INF-CAUS.load ACC
ha=masa'it 'it-an
 the=truck with-3PL.F
 ‘What Hila did with the bags is load the truck with them.’ (constructed)

Therefore, the pseudo-cleft construction is not targeting a separate use of *'im* as an instrument oblique, but rather is targeting the locatum itself functioning as an instrument.

Second, just like an instrumental oblique can never hold a preconditioning or a helping role, a locatum oblique cannot either. Intuitively, a locatum is a core participant of a locative verb, being one of the three main participants of a locative event. This status is evident from a causal paraphrase, which Rissman (2013) uses to distinguish core participants from peripheral participants.

- (210) a. Chloe cut the bread with a knife. (Rissman 2013, p. 137, (1a))
 b. i. Chloe acting on the knife caused the knife to make contact with the bread.
 ii. The contact between the knife and the bread caused the bread to be cut. (Rissman & Rawlins 2017, p. 139, (7))

Namely, a causal paraphrase identifies all of the participants with causal force along the causal chain, and therefore all of the participants which are core to the unfolding

of the event, and the causation relationships between them. The causal paraphrase of the locative sentence with *mile* ‘fill’ in (211) demonstrates that the locatum is a causer of an event of the location filling up.

- (211) a. *Noa mil’-a et ha=masa’it b=a=saki-ot*
 Noa INTNS.fill-3SG.PST.F ACC the=truck in=bag-PL.F
 ‘Noa filled the truck with the bags.’
- b. *Noa garm-a l=a=saki-ot la-zuz v=ha=saki-ot*
 Noa cause-3SG.F.PST to=the=bag-PL INF-move and=the=bag-PL
garm-u l=a=masa’it l-hitmale
 cause-3PL.PST to=the=truck INF-REFL.fill
 ‘Noa caused the bags to move, and the bags caused the truck to fill up.’
 (constructed)

Here, (211b) is a causal paraphrase of the sentence in (211a), which includes the locative verb *mile* ‘fill’. In that causal paraphrase, the locatum *ha=saki-ot* is the subject of the verb of causing *garam* ‘cause’, and as such, is a core participant with causal force, just as an instrument is. Therefore, just as a *with* instrumental is a core participant and a causal intermediary, so too is a locatum.

Additionally, locata are able to appear in other contexts which introduce an instrument, while still retaining its function as a locatum. Namely, under embedding in a periphrastic instrument introduced by *hiftimef*, it should still be able to function as a locatum. Given that, consider the sentence in (212).

- (212) *kayom ha=adam miftamef b=a=’oxel l-male’*
 to.date the=man REFL.use[3SG.M.PRS] in=the=food INF-INTNS.fill
xasax-im rab-im b=xaya-v
 void-PL many-PL.M in=life-POSSPL.3SG.M
 ‘Even today, man uses food to fill many holes in his life.’ (corpus)

Here, the locative verb *mile* ‘fill’ is in a structure with the verb *hiftimef* ‘use’, which introduces an instrument. Specifically, following Rissman & Rawlins (2017), *hiftimef*

here introduces an instrument which is involved in the event of filling by the agent's intentions (cf. the denotation for *use* in (204)). However, the instrument *'oxel* 'food', is also serving as the locatum, in that the object filling the metaphorical hole is the food. Thus, under embedding in another structure which introduces a type of instrument, the locative verb *'mile*' is able to have its locatum serve as an instrument while also preserving its status as a locatum.

Before turning to the consequences this has on the locative alternation, I present a few caveats about the data above, specifically related to their general acceptability. In all three of the diagnostic sentences in (208)-(212), the locative verb being probed in the location object frame is *'mile*', not any of the HIF'IL locative verbs under consideration here. Because these diagnostics, especially the pseudo-cleft in (208) and the *use* periphrasis in (212), involve the positioning of the locative in a highly non-canonical position, judgments on their validity can vary.¹² Additionally, acceptability of the locatum in these non-canonical positions can vary by verb, i.e. (213).

- (213) *nitan* *l-hiftamef* *b=solela-t* *ha=maxfir*
 MID.be.possible[3.SG.IMPERS] INF-REFL.use in=battery-POSS the=tool
kday l-hat'in maxfir-im axer-im
 PURP INF-CAUS.load tool-PL other-PL.M
 ('*im xafmal/?'im ha=solela*)
 (with electricity/with the=battery)
 'It is possible to use the device's battery to charge other devices (with power/?with the battery).'

Here, the most readily available reading for the instrumental argument of *hifamef*, *solela* 'battery', is as a instrument holding a helping in an event of charging a device

¹²Extrasentential supporting factors, such as contrastive focus, have been noted to support instruments in non-canonical positions (DeLancey 1991; Alexiadou et al. 2006), and as such, a more systematic assessment of under what conditions these diagnostics are acceptable might be warranted. For now, I note that for some speakers and in some contexts the sentences in (208)-(212) are acceptable, and take that as their general availability.

with electricity. A reading where the battery itself is a locatum, in that it moves to be under the control of the device, is highly dispreferred. This is in contrast to (212), wherein a reading where the instrumental argument is also the locatum is highly available.

I take this difference in acceptability by verb to be an outgrowth of individual semantic differences between locative verbs which make some of them acceptable with their locata in non-canonical positions and some of them not. Importantly, in general, locative verbs in Hebrew can vary by which of their argument structural frames they can appear in (cf. (151)-(152)), which Segal (2020) derives from specific lexical-semantic constraints which make the verb compatible with the semantic contribution of the argument structural frame (see e.g. the discussion in section 4.2). *mile* ‘fill’, the verb used in the diagnostics above, is one of the few verbs which can appear in all three argument structural frames available for locative verbs in Hebrew, show in (214).

- (214) a. Locatum object frame:

hu mile *mayim b=tox* *ha='ambatiya*
he INTNS.fill[3SG.M.PST] water in=inside the=bathtub
'He filled water into the bathtub.'

- b. Location object frame:

hu mile *et* *ha=’ambatiya b=mayim*
 he INTNS.fill[3SG.M.PST] ACC the=bathtub in=water
 ‘He filled the bathtub with water.’

- c. Locatum subject frame:

ha=mayim mil'-u *et ha=ambatiya*
the=water INTNS.fill-3PL.PST ACC the=bathtub
'Water filled the bathtub.'

I take the general syntactic flexibility of *mile'* as indicating its compatibility with these structures which diagnose a locatum oblique as an instrument. Loading verbs

like *hit'in* are comparatively more syntactically specific, and I take that as the reason it is less felicitous in these diagnostic structures. I remain agnostic as to what this property is that permits *mile'* to appear with a locative in a non-canonical position while *hit'in* does not.

Thus, the data above demonstrate that diagnostics which pick out an oblique as a semantic instrument can also pick out a locatum oblique, and additionally preserves its function as a locatum as a general fact, even if acceptability varies from verb to verb. They therefore cast a locatum into an instrumental role, contra the results from Segal (2020). Importantly, because the locatum oblique is an instrumental, it is a core participant in the event, and additionally one with causal force, in that it is the causer of a subevent of change as part of the larger event represented by the locative verb, demonstrated in (211). In the next section, I will discuss the consequences this has for *hit'in* as compared to *hexnah*, and point towards the conclusion that the HIF'IL template is generating additional entailments of one of the arguments being an causer relative to their PA'AL counterparts, with the only difference being whether the effector is initial or medial.

4.5 HIF'IL as the causative-instrumental syncretism

In section 4.3, I demonstrated that the HIF'IL loading verb *hit'in* ‘load’ significantly preferred the location object frame, with a locatum oblique, over the locatum object frame. Namely, the felicity pattern from (153), repeated here, holds.

- (153) a. *Noa hit'in-a* *et ha=masa'it b=a=saki-ot*
 Noa CAUS.load-3SG.PST.F ACC the=truck in=the=bag-PL.F
 ‘Noa loaded the truck with the bags.’
- b. ?*Noa hit'in-a* *et ha=saki-ot al ha=masa'it*
 Noa CAUS.load-3SG.PST.F ACC the=bag-PL.F on the=truck
 ‘Noa loaded the bags onto the truck.’ (constructed)

However, not all HIF'IL locative verbs prefer the location object frame. Namely, *hexnah*

‘park’ is ungrammatical in the location object frame, demonstrated in (180) repeated here.

- (180) **Yosi hexnah* *et ha=midraxa b=a=rexev*
 Yosi CAUS.park[3SG.M.PST] ACC the=car on=the=car
 (intended) ‘Yosi parked the sidewalk on the car.’

Here, I will suggest that the difference between these two verbs such that they prefer different argument structural frames despite being the same template derives from the HIF’IL form having additional causal entailments which their PA’AL form do not. However, based on the entailments of causation which are present in the simpler PA’AL form, the additional causers are realized differently, either as an initial causer in the case of *hexnah* or a medial causer in the case of *hit’in*. In this way, HIF’IL is realizing the causative-instrumental syncretism, where a single morphological marker can add either a causer or an instrument.

Descriptively, the causative-instrumental syncretism describes a phenomenon where a single morpheme, typically an applicative morpheme, can introduce both a causer and an instrument. I demonstrate this in its most canonical form here in (216) from Jerro (2017).

- (215) *Habimana y-a-men-a igi-kombe*
 Habimana 1.SBJ-PST-break-IPFV 7-cup
 ‘Habimana broke the cup.’
- (216) a. *Habimana y-a-men-esh-eje umw-ana igi-kombe*
 Habimana 1.SBJ-PST-break-ISH-PFV 1-child 7-cup
 ‘Habimana made the child break the cup.’
- b. *Habimana y-a-men-esh-eje igi-kombe in-koni*
 Habimana 1.SBJ-PST-break-ISH-PFV 7-cup 9-stick
 ‘Habimana broke the cup with a stick.’

(Kinyarwanda, Jerro 2017, p. 753, (3)-(4))

In both examples in (216), the verbal affixal morpheme *-ish*, realized phonetically here as *-esh* and glossed as *-ISH*, attaches to the verb stem *men* ‘break’ and introduces a new argument. In the case of (216a), the new argument is a causer, Habimana, who causes a second participant to break the cup. However, the same morpheme affixing to the same verb in (216b) can also add an instrument, *in-koni* ‘stick’, wherein Habimana remains the causer, but the new argument is instead an instrument which serves as the means which Habimana performs the breaking by.

Intuitively, if the affix in (216) is taken as syntactic, modifying the argument structure of the verb, these two functions, adding a causer and adding an instrument, are at odds with each other. Namely, the addition of a causer as in (216a) is an instance of the addition of a new subject, with the demotion of the subject of the non-derived verb, while the addition of an instrument is the addition of an object; the syntactic effects are fundamentally different. However, Jerro (2017) unifies the two functions under a single semantic process of adding a causer, but underspecifying for the new causer is initial, initiating the whole event, or medial, initiating a change in a later part of the event having been initiated by an earlier causer. Jerro treats the added participant as an effector, following Van Valin & Wilkins (1996), a generic role covering anything which has causal force, namely agents, causes and instruments. Specifically, he gives a denotation for *-ish* which follows the denotation for English *with* from (Rissman 2011, cf. (205)):

$$(217) \quad \llbracket -ish \rrbracket = \lambda P \lambda y \lambda x_1 \dots \lambda x_n \lambda e_1 \dots \lambda e_m [P(x_1 \dots x_n, e_1 \dots e_m) \wedge \exists e' [e' \subset e_m \wedge ag'(e', y)]] \quad (\text{Jerro 2017, p. 771, (39)})$$

As such, *-ish* introduces a new effector in a proper subevent of the whole event predicated by the verb which it attaches to. If the new effector is a participant in an initial subevent, it is realized as a causer and produces causativization, and if the new effector is medial, it is realized as an instrument.

Given that the locatum oblique also patterns like an instrument, derivation of a HIF’IL verb with a contrasting PA’AL verb also manifests as the causative-instrumental

syncretism. Namely, the contrast between *hexnah* and *hit'in* is such that the former is a causative alternant of its PA'AL correlate and *hit'in* is an instrumental correlate:

- (218) a. *Yosi hexnah* *et ha=rexev b=a=midraxa*
 Yosi CAUS.park[3SG.M.PST] ACC the=car in=the=sidewalk
 'Yosi parked the car on the sidewalk.'
- b. *Noa hit'in-a* *et ha=masa'it b=a=saki-ot*
 Noa CAUS.load-3SG.PST.F ACC the=truck in=the=bag-PL.F
 'Noa loaded the truck with the bags.' (constructed)

I take the difference between the two HIF'IL verbs in (218) to be conditioned by the semantics of the simpler PA'AL correlate. I understand the HIF'IL template here as specifying for a participant with causal force, which in turn can be satisfied either by adding one to the denotation of the simpler PA'AL correlate, or by one already being present its denotation. The additional causal entailments which are present in the HIF'IL forms are similar to those added by *-ish* in (217). Specifically, a participant unspecified for causer or instrument is added as a participant in a proper subevent of the main event denoted by the verb. However, to allow for compatibility with statives, which do not entail an event, I represent this kind of entailment as requiring a participant in a subevent rather than a proper subevent, thus allowing the addition of an event where one is not entailed. I give this type of entailment schematically in (219).

- (219) Effector entailments, schematically:
 $\forall x, e [effector'(e, x) \rightarrow \exists e' [e' \subseteq e \wedge ag'(e', x)]]$

This type of entailment simply requires there to be an event *e* in the denotation of the HIF'IL verb, and a subevent *e'* in which there is a participant *x* that is an effector.

hexnah is related to the stative *xanah*, which I give a denotation for in (220).¹³

¹³I presume that the spatial preposition builds a spatial relationship orienting the subject as the figure to the ground which is the argument of the preposition, with the specific orientation provided

- (220) a. *ha=rerev xanah* *b=a=midraxa*
 the=car park[3SG.M.PST] in=the=sidewalk
 ‘The car is parked on the sidewalk.’ (constructed)
- b. $\llbracket xanah \rrbracket = \lambda x \lambda s [\text{parked}'(x, s)]$

Because *xanah* is a stative predicate, it does not entail an event, just a state which holds of the theme. Accordingly, to satisfy the causal requirements imposed by HIF'IL, *hexnah* needs to have an effector of the kind in (219) added, and accordingly, an event which the effector can cause. As such, *hexnah* has a denotation as in (221).

$$(221) \quad \llbracket hexnah \rrbracket = \lambda x \lambda y \lambda e \lambda s [\text{cause}'(e, s) \wedge \text{parked}'(x, s) \wedge \exists e' [\text{ag}'(e', y) \wedge e' \subseteq e]]$$

Because the stative *xanah* does not have a causer in the first place, any new causer added will by definition be the initial causer. As such, the effect here is causativization. Additionally, by adding an entailment of causation and subsequently a causer, the VCMP additionally holds. Namely, the addition of this causer produces a strictly stronger meaning for the more marked HIF'IL *hexnah* than the less marked PA'AL *xanah*:

$$(222) \quad \left\{ \begin{array}{c} \underline{xanah} \\ \text{Locatum parked} \end{array} \right\} <_s \left\{ \begin{array}{c} \underline{hexnah} \\ \text{Locatum parked} \\ \text{Effector causes state of locatum parked} \end{array} \right\}$$

To contrast, HIF'IL *hit'in* is related to PA'AL *ta'an*, which already entails an event, namely one of loading:

- (223) a. *Noa ta'an-a* *et ha=masa'it b=a=saki-ot*
 Noa load-3SG.PST.F ACC the=truck in=the=bag-PL.F
 ‘Noa loaded the truck with the bags.’
- b. $\llbracket ta'an \rrbracket = \lambda y \lambda x \lambda e \exists s_1 \exists s_2 [\text{load}'(x, y, e) \wedge \exists g_1 [\text{result}'(x, s_1, g_1, e)] \wedge \exists g_2 [\text{result}'(y, s_1, g_2, e)]]$

by the preposition. I follow Kastner (2020, p. 91, (46)) in composing the prepositional phrase with the verb through event identification, and thus it is not represented. As the specific locative semantics are not relevant to this issue, I gloss over it here.

Unlike *xanah*, *ta'an* already specifies for a causer as well as an instrument, each causing a portion of the event. Particularly, the locatum *saki-ot* ‘bags’ in (223a) is marked as an oblique, which, as argued in the previous section, marks an instrumental *qua* intermediary causer. As such, (223a) can have the same causal paraphrase as *hit'in* from (211b), repeated here.

- (224) *Noa garm-a l=a=saki-ot la-zuz v=ha=saki-ot*
 Noa cause-3SG.F.PST to=the=bag-PL INF-move and=the=bag-PL
garm-u l=a=masa'it l-hitmale
 cause-3PL.PST to=the=truck INF-REFL.fill
 ‘Noa caused the bags to move, and the bags caused the truck to fill up.’
 (constructed)

As such, the locatum marked as an oblique is already an effector, and *ta'an*, fully specified with its arguments, has a denotation as in (225).

- (225) $\llbracket ta'an \text{ et } ha=masa'it \text{ } b=saki'ot \rrbracket = \lambda e \exists s_1 \exists s_2 [load'(\mathbf{b}, \mathbf{t}, e) \wedge \exists g [result'(\mathbf{b}, s_1, g, e)] \wedge result'(\mathbf{t}, s_1, \mathbf{g}_{FULL}, e)]$

This denotation subsequently entails that the locatum is an effector, as shown in (226).

- (226) $\forall e [\llbracket ta'an \text{ et } ha=masa'it \text{ } b=saki'ot \rrbracket(e) \rightarrow \exists e' [e' \subset e \wedge ag'(e', \mathbf{b})]]$

Accordingly, for the HIF'IL correlate *hit'in*, the requirement for an effector as imposed by the template is already satisfied. Namely, because *hit'in* is lexically specified to entail holistic change of state, its locatum will always be marked as an oblique and therefore will always be an instrument. As such, it is already specified for an effector, and so one does not need to be added in the course of forming the HIF'IL correlate *hit'in*. The effect here is thus vacuous, without any causativization effect at all, but the creation of a form in HIF'IL is still licit because the requirement to entail an effector is still satisfied.

Thus, just as *ta'an* with all of its arguments entails an effector, *hit'in* lexically entails one as well.

- (227) a. $\llbracket hit'in \rrbracket = \lambda y \lambda x \lambda e \exists s_1 \exists s_2 [load'(x, y, e) \wedge \exists g [result'(x, s_1, g, e)] \wedge result'(y, s_1, \mathbf{g}_{FULL}, e)]$
 b. $\forall x, y, e [\llbracket hit'in \rrbracket(y, x, e) \rightarrow \exists e' [e' \subset e \wedge ag'(e', y)]]$

As both *ta'an* and *hit'in* entails that one of their arguments is an effector, their entailments do not differ from one another, and so this distinction does not factor into the VCMP. However, the distinction follows directly from how *hit'in* specifies for holistic change of state, as necessitated by the VCMP. I remain agnostic as to why specifically these additional causal entailments are realized syntactically in the way they do. For *hit'in*, the argument structural frame which forces quantized change of location, i.e. marking the locatum as an oblique, is the same one which coidentifies the locatum as an instrument. I pose it as an empirical fact that here quantized change of location and an additional instrument are linked and thus realized through the same grammatical means and leave it open how to derive this connection.

In summary, *hexnah* and *hit'in* both require for one of their participants to be an effector where their PA'AL counterparts are not by virtue of their membership in the HIF'IL template. They differ in whether the effector is the causer initiating the event, or an instrument causing an intermediary subevent. This distinction is derived directly from whether their simple PA'AL counterparts are specified for an effector or not, and thus whether the requirements for a causer as imposed by HIF'IL are already satisfied by the PA'AL verb. PA'AL *xanah* is a stative and so does not entail an effector; alternation into HIF'IL adds one which, as the sole effector, is a causer. PA'AL *ta'an* does entail an effector already, which satisfies the requirement imposed by the HIF'IL template, and as such, the effector need only be an instrument.

4.6 Conclusion

Here, I have assessed the effect the HIF'IL template has on the range of locative verbs which it realizes. I have demonstrated that *hit'in* significantly prefers the location object frame over the locatum object frame. This preference in turn indicates an entailment of quantized change of state which does not exist in its PA'AL counterpart. Additionally, I have given evidence that the oblique in the location object frame is an instrument, as it patterns like an intermediary causer with respect to several diagnostics which pick it out. Therefore, *hit'in* as it contrasts with *ta'an* satisfies the VCMP, as the former has several additional entailments which the latter does not, making its truth conditions stronger.

This specific case study also demonstrates how the VCMP interacts with argument marking. Namely, the additional entailments in the stronger and more marked verb force a specific argument structural frame which is compatible with those entailments. The less marked form is compatible with more argument structural frames, following its weaker entailments. In the next chapter, I will expand how the VCMP can determine argument marking by investigating how differences in template determine differences in the semantics of the causative-inchoative alternation, the most canonical argument structural function of the templates.

Chapter 5: The reflexivity of HITPA'EL and the causative-inchoative alternation

5.1 Introduction

In this chapter, I will be returning to the original, canonical function of the Hebrew template system, realizing external argument alternations, to demonstrate that even under the most syntactic functions of the template system, the templates contrast with one another to generate semantic differences. Specifically, I will be investigating how the causative-inchoative alternation is realized using different templatic pairs, and whether those differences in templatic membership for the alternants reflect semantic differences in how the alternation is derived.

The causative-inchoative alternation in Hebrew is realized as differences in templatic membership, with each alternant in an alternating pair being realized in a different template.¹ I demonstrate this in (228) with the root $\sqrt{\text{B'R}}$ related to burning.

(228) a. **PA'AL inchoative:**

ha=sade' ba'ar
the=field burn[3SG.M.PST]
'The field burned.'

b. **HIF'IL causative:**

ha=xakla'i hiv'ir *et ha=sade'*
the=farmer CAUS.burn[3SG.M.PST] ACC the=field
'The farmer burned the field.' (constructed)

Important to note is that the causative-inchoative alternation is not constrained to just one template or template pair. Arad (2005) identifies at least six different pairs

¹Hebrew has one labile alternation, with both alternants in the HIF'IL template. I will not be addressing why the only labile alternation in Hebrew occurs with HIF'IL specifically, but for details, see Lev (2016) and Kastner (2019).

of templates which generate the the causative-inchoative alternation, which she formalizes as conjugation classes. I exemplify each of these in (229).

- (229) a. **NIF'AL inchoative – PA'AL causative:**
niftax 'open (intrans.)' – *patax* 'open (trans.)'
- b. **PA'AL inchoative – HIF'IL causative:**
ba'ar 'burn (intrans.)' – *hiv'ir* 'burn (trans.)'
- c. **NIF'AL inchoative – HIF'IL causative:**
ne'elam 'disappear' – *he'elim* 'make disappear'
- d. **HITPA'EL inchoative – PI'EL causative:**
hitmale' 'fill (intrans.)' – *mile'* 'fill (trans.)'
- e. **HITPA'EL inchoative – HIF'IL causative:**
hitbaher 'become clear' – *hivhir* 'make clear'
- f. **HIF'IL inchoative – HIF'IL causative:**
he'edim 'become red' – *he'edim* 'make red'

Following Arad (2005), the root selects which conjugation class *qua* templatic pair it is realized in, and thus the root determines the morphological strategy for realizing the causative-inchoative alternation. However, because Hebrew uses so many different morphological strategies to realize the causative-inchoative alternation, the question arises of whether there are any principled differences between each of these template pairs which might guide how roots select which one they appear in. Here, I will be focusing specifically on the contrasts in (229b), (229c), and (229d). These three template pairs represent the range of argument alternation types based on the markedness relationships between the causative and the inchoative (to be discussed in the next section), and thus will be able to assess the extent to which different alternation types have different semantics.²

²I do not consider (229a) and (229f) primarily due to constraints in the experimental methodology mobilized in this chapter.

Notably, each of these templatic pairs additionally represent a markedness difference between both the causative and the inchoative forms, as well as between different realizations of the inchoative. This markedness difference determines the markedness hierarchy developed in Chapter 2, repeated here:

(230) **Markedness hierarchy for the Hebrew templates:**

PA'AL << HIF'IL, PI'EL << NIF'AL << HITPA'EL

For example, in (229a), the NIF'AL inchoative *niftax* is more marked than the PA'AL causative *patax*, following directly from their templatic membership. In this specific case, the markedness difference derives from the additional affixal material *ni-* on the NIF'AL verb which is not present on the PA'AL verb.

Following the principles developed here thus so far, the difference in templates would additionally reflect a difference in truth conditional strength. Importantly, that truth conditional strength difference would follow the markedness relationship between the templates, with the more marked template realizing a stronger meaning compared to the less marked template. Consider, for example, the root $\sqrt{\text{RTV}}$, related to wetness, which forms an inchoative both in NIF'AL and HITPA'EL:

- (231) a. *nirtav-ti* *b=a=gefem*
 MID.wet-1SG.PST in=the=rain
- b. *hitratav-ti* *b=a=gefem*
 REFL.wet-1SG.PST in=the=rain
 'I got wet in the rain.' (constructed)

Laks (2013) approaches the difference between these two seemingly synonymous inchoative forms as simply one of morphophonology, with no necessary effect on the semantics. However, on the basis of the markedness differences between the NIF'AL template and the HITPA'EL template, the VCMP would predict that these two options would reflect a semantic difference, with the more marked *hitratev* having stronger truth conditions than the less marked *nirtav*.

In this chapter, I will demonstrate that the causative-inchoative alternation in Hebrew is semantically diverse, in that different morphological strategies for realizing the alternation reflect different types of semantic contrasts between the causative and the inchoative. Specifically, using an experimental paradigm from Lundquist et al. (2016), I will demonstrate that the NIF'AL – HIF'IL pair realizes decausativization, where the inchoative has the covert agent existentially bound, while the HITPA'EL – PI'EL pair realizes reflexivization, where the inchoative has the covert agent identified with the theme. I will show that under this paradigm, HITPA'EL inchoatives prefer depictions of change where the theme undergoing the change is feasibly construed as the initiator of the change as well, while NIF'AL inchoatives do not have such a preference. In turn, the reflexive semantic form associated with HITPA'EL is a stronger meaning than the decausal semantic form associated with NIF'AL, concordant with the markedness difference between the two templates.

In the next section, I will survey the literature on the morphological typology of the causative-inchoative alternation, and present the two different semantic strategies possible for the causative-inchoative alternation, decausativization and reflexivization, and present in short the evidence for both. In section 3, I will introduce the experimental paradigm from Lundquist et al. (2016) for adjudicating between the two semantic strategies, and present initial data indicating that different Hebrew template pairs might realize different strategies. Section 4 will present the experiment as applied to Hebrew and section 5 will show the results. Finally, section 6 will discuss how these results indicate that the more marked template realizes a stronger meaning.

5.2 The morphosemantics of the causative-inchoative alternation

5.2.1 Morphological markedness patterns and the semantic diversity hypothesis

The causative-inchoative alternation describes an alternation between a pair of change-of-state verbs where one is transitive and the other intransitive. This is

demonstrated in (232) with English alternating *cool*:

- (232) a. Causative: John cooled the soup (down).
b. Inchoative: The soup cooled (down).

The alternating argument (the subject in the transitive case) usually fills a causer role, while the shared argument is typically a theme in a change of state event (see e.g. Levin 1993, sect. 1.1.2.1).

Crosslinguistically, this alternation can take on multiple different forms depending on the relationship between the causative form and the inchoative form (Nedyalkov & Silnitsky 1973; Haspelmath 1993; Piñón 2001; Nichols et al. 2004; Haspelmath et al. 2014; Nichols 2018, *inter alia*). Specifically, languages differ in the markedness relationships between the two forms, depending on which one, if either, has more morphological complexity:

- (233) a. Causativizing (causative more marked) – Khalka Mongolian:
i. *ongoj-x* – ‘open (intrans.)’
ii. *ongoj-lg-ox* – ‘open (trans.)’ (Piñón 2001, p. 348, (13b))
b. Anticausativizing (inchoative more marked) – Russian:
i. *rasplavit’-sja* – ‘roll (intrans.)’
ii. *rasplavit’* – ‘roll (trans.)’ (Haspelmath 1993, p. 89, (2a))
c. Equipollent (both equally marked) – Hungarian:
i. *felébre-d* – ‘wake up (intrans.)’
ii. *felébre-zt* – ‘wake up (trans.)’ (Piñón 2001, p. 348, (14a))
d. Labile (neither marked) – English:
i. *open* – intrans.
ii. *open* – trans.
e. Suppletive (two forms unrelated) – English:

- i. *die* – intrans.
- ii. *kill* – trans.

Here, the bolded morpheme represents extra affixal material providing additional morphological complexity.

Under this kind of approach, the fact that the English forms in (228) have the same form is actually an odd outgrowth of the lack of morphology marking causation or change of state. Instead, *a priori*, because of the different morphological derivational directions between the causative and the inchoative, each of the different patterns in (233) would be derived by different semantic derivational processes. This approach would be a direct outgrowth of some principle of iconicity (Croft 2002, sect. 4.2), such as the ‘iconicity of complexity’ from Haspelmath (2008, p. 2, (2)). Causativizing verbs, with the inchoative alternant as basic, would produce causal semantics from a plain change of state predicate. Anticausativizing verbs, with the causative alternant as basic, would reduce a causal structure into a plain change of state predicate. Finally, because both alternants of equipollent verbs are equally complex, they would derive both the causative and the inchoative from a underspecified base.

The view that the meaning of the causative-inchoative alternation changes with the form constitutes the *semantic diversity* hypothesis, that the causative-inchoative alternation forms multiple different semantically coherent classes, which reflects in the morphology. As a hypothesis, it predicts that any difference in how the causative-inchoative alternation is derived morphologically represents a difference in the semantic contrast between the causative and the inchoative. Concretely, it would predict, for example, a causativizing pair like that in (233a) to have causativizing semantics, with the addition of a causer to the inchoative to form the causative, while an anticausativizing pair like that in (233b) to have anticausativizing semantics, with the removal of a causer from the causative to form the inchoative. Haspelmath (1993, 2016) exemplifies this approach typologically by linking complexity in the verb form

to spontaneity in the event which the verb represents. An event type which is more naturally spontaneous, in that it does not require direct involvement of an external agent, is more likely to be realized as a causativizing pair like (233a). Conversely, a less spontaneous type of event is more likely to be realized as an anticausativizing pair like (233b). Thus, the property of spontaneity is mediating the morphological realization of the alternation, where differing amounts of spontaneity reflect in differing realization patterns for the causative-inchoative alternation.

The semantic diversity hypothesis has been noted in individual languages as well. For example, Labelle & Doron (2010) argued for a correlation between markedness and agency in the causative-inchoative alternation in French. Specifically, French has two strategies for forming an inchoative, one marked with a verbal clitic *se* and another without the clitic:

- (234) a. *La branche a cassé*
 the branch AUX break
 b. *La branche s' est cassé*
 the branch SE AUX break
 'The branch broke.' (Labelle & Doron 2010, p. 304, (1))

Under their analysis, the clitic *se* in (234b) reflects a kind of middle voice (Alexiadou & Doron 2012), which lacks entailments of agency for the covert causer. Lacking *se* as in (234a) is instead an active derivation, where the covert causer is strictly agentive. Thus, depending on the morphological form of the alternation, the semantics of the alternation differs, between having agency entailments and not.

5.2.2 Two semantic processes for inchoative formation

Here, I am going to explore whether the semantic diversity hypothesis holds for Hebrew and its different morphological strategies for the causative-inchoative alternation. Specifically, I am going to investigate whether different morphological means of realizing the causative-inchoative alternation reflect two different common deriva-

tional processes proposed for the alternation: existential binding and reflexivization.³ In essence, I will be probing what the semantic difference is between the causative and inchoative forms to determine how one is formed from the other. In the case where two different alternating template pairs have one reflecting existential binding and the other reflexivization the semantic diversity hypothesis will hold.

Semantic diversity in the causative-inchoative alternation is a critical fact for the VCMP, as both necessitate that different morphological forms are realized with different semantics. In fact, the VCMP not only predicts that the causative-inchoative alternation in Hebrew will be diverse, but also the kind of diversity it should have, with more marked templates having a stronger meaning. Namely, given the two templatic options for the inchoative in (231), VCMP predicts that *nirtav* and *hitratev* will not be synonymous, but instead differ in that *nirtav*, the verb in the less marked template, should have a weaker meaning than that of *hitratev*, the verb in the more marked template. Here, I will focus on how the entailments associated with the agent differ between these two classes of verbs, and whether the template correlate with a different set of entailments such that the former is weaker than the latter.

At first glance, the alternants seem to exist in an embedding relationship with one another, where the meaning of the inchoative exists fully in the causative, as shown by the rough paraphrases of (232) in (235) (McCawley 1968; Lakoff 1970; Dowty 1979):

- (235) a. The soup cooled (down).
 ‘The soup became cool(er).’

³Other processes have been posited, such as causer deletion or causer addition. I will not be addressing them here, as for the sake of the experimental paradigm mobilized in this chapter, they all generate the same predictions for the results of the experiment (see Lundquist et al. 2016, p. 2, (5-6)). The question of which non-reflexive semantic process is under consideration here is therefore orthogonal to the main issue of whether a given inchoative is reflexive or not. As such, I will be presuming the process is existential binding.

- b. John cooled the soup (down).
‘John caused that the soup became cool(er).’

The inchoative form of the verb roughly has a meaning of ‘x changes from not having state y to having state y’, while the causative form roughly encodes ‘z causes x to change from not having state y to having state y.’ Morphologically, this kind of embedding would naturally correlate with a causativizing pair, such as in the one in (233a) repeated below.

(233a) Causativizing (causative more marked) – Khalka Mongolian:

- a. *ongoj-x* – ‘open (intrans.)’
- b. *ongoj-lg-ox* – ‘open (trans.)’ (Piñón 2001, p. 348, (13b))

Just as the inchoative verb is embedded in the causative verb in (233a), the meaning of the inchoative is fully embedded in the causative.

In classic event structural approaches, this kind of embedding was represented as the inchoative event structure inside the causative template, as shown in (236) from Dowty (1979, p. 124):

- (236) a. State: $\pi_n(\alpha_1, \dots, \alpha_n)$
- b. Inchoative: $\text{BECOME}[\pi_n(\alpha_1, \dots, \alpha_n)]$
- c. Causative: $(\text{DO}(\alpha_1[\pi_n(\alpha_1, \dots, \alpha_n)])) \text{ CAUSE } (\text{BECOME } [\pi_n(\alpha_1, \dots, \alpha_n)])$

(236c) is produced through a process which embeds (236b) as an argument of the CAUSE operator, and as such derives the causative from the inchoative. This sort of derivational treatment was expanded by Rappaport Hovav & Levin (1998) with Template Augmentation in the lexicon, stated in (237), which freely permits precisely this kind of derivation of one event structural template from another.

- (237) **Template Augmentation:** Event structure templates may be freely augmented up to other templates in the basic inventory of event structure templates. (Rappaport Hovav & Levin 1998, p. 111, (23))

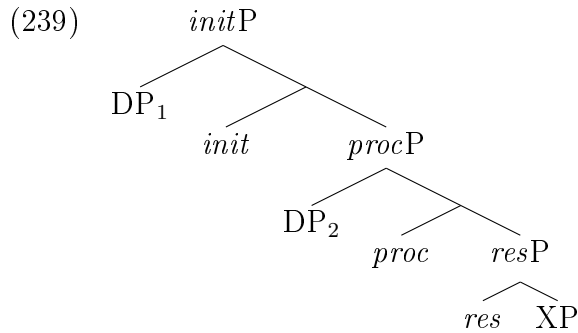
For them, this principle operates over a fixed set of event structural templates, which under Template Augmentation combine to create a range of verb types:

- (238)
- a. Activity: $x\text{ACT}_{\langle \text{MANNER} \rangle}$
 - b. State $x < \text{STATE} >$
 - c. Achievement $\text{BECOME}[x < \text{STATE} >]$
 - d. Accomplishment $[x\text{ACT}_{\langle \text{MANNER} \rangle}]\text{CAUSE}[\text{BECOME}[x < \text{STATE} >]]$
 - e. Accomplishment: $x\text{CAUSE}[\text{BECOME}[x < \text{STATE} >]]$

(Rappaport Hovav & Levin 1998, p. 108)

Thus, an inchoative event structure can embed into a causative event structure, generating a form of causativization as shown in (236c).

For some, this kind of process takes place in the syntax, with different syntactic heads corresponding to different operations in the event structure, with a change-of-state head embedded underneath a head introducing a causer. For example, First Phase Syntax (Ramchand 2008) assumes the syntactic hierarchy of event structural heads shown in (239):



(Ramchand 2008, p. 46, (44))

Here, *proc* encodes a change-of-state event (into the state encoded by *res* and its complement). Realizing just up to *procP* produces a sentence with an inchoative verb, but embedding it underneath *initP*, where *init* encodes a causing event and

its specifier the causer, produces a sentence with a causative verb. This syntactic structure is a form of causativization which embeds the inchoative in the causative, reflecting the event structures in (236).

The converse of causativization is anticausativization, where the causative alternant is basic and the inchoative alternant is derived by a process which removes the causer. This kind of process is reflected in the morphology by an anticausativizing paradigmatic pair, as in (233b) repeated here.

(233b) Anticausativizing (inchoative more marked) – Russian:

- a. *rasplavit’-sja* – ‘roll (intrans.)’
- b. *rasplavit’* – ‘roll (trans.)’ (Haspelmath 1993, p. 89, (2a))

In its most basic form, anticausativization takes the form of existential binding (Levin & Rappaport Hovav 1995; Rappaport Hovav & Levin 1998), where an open causer argument in the basic causative alternant is bound existentially in the inchoative. Such is the approach taken in Theta Theory and the Generative Lexicon (Reinhart 2002; Reinhart & Siloni 2005; Horvath & Siloni 2011b,a; Reinhart et al. 2016), which mobilizes the derivational process of ‘expletivization’, which I represent in (240).

(240) **Expletivization:** Reduction of an external [+c] role (semantically null function)

- a. $V_{acc}(\theta_{1[+c]}, \theta_2) \rightarrow R_e(V)(\theta_2)$
- b. $R_e(V)(\theta_2) = V(\theta_2)$ (Reinhart 2002, p. 241, (15))

In (240a), expletivization is specified to operate over a transitive verb V_{acc} whose first argument θ_1 is a causer specified with [+c]. This verb is input to a reduction operation R_e which takes two total inputs, a verb and one of its arguments. Its output, as specified in (240b), is a reduced version of its input verb preserving only the specified argument.

Schematically, I represent existential binding as existential quantification of an effector argument in the inchoative which is open through lambda abstraction in the causative, shown in (241). I remain agnostic to the specific process which relates these forms, as to not assume how the two forms are derived from one another:

(241) **Existential binding:**

Causative $\lambda Q\lambda x\lambda y\lambda e\lambda e'\exists s[effector'(y, e') \wedge cause'(e', e) \wedge become'(e, s) \wedge Q(s, x)]$

Inchoative $\lambda Q\lambda x\lambda e\lambda e'\exists y\exists s[effector'(y, e') \wedge cause'(e', e) \wedge become'(e, s) \wedge Q(s, x)]$

(based off of Beavers & Koontz-Garboden 2020, p. 14, (22))

Causativization is technically a distinct process from anticausativization, as in the inchoative, there is no causer under the former, while in the latter, there is a causer, but bound off. However, this difference between causativization and anticausativization is irrelevant for the purposes of the study here, which hinges specifically on the entailment relationship between the causative and the inchoative. Namely, under reflexivization, the causative only entails the inchoative in exactly the situation where the theme is its own agent, where under both causativization and anticausativization, the causative always entails the inchoative (see Lundquist et al. 2016, p. 2, (5)). Thus, because the two have the same entailment relationship between the causative and the inchoative, they are equivalent processes for the purpose of this study. For consistency with the study in Lundquist et al. (2016), I will consider the process to be anticausativization, but nothing hinges on this decision.

Both causativizing morphology and anticausativizing morphology contrast with equipollent morphology, where both the causative and the inchoative are derived from morphology operating over a common base, making them equally complex, as in (233c) repeated here.

(233c) Equipollent (both equally marked) – Hungarian:

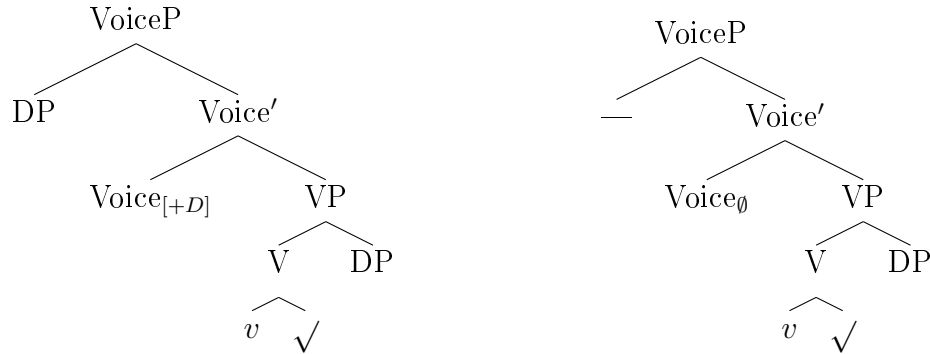
a. *felébre-d* – ‘wake up (intrans.)’

b. *felébre-zt* – ‘wake up (trans.)’ (Piñón 2001, p. 348, (14a))

The syntactic layering approach from Schäfer (2008) and Alexiadou et al. (2015), also utilized in Kastner (2020), represents an equipollent derivation, as the functional head Voice specifies for the semantic and syntactic status of the external argument.⁴ Different types of Voice determine different types of external arguments. As such, a Voice head requiring an external argument produces a causative verb, while a Voice head restricting one produces an inchoative verb. The structures for both are those given in (242), from Alexiadou et al. (2015).⁵

(242) a. Causative structure:

b. Inchoative structure:



In these structures, *v* introduces causation upon deriving the root into a verb, and thus Voice determines the identity of the causer, if there is one. Voice_[+D] is specified with an EPP-feature which requires a DP to Merge in its specifier. This produces a sentence with a transitive, causative verb. To contrast, Voice_∅ prevents such an argument from Merging, thus producing a sentence with an intransitive, inchoative verb. In both cases, Voice scopes over a core of the root and a verbalizing head

⁴See section 2.2.2 for a short overview of this approach

⁵Schäfer (2008) and Alexiadou et al. (2015) also include semantic information in their Voice heads, specifying the thematic role which the external argument, overt or suppressed, would hold.

providing change-of-state semantics. Voice_[+D] introduces a causer and so creates a caused change-of-state verb, while Voice_∅ has no causer, and so creates a inchoative verb with no causer.

In some cases, the morphology which indicates anticausativization is syncretic with a reflexivizing marker, as in Spanish examples in (243) from Koontz-Garboden (2009).

- (243) a. *Kim se cortó.*
 Kim REFL cut
 ‘Kim cut herself.’ (true reflexive)

(Koontz-Garboden 2009, p. 89, (28))

- b. *El vaso se rompió.*
 the vase REFL broke
 ‘The vase broke.’ (inchoative)

(Koontz-Garboden 2009, p. 84, (13a))

This syncretism has led to the development of a reflexivization analysis of anticausatives, starting with Chierchia (2004) but further developed in Koontz-Garboden (2009), Beavers & Zubair (2013), and Beavers & Koontz-Garboden (2013). Here, instead of removing or uncontentfully binding the causer participant, a reflexivization operator such as in (244) coidentifies it with the theme, thus producing a meaning similar to a true reflexive.⁶

⁶Importantly, this process is not identical to the introduction of a reflexive pronoun as the object of the verb, deriving a true reflexive. For example, true reflexives can appear with reflexive pronouns, while a reflexive inchoative cannot, demonstrated with *a si mismo* for the Spanish sentences in (243).

- i. a. *Kim se cortó a si misma.*
 Kim REFL cut by self same
 ‘Kim cut herself.’ (true reflexive)
 b.#*El vaso se rompió a si mismo.*
 the vase REFL broke by self same
 (intended) ‘The vase broke itself.’ (inchoative)

The introduction of a reflexive pronoun such as *itself* has been taken to introduce a distinct set of

- (244) **Reflexivization operator:** $\lambda R\lambda x.R(x, x)$ where R is a two place relation (i.e. corresponding to a transitive, causative verb).

(based off of Chierchia 2004, p. 29, (13))

Where this process differs is the thematic role of the causer under this process is a generic effector, while for a true reflexive, the thematic role is an agent. Thus, a reflexivized inchoative has a different set of entailments than a true reflexive: where latter requires some form of agency, the former is much more generic.

Application of a reflexivization operation like that in (244) would result in representations for the causative and the inchoative as follows:

- (245) **Reflexivization:**

Causative $\lambda Q\lambda x\lambda y\lambda e\lambda e'\exists s[effector'(y, e') \wedge cause'(e', e) \wedge become'(e, s) \wedge Q(s, x)]$

Inchoative $\lambda Q\lambda x\lambda e\lambda e'\exists s[effector'(x, e') \wedge cause'(e', e) \wedge become'(e, s) \wedge Q(s, x)]$

Under this analysis, the inchoative has the same semantic structure as a true reflexive, with the causer identical to the theme.

Both existential binding and reflexivization crucially preserve the causal structure in the inchoative in the causative.⁷ Koontz-Garboden (2009) and Beavers & Koontz-Garboden (2013) have shown that negation is able to target causal meaning even in inchoative verbs, *pace* Schäfer & Vivanco (2016). This is shown in the Spanish example in (246) from Koontz-Garboden (2009), where *not* here is targeting the identity of the causer, not the existence of the event:⁸

agency inferences from reflexive inchoative marking which explain this divergent behavior; see the literature cited here for more details.

⁷I will not address all of the arguments for covert causation in the inchoative here, as the relevant predictions for the study are the same regardless of which form is more basic. For some more evidence, see e.g. Alexiadou et al. (2006, 2015), but also Horvath & Siloni (2011b) and Fadlon (2016) to the contrary.

⁸The sentence includes the negative polarity item *ningún* – ‘any.NEG’ – in order to rule out metalinguistic negation and to ensure that it can target only the causer.

- (246) (A boy breaks a vase and his father comes in to investigate what happened.
The boy asserts that the vase just broke, and the father responds.)

No se rompió ningún vaso; lo rompiste tú.

(lit.) ‘No glass didn’t break – you broke it!’

(based off of Koontz-Garboden 2009, p. 103, (57))

Such negation is only possible if the causer participant (or more generally, causation) still exists within the semantic structure of the inchoative. Otherwise, the negation would not be able to target any potential causer independent of the event.

The crucial difference between existential binding and reflexivization as processes underlying the causative-inchoative alternation lies within the entailment relationship between the two alternants. Under existential binding, the causative is monotonically more contentful than the inchoative. Specifically, the only difference between the inchoative and the causative is whether the causer is specified. Regardless of what saturates the open causer argument, that causer will satisfy the requirements of the inchoative, namely that some causer exists. As such, a strict entailment relationship should hold between the two alternants, such that the causative entails the inchoative (see e.g. Martin & Schäfer 2014; Schäfer & Vivanco 2016). To contrast, no such strict entailment relationship holds between the two alternants under reflexivization, as there is no such monotonically additive relationship. Instead, the causative is only equivalent to the inchoative in exactly the cases where the theme serves as the causer in its own event.

Importantly, there is a difference in strength between these two semantic forms. Reflexivization specifies the identity of the causer in the event, namely coidentifying it with the theme. To contrast, existential binding only requires there to be a causer in the event at all, with no commitment to the identity of the causer. The former is stronger than the latter, as having a specified causer entails having some non-specific causer, shown schematically in (247).

$$(247) \quad \left\{ \frac{\text{Existential binding}}{\exists x \exists e [\text{effector}'(e, x)]} \right\} \subset \left\{ \frac{\text{Reflexivization}}{\exists x \exists e [\text{effector}'(e, x)]} \right\} \\ \left\{ \exists e [\text{theme}'(e, \iota x [\text{effector}'(e, x)])] \right\}$$

If Hebrew is a semantically diverse language, based on the markedness difference between the two templates, the VCMP predicts that the inchoatives in the more marked template will be reflexive, and those in the less marked template will be existential. Therefore, I will mobilize this difference in the entailment relationship between the causative and the inchoative to adjudicate between these two analyses, and specifically, to assess whether Hebrew mobilizes both, and whether those differences fall along templatic lines following the markedness differences between them.

5.3 Testing the semantic diversity hypothesis

5.3.1 An experimental paradigm for inchoative semantics

In order to determine which relationship between the causative and the inchoative holds in Hebrew, and whether that relationship is different by template pair, I will be utilizing an experimental paradigm from Lundquist et al. (2016). Their paradigm is a felicity test, where they showed participants depictions of caused events and asked whether the event happened using the inchoative. The paradigm specifically mobilizes the fact that under reflexivization, the inchoative is only equivalent to the causative in the case where the theme is equally the causer. Thus, their paradigm assesses whether depictions of caused events where the theme is feasibly also the causer can be felicitously described with the appropriate verb. If such a context creates a more felicitous description, then that inchoative verb likely has reflexive semantics.

To understand the distinction between existential binding and reflexivization which is being mobilized here, consider the following two contexts for the single sentence in (248), with the inchoative verb *nirtav* ‘get wet.’

- (248) *ha=magevet nirtv-a* *b=mayim*
the=towel MID.wet-3SG.F.PST in=water
‘The towel got wet with water.’

- a. Context: A person dunks a towel into a bucket of water and pulls it up dripping wet.
- b. Context: A person accidentally drops a towel on the floor onto a puddle of water, which the towel subsequently absorbs.

Though the sentence does not express a causer, these two contexts differ mainly in how involved a putative causer is in the event. In the former context, the causer is actively causing the change in wetness of the towel, in that through their active intention and force, the towel enters the bucket and becomes wet. To contrast, in the second context, the causer only tangentially causes the towel to become wet. While they did cause the towel to become wet through the fact that they dropped it in a wet place, it was primarily due to the absorbent properties of the towel that the towel got wet. In the latter context in (248), where the primary force behind the towel becoming wet is its own inherent properties, the theme is reasonably construed as causing its own change of state (i.e. through its natural absorptiveness).

To operationalize this notion of the theme serving as its own cause, Lundquist et al. established two types of conditions based on whether the agent or the theme is more in focus. An agent focus context involves an external agent who is the primary initiator and by whose force the event successfully unfolds. To contrast, a theme focus context involves an external agent as only the initiator of the event, but the event unfolds due to some properties of the theme itself, with the external agent not involved throughout. For example, consider the example in (249) along with its associated context.

(249) a. **Agent focus:**

(Context: A woman rolls a barrel across the road. Because of how heavy the barrel is, the woman needs to push continuously to get the barrel to roll across the road.)

‘Did the barrel roll across the road?’

b. **Theme focus:**

(Context: A woman rolls a ball across the road. Because the ball is pre-disposed to roll, all it takes it one initiating movement, such as a little nudge, to get the ball to roll across the road.)

‘Did the ball roll across the road?’

(Lundquist et al. 2016, p. 19, (41))

In the case of the agent focus context above, because the theme, i.e. the barrel, is so heavy, the external causer, i.e. the woman, needed to participate in the act of rolling throughout the whole course of the event. To contrast, in the theme focus context, because the ball is so light and easy to roll, only a small initiating action is required on the part of the agent, after which the event unfolds without her participation.

Thus, if reflexivization is the process underlying a causative-inchoative pair, then the inchoative should be able to describe caused events in theme focus contexts better than agent focus contexts. To contrast, under existential binding, the causative entails the inchoative. Thus, every caused event which can be described with a causative can equally be described as an inchoative. Therefore, under existential binding, differences in focus should not have an effect on the ability for the inchoative to describe caused events. As applied to the contexts in (248), the former context would constitute an agent focus context, and the latter context would be a theme focus context. If the semantics of *nirtav* reflects existential binding, then it should be felicitous in both contexts, i.e. the pattern in (250).⁹

- (250) *ha=magevet nirtv-a* *b=mayim*
 the=towel MID.wet-3SG.F.PST in=water
 ‘The towel got wet with water.’

⁹The felicity patterns here are idealized patterns intended to illustrate how to identify which semantic strategy a given inchoative uses. In line with how this experimental paradigm addresses felicity and naturalness and in the absence of any traditional linguistic judgments, I use the diacritic #/? when showing examples with contexts to indicate that a given context is dispreferred across the participants’ responses, and ✓ to indicate it is preferred.

- ✓ Context (agent focus): A person dunks a towel into a bucket of water and pulls it up dripping wet.
- ✓ Context (theme focus): A person accidentally drops a towel on the floor onto a puddle of water, which the towel subsequently absorbs.

Comparatively, if the semantics of *nirtav* reflects reflexivization, only the theme focus context should be licit, with the agent focus context dispreferred:

- (251) *ha=magevet nirtv-a b=mayim*
 the=towel MID.wet-3SG.F.PST in=water
 ‘The towel got wet with water.’

#/? Context (agent focus): A person dunks a towel into a bucket of water and pulls it up dripping wet.

- ✓ Context (theme focus): A person accidentally drops a towel on the floor onto a puddle of water, which the towel subsequently absorbs.

The fundamental question being probed in this paradigm is which of these two felicity patterns holds, and whether different classes of verbs pattern differently.

On the backdrop of these predictions, Lundquist et al. applied this paradigm to Norwegian and English. English, as demonstrated in (233d) and (235), uses labile marking to realize the causative-inchoative alternation. Norwegian instead is an anticausativizing language (cf. (233b)), demonstrated in (252), where the inchoative is marked with *seg*.

- (252) a. *Peter åpnet vinduet*
 Peter opened window.DEF
 ‘Peter opened the window.’

- b. *Vinduet åpnet seg*
 window.DEF opened REFL
 ‘The window opened.’

(Lundquist et al. 2016, p. 15, (8))

The anticausativizing marker *seg* is additionally syncretic with a reflexivizing marker, shown in (253).

- (253) *Per barberet seg*
 Per shaved REFL
 ‘Per shaved himself.’ (Lundquist et al. 2016, p. 11, (26a))

Additionally, some Norwegian verbs are labile, with both the causative and the inchoative marked realizing completely unmarked.

- (254) a. *Peter smeltet smøret*
 Peter melted butter.DEF
 ‘Peter melted the butter.’
 b. *Smøret smeltet*
 butter.DEF melted
 ‘The butter melted.’ (Lundquist et al. 2016, p. 8, (16))

Therefore, based just on the marking patterns, English, lacking any anticausativizing marker syncretic with a reflexive, should mobilize only existential binding, while Norwegian should realize reflexivization for the marked inchoatives, in accordance with the anticausativizing marker’s syncretism with a reflexive marker, and existential binding for the unmarked inchoatives. Norwegian might validate the semantic diversity hypothesis, and show that languages can mobilize several different derivational strategies based on the morphological form of the alternation.

Just like Norwegian, Hebrew also has a variety of means of marking inchoatives, i.e. the templates surveyed in the previous section. Example (231), repeated here, show two, both derived from the root $\sqrt{\text{RTV}}$, related to wetness. The former includes a verb in NIF’AL, *nirtav*, and the latter includes a verb in HITPA’EL, *hitratev*.

- (231) a. *nirtav-ti* *b=a=gefem*
 MID.wet-1SG.PST in=the=rain
 b. *hitratav-ti* *b=a=gefem*
 REFL.wet-1SG.PST in=the=rain

‘I got wet in the rain.’ (constructed)

Semantic diversity in Hebrew would manifest as these two verbs having different felicity patterns with respect to agent focus or theme focus contexts. Just as with Norwegian, HITPA’EL also serves as a canonical reflexive, and regularly forms reflexive verbs.

- (255) a. *Yosi raxac et yada-v*
 Yosi wash[3SG.M.PST] ACC hand-PL:3SG.M.POSS
 ‘Yosi washed his hands.’
 b. *Yosi hitraxec*
 Yosi REFL.wash[3SG.M.PST]
 ‘Yosi washed himself.’

Thus, *prima facie*, *hitraxec*, as an inchoative verb in HITPA’EL, should pattern as if it has reflexive semantics, and prefer theme focus contexts and disprefer agent focus contexts:

- (256) *ha=magevet hitratv-a b=mayim*
 the=towel REFL.wet-3SG.F.PST in=water
 ‘The towel got wet with water.’
 #/? Context (agent focus): A person dunks a towel into a bucket of water and pulls it up dripping wet.
 ✓ Context (theme focus): A person accidentally drops a towel on the floor onto a puddle of water, which the towel subsequently absorbs.

nirtav on the other hand would pattern as if it reflected existential binding, and prefer both contexts equally:

- (257) *ha=magevet nirtv-a b=mayim*
 the=towel MID.wet-3SG.F.PST in=water
 ‘The towel got wet with water.’

- ✓ Context (agent focus): A person dunks a towel into a bucket of water and pulls it up dripping wet.
- ✓ Context (theme focus): A person accidentally drops a towel on the floor onto a puddle of water, which the towel subsequently absorbs.

If such a pattern holds, then two different classes of Hebrew verbs reflect two different types of semantics, thus validating the semantic diversity hypothesis.

Ultimately, Lundquist et al. demonstrated that on a language level, English and Norwegian differed in how their verbs could describe theme focus vs. agent focus depictions. As anticipated, English did not discriminate between the two focus conditions and accepted both roughly equally, and as such, demonstrated the pattern indicative of existential binding. Norwegian demonstrated the opposite pattern and discriminated between the two focus conditions, preferring theme focus contexts over agent focus contexts. As such, Norwegian demonstrated the pattern indicative of reflexivization. However, Norwegian did not demonstrate a difference within the language, as might have been anticipated by the fact that some of its verbs are labile. Thus, despite the fact that Norwegian uses two different marking strategies to realize the causative-inchoative alternation, it does not utilize two different semantic derivational strategies and did not validate the semantic diversity hypothesis.

5.3.2 The distribution of alternation patterns in Hebrew

The results from Lundquist et al. (2016) did not validate the semantic diversity hypothesis, as a language with different morphological realizations of the causative-inchoative alternation, namely Norwegian, does not have different semantic forms for the inchoatives along those morphological lines. As such, the semantic diversity hypothesis may not be valid, and languages have to select one of the two derivational processes regardless of the morphological markings which exist in the language. However, Lundquist et al. mused on a general principle which might explain why Norwegian failed to pattern as a semantically diverse language:

One could speculate that the language learner simply does not get enough evidence for a productive (anti)-causative rule from exposure to the language, whereas she does get evidence for a reflexivization operation identifying one argument with another. (Lundquist et al. 2016, p. 27)

Based on this principle, a language which has multiple different morphological strategies for realizing the causative-inchoative might not attach different processes to each if there is not enough prominence for one of them. Already, Amaral et al. (2023) has demonstrated that Brazilian Portuguese, which also has reflexive and labile strategies for realizing the causative-inchoative alternation, is semantically diverse, and note that both strategies are highly productive. Therefore, Norwegian might not be a semantically diverse language because of the rarity of the labile alternation within the language.¹⁰

If Hebrew were to be a potential candidate for a semantically diverse language and thus demonstrate differences in how it semantically derives the causative-inchoative alternation based on morphological strategy, it should have enough of a distinction between the different templatic means of realizing the causative-inchoative alternation to warrant attaching different semantics to each. Across a sample of verbs, there should be a relatively flat distribution of alternation patterns, with no one pattern more prominent than another. Prominence here can be understood as having an alternation pattern with an outsized majority of the verbs, which I will quantify here as how spread out a representative set of verbs is across alternation patterns. The less spread out the counts are across the alternation patterns, the more prominent a single class will be. Norwegian should have a very low spread, while Hebrew should have a very high spread.

To assess prominence of alternation patterns in both languages, I collected the 37 most frequent Hebrew verbs which participate in the causative-inchoative alternation in heTenTen 2014, a corpus of web-scraped Hebrew text from 2014 (Baroni

¹⁰Lundquist et al. notes that only a dozen verbs in Norwegian utilize a labile alternation (p. 8).

et al. 2009), and compared their morphological form to that of their translational equivalents in Norwegian.¹¹ Each of the verbs in the sample were coded for their morphological derivational direction among the ones in (233). For the Hebrew forms, I did not distinguish between a reflexive anticausativizing strategy (i.e. one which has HITPA'EL as its inchoative, cf. (229d)) and a non-reflexive anticausativizing strategy (i.e. one which has a more complex template other than HITPA'EL as its inchoative, cf. (229a)). This was to allow for consistency in comparing the patterns crosslinguistically; because Norwegian does not have two anticausativizing strategies, I treated Hebrew as if it did not as well. For the Hebrew forms, I relied only on affixal material to inform which one is more complex, such that if a template has affixal material, it is more complex than one without; this strategy derived contrasts which accorded with the markedness hierarchy from Section 2.4.2.

Figure 5.1 show the proportion of marking types for the verbs in the sample for both Norwegian and Hebrew. As expected, an overwhelming majority of the forms in Norwegian utilize anticausative marking as in (252) (33 in total), with a marginal number of the forms utilizing a labile strategy as in (254) (just 1). This pattern produces a standard deviation across the counts of 22.4, indicating a very low spread across the alternation patterns, with one pattern, namely the anticausative, substantially more prominent than the other. To contrast, though anticausative marking is also the most common marking pattern in Hebrew (15), it also uses three other patterns, each of which are more common than labile marking in Norwegian. In addition, the number of verbs in the second most common type in Hebrew, equipollent marking, is only approximately half the amount of the number of verbs which use anticausative marking (6). This patterns produces a standard deviation of 4.79, indicating a much wider spread across alternation patterns. Therefore, compared to Norwegian, Hebrew does have enough prominence across its utilization of alternation patterns which might warrant different semantic derivational strategies. In the

¹¹Thanks to Dag Lindeberg for providing the Norwegian translations and data on the morphological forms.

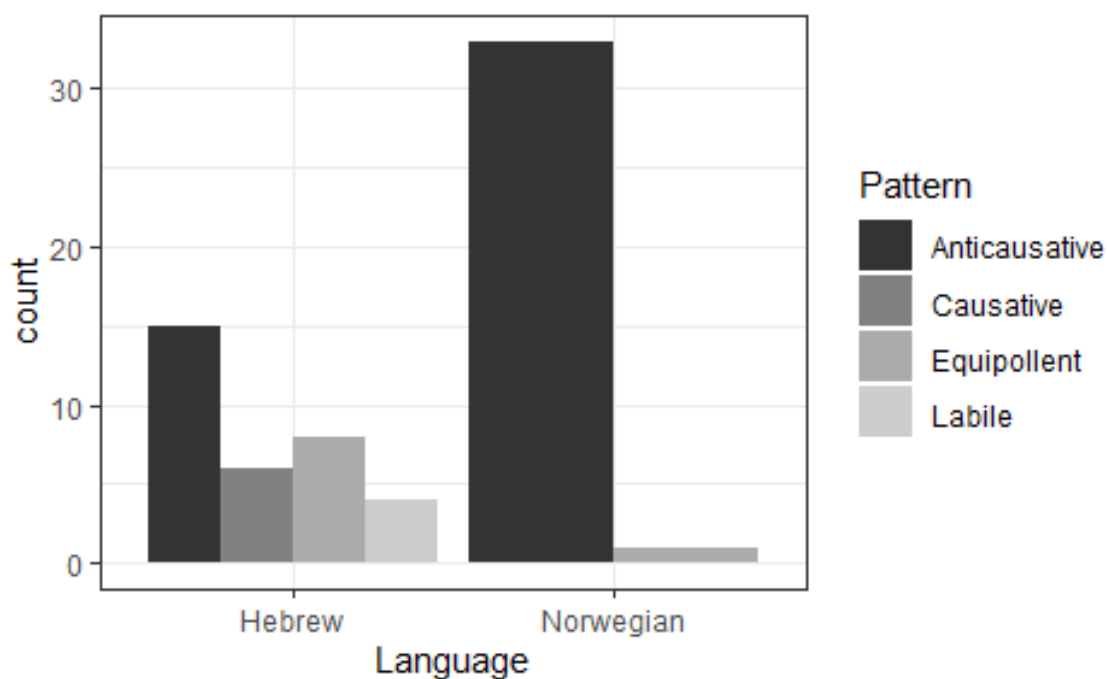


Figure 5.1: Hebrew and Norwegian by alternation pattern

next section, I will apply the paradigm from Lundquist et al. (2016) to three classes of verbs in Hebrew and demonstrate that Hebrew is in fact a semantically diverse language.

5.4 Assessing the semantic diversity of Hebrew

5.4.1 Design and materials

Following the original study from Lundquist et al., I tested the entailment relationship between the causative and inchoative forms of the Modern Hebrew verbs which mobilize three different pairs of templates to alternate. I give these classes in (258):

	Template pair	Root	Example	Translation
(258)	a. PA'AL – HIF'IL	$\sqrt{B'R}$	<i>ba'ar – hiv'ir</i>	burn
	b. HITPA'EL – PI'EL	$\sqrt{ML'}$	<i>hitmale' – mile'</i>	fill
	c. NIF'AL – HIF'IL	$\sqrt{'LM}$	<i>ne'elam – he'elim</i>	disappear

Each class had a representative sample of seven verbs, the inchoatives of which are given in Table 5.1. For each of these verbs, two video stimuli were produced, 5-10

PA'AL – HIF'IL	NIF'AL – HIF'IL	HITPA'EL – PI'EL
<i>zaz</i> 'move'	<i>nirdam</i> 'fall asleep'	<i>hitmale</i> 'fill'
<i>ba'ar</i> 'burn'	<i>ne'elam</i> 'disappear'	<i>hitmotet</i> 'topple'
<i>ma'ad</i> 'trip;	<i>nidlak</i> 'light'	<i>hitnape'ax</i> 'blow up'
<i>panah</i> 'turn'	<i>names</i> 'melt'	<i>hitpazer</i> 'scatter'
<i>shatak</i> 'quiet'	<i>nish'an</i> 'lean'	<i>hityabesh</i> 'dry'
<i>davak</i> 'stick'	<i>ne'eram</i> 'pile up'	<i>hitkarev</i> 'bring closer'
<i>alah</i> 'go up'	<i>nifrad</i> 'separate'	<i>hitkapel</i> 'fold'

Table 5.1: Verbs used in the study, by template class

seconds each, one for an agent focus context and one for a theme focus context, for a total of 42 test stimuli. All test stimuli portrayed a caused event with an external causer. In the agent focus context, this external causer was the proximal cause of the event as well as directly responsible for its complete unfolding. In the theme focus context, the external causer was the proximal cause of the event, in that the external causer initiated the event, but properties inherent to the theme were responsible for the event unfolding.

All agent focus stimuli were produced by showing some external causer initiating the event and sustaining its involvement throughout the whole event. For example, *hitkapel* 'fold' in the agent focus context showed a person slowly and deliberately making folds in a towel multiple times. The primary strategy for constructing theme focus contexts was relying on inherent properties of the theme as the primary cause of the event. *hitkapel* in the theme focus context showed a person dropping a towel from a height over the back of a chair, thus serving as an immediate causer. Due to the natural structural properties of the towel, on contact with the chair back the towel folded. In the cases where this strategy could not be relied upon, the theme focus context was portraying the immediate cause as accidental. For example, *ma'ad* 'trip' in the theme focus context showed one participant inadvertently tripping another by

virtue of sitting in a chair and having their legs sticking out. To contrast, *ma'ad* in the agent focus context had that first participant intentionally stick their legs out to trip the second participant.

In addition to the 42 test stimuli, 26 filler stimuli were produced. The filler stimuli depicted simple events, such as a person jumping up and down several times, or a scene with multiple different items in it. The participant would then be prompted to answer something about the nature of the event or the items in the scene. Half the stimuli were accompanied by questions where the answer was clearly yes, and the other half by questions where the expected answer was no; there were 13 of each which were shown to each participant. These fillers were to ensure that the participant was attentive to the task, as well as reinforce that the task was not to judge the grammaticality of the sentences presented, but rather to determine whether the sentence felicitously describes the video in the stimulus.

Data from 60 participants were collected; all the participants were found through experiment recruitment Facebook pages at Tel Aviv University and Hebrew University of Jerusalem, were self-attested native speakers of Hebrew of approximately college age, and were at the time living in Israel. Each participant was shown 47 total stimuli: 21 target stimuli and 26 filler stimuli. For the target stimuli, participants were shown a stimulus from each verb, but only one verb in each condition; this was to prevent comparison across stimuli. Each stimulus was presented to the participant, and after seeing the stimulus, the participant was presented with a yes-no question. In the case of the test stimuli, the question probed whether the inchoative alternant of the verb happened, as exemplified with *hitkapel* in the theme condition:

- (259) Video: Towel gets dropped over the back of a chair, hits the back of the chair and folds in half over it.

Question: *ha'im ha=magevet hitkapl-a*
 Q the=towel REFL.fold-3SG.F.PST

‘Did the towel fold over?’

An expected-yes filler video stimulus would be accompanied by a question where the answer was clearly yes, as exemplified below:

- (260) Video: A person sits at a table with a cell phone on it, which rings and the man answers.

Question: *ha'im hu anah* *l=a=telefon*
Q he answer[3sg.m.pst] to=the=telephone

‘Did he answer the phone?’

Alternatively, an expected-no filler stimulus would be accompanied by a question where the answer was clearly no:

- (261) Video: A man sits at a table with a book and a full glass of water on it.

Question: *ha'im haya* *'iton* *'al ha=shulxan*
Q is[3SG.M.PST] newspaper on the=table

‘Was there a newspaper on the table?’

As this experiment was crucially not probing acceptability, but rather the felicity of the sentence to describe the event, all questions were grammatical sentences of Hebrew.

The experiment was conducted using PsychoPy3 (Peirce et al. 2019) running on Windows 10 on a Lenovo Yoga 720 with a 15” screen. After showing the video stimulus, the participant was prompted to answer the question by pressing 0 for no and 1 for yes. Throughout the analysis of the results, I will be framing the data in terms of ‘response rate’, which should be understood as percentage of ‘yes’ (or ‘1’) responses out of total responses. There was no delay between the presentation of the video stimulus and its question, and the question and the next video stimulus.¹²

¹²All target and filler stimuli, and the PsychoPy experiment code, can be found at <https://osf.io/q4gr3>.

5.4.2 Hypotheses

In order to determine which templates are anticipated to realize which semantic process for forming the inchoative, I review here the reflexivity of HITPA'EL, and suggest that because it is a reflexive template, it should realize reflexivization when forming inchoatives. HITPA'EL robustly realizes a reflexive alternation with multiple other templates. As shown in (262)–(264), HITPA'EL reflexively alternates with transitive verbs in PA'AL, HIF'IL, and even PI'EL, the template which it forms a causative-inchoative alternating pair with.

(262) PA'AL – HITPA'EL reflexive:

- a. *Yosi raxac et yada-v*
Yosi wash[3SG.M.PST] ACC hand-PL:3SG.M.POSS
'Yosi washed his hands.'
- b. *Yosi hitraxec*
Yosi REFL.wash[3SG.M.PST]
'Yosi washed himself.'

(263) HIF'IL – HITPA'EL reflexive:

- a. *Yosi herim et ha=gavi'a shel-o*
Yosi lift[3SG.M.PST] ACC the=trophy POSS-3SG.M
'Yosi lifted up his trophy.'
- b. *Yosi hitromem*
Yosi REFL.lift[3SG.M.PST]
'Yosi lifted himself up.'

(264) PI'EL – HITPA'EL reflexive:

- a. *Yosi gile'ax et pana-v*
Yosi shave[3SG.M.PST] ACC face-PL:3SG.M.POSS
'Yosi shaved his face.'
- b. *Yosi hitgale'ax*
Yosi REFL.shave[3SG.M.PST]
'Yosi shaved himself.'

Here, a transitive verb has its object reduced and its latent theme argument coidentified with its surface subject, producing a meaning where the agent is explicitly acting on itself.

Importantly, the reflexives in (262)–(264) are distinct from the inchoatives which HITPA'EL can also derive. The reflexive in Hebrew patterns like an unergative, as shown in (265) and (266) by the failure to license a possessive dative or unconditioned VS word order, two traditional unaccusativity diagnostics for Hebrew (Borer & Grodzinsky 1986).¹³

(265) Possessive dative:

- a. *ha=kos hitmal'-a le=Yosi*
the=cup REFL.fill-3SG.F.PST to=Yosi
‘Yosi’s cup filled up.’
- b. * *ha=ax hitraxec le=Yosi*
the=brother REFL.wash[3SG.M.PST] to=Yosi
(intended) ‘Yosi’s brother washed himself.’

(266) Unconditioned VS word order:

- a. *hitmal'-a ha=kos*
REFL.fill-3SG.F.PST the=cup
‘The cup filled up.’
- b. * *hitraxec Yosi*
REFL.wash[3SG.M.PST] Yosi
(intended) ‘Yosi washed himself.’

As such the subject of a HITPA'EL reflexive is a true deep subject filling an agent role (see e.g. Reinhart & Siloni 2004; Siloni 2008), unlike the subject in HITPA'EL inchoatives, which pattern like an unaccusative with respect to these diagnostics and so whose subject fills a theme role. Therefore, because HITPA'EL functions as a true

¹³There are reasons to doubt the diagnostic utility of these two tests, see Gaftor (2014), but as the traditional probes for Hebrew unaccusativity, I rely on them here.

reflexivizer across templates, of the three template pairs in (258) under investigation here, HITPA'EL – PI'EL should be the template to realize reflexivization for its inchoatives. Additionally, if Hebrew is a semantically diverse language, it should do such to the exclusion of the other two pairs.

Therefore, in terms of the paradigm used here, semantic diversity in Hebrew with a reflexive HITPA'EL should manifest as a general preference for theme focus contexts where the theme is construed as the effector of its own event of change over agent focus contexts for the verbs in the HITPA'EL – PI'EL class, where the other two pairs do not display such a preference. Specifically, it predicts the two following patterns in the data.

Prediction 1: Between classes, HITPA'EL – PI'EL has a significantly greater difference in response rate between theme focus and agent focus contexts which is greater than the other two classes.

Prediction 2: Class internally, only HITPA'EL – PI'EL has a significant difference between the response rate for agent focus contexts and theme focus contexts such that theme focus contexts are greater.

Taken together, these three patterns in the data will demonstrate that 1) HITPA'EL – PI'EL has the greatest preference for theme focus contexts over agent focus contexts, and 2) only HITPA'EL – PI'EL prefers theme focus contexts over agent focus contexts, with no internal preferences in the other template pairs. In turn, HITPA'EL – PI'EL will demonstrate a general preference for theme focus contexts not present in the other two classes, pointing towards HITPA'EL realizing reflexivization where the other inchoatives do not. This is a semantically diverse language, with two different morphological classes of verbs realizing two different semantic alternation strategies.

5.5 Results

5.5.1 Overview

First, data from no participants had to be removed from the dataset due to poor performance on the fillers. However, all the data from one stimulus, *hityabesh* ‘dry’ in the theme focus had to be removed due to poor performance as a stimulus. It received only a 3% response rate, which is 4 standard deviations from the mean response rate across all stimuli ($\sigma = 20.0$; $\bar{x} = 82.4\%$) and 1.7 standard deviations from the stimulus with the next lowest response rate, *ne’elam* ‘disappear’ in the theme focus ($\bar{x} = 36.7\%$). Upon reviewing the stimulus, it was clear that it failed to meet a result state entailment associated with the verb, namely that the thing drying out (in the case of this stimulus, a towel) had to be completely dry.

With that adjustment, Figure 5.2 shows the mean response rate by template class-condition pair. On first glance, HITPA’EL – PI’EL has the highest response rate for theme focus contexts compared to the next highest in PA’AL – HIF’IL (94.4% vs 88.6%). This indicates the validity of prediction 1, that HITPA’EL – PI’EL has the highest response rate for theme focus contexts compared to the other classes. However, HITPA’EL – PI’EL is not the only template pair which prefers theme focus context over agent focus contexts. PA’AL – HIF’IL also displays a preference for theme focus contexts over agent focus contexts. As such, it is possible that prediction 2 will not hold. Namely, it is possible that PA’AL – HIF’IL also realizes reflexivization as demonstrated through a general preference for theme focus contexts over agent focus contexts. NIF’AL – HIF’IL is the only class among the three with a preference for agent focus contexts over theme focus contexts, and additionally has the lowest response rate for theme focus contexts (81.0%).

The data initially indicates that HITPA’EL – PI’EL does have reflexive semantics, but that PA’AL – HIF’IL does as well. These two classes pattern differently than NIF’AL – HIF’IL, which in turn would realize existential binding. This pattern would still indicate that Hebrew is semantically diverse, but would leave open the question

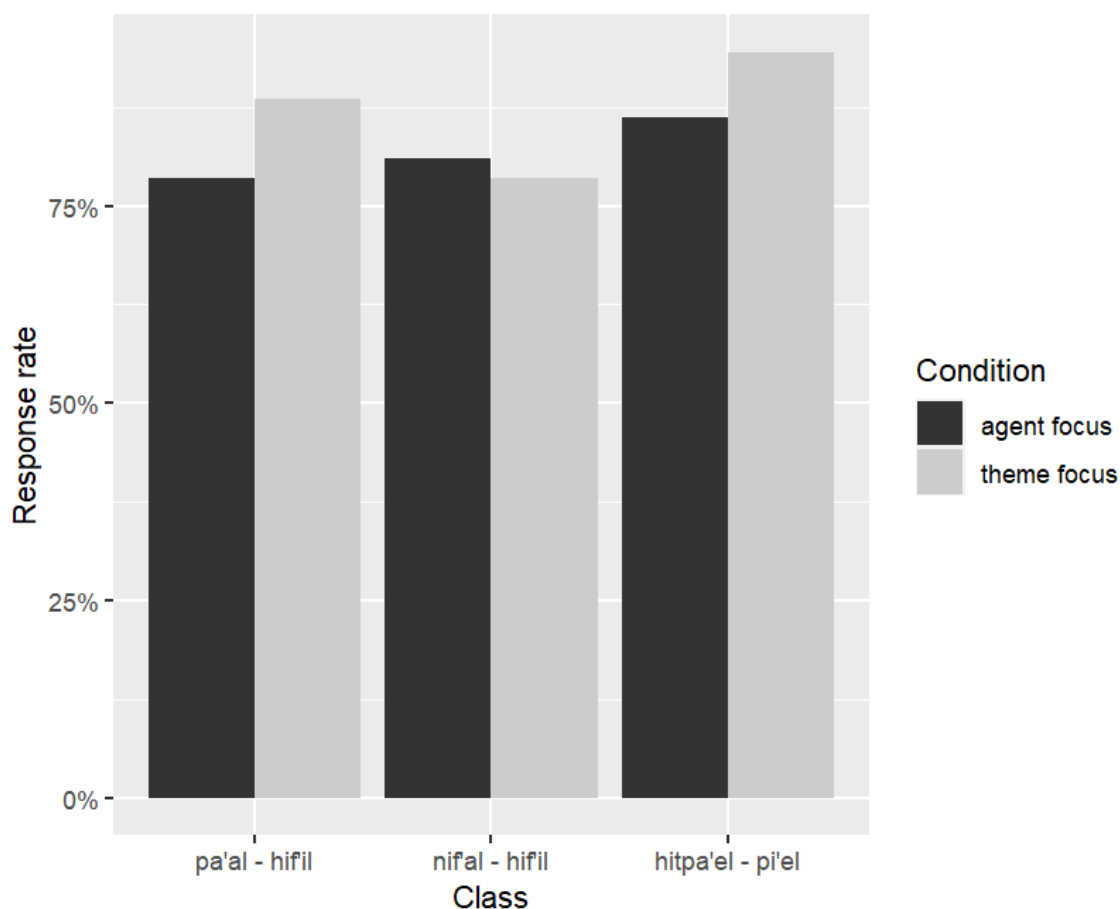


Figure 5.2: Mean response rate by class-condition pair

of why a non-reflexively marked inchoative utilizes reflexivization. To confirm the pattern demonstrated in Figure 5.2, I will walk through each of the predictions in turn, assessing the significance of each of the relevant differences.

5.5.2 Prediction 1: differences in magnitude of differences

First, prediction 1 states that HITPA'EL – PI'EL should have the largest preference for theme focus contexts over agent focus contexts when compared to the other two classes. To assess this prediction, I fit a logistic mixed effect model over the whole data set. Response was the dependent variable, and condition, class, and their inter-

action were the dependent variables. Both class and conditions were coded as factors, with both focus contexts and all three classes as levels. NIF'AL – HIF'IL was selected as the intercept for class and agent focus was selected as the intercept for condition. In addition, this model included a random intercept for both participant and verb.¹⁴

	Log odds	Error	z value	p-value
Intercept	2.29	0.55	4.17	0.00***
Theme condition	−0.18	0.30	−0.59	0.55
PA'AL – HIF'IL class	−0.03	0.73	−0.04	0.97
HITPA'EL – PI'EL class	0.14	0.72	0.19	0.85
Theme condition : PA'AL – HIF'IL class	0.82	0.47	1.75	0.08
Theme condition : HITPA'EL – PI'EL class	1.43	0.52	2.73	0.01**

Table 5.2: Effects of class and condition on response rate

Table 5.2 shows the result of this regression. The only significant effect is that of the interaction between the theme focus condition and the HITPA'EL – PI'EL class ($p > 0.05$), which is positive (log odds > 0). This in turn indicates that, compared to the intercept class of NIF'AL – HIF'IL, HITPA'EL – PI'EL has a greater difference in response rate between theme and agent focus contexts. In addition, the by-condition difference in PA'AL – HIF'IL as compared to the intercept is not significant, despite PA'AL – HIF'IL having a higher response rate for theme focus contexts than agent focus contexts. Therefore, this regression confirms prediction 1, demonstrating that HITPA'EL – PI'EL has a significantly higher preference for theme focus contexts over agent focus contexts compared to the other two classes.

¹⁴This random effect structure was selected by a process of model selection through a series of chi-squared goodness of fit tests comparing all of the models which converged (Bates et al. 2015). In addition, this random effect structure represents reasonable assumptions about the data. In particular, the fact that neither of the random variables include a slope presumes that both the participants, as speakers of the same language, and the classes, as independent morphological objects, are internally homogeneous with respect to the task. Yet, random intercepts for both control for individual differences between each of the participants and verbs.

5.5.3 Prediction 2: class internal differences by condition

Given that prediction 1 holds for HITPA'EL – PI'EL, prediction 2 anticipates that the preference for theme focus contexts over agent focus contexts within HITPA'EL – PI'EL is significant, and no other class shows such a significant preference. Clearly, as the data shows in Figure 5.2, HITPA'EL – PI'EL does have a higher preference for theme focus contexts over agent focus contexts. However, so does PA'AL – HIF'IL, though Table 5.2 shows that that difference is not significant.

	Log odds	Error	z value	p-value
PA'AL – HIF'IL class	0.6470	0.3495	1.851	0.0641
NIF'AL - HIF'IL class	–0.1766	0.2990	–0.591	0.5547
HITPA'EL – PI'EL class	1.2506	0.4341	2.881	0.004***

Table 5.3: Class-internal differences in response rate by focus condition

As such, I computed the pairwise estimated marginal means from the regression model in Table 5.2. Class was the grouping variable, and differences in condition were computed for each class. A positive log odds difference (log odds > 0) indicate a greater preference for theme focus contexts over agent focus contexts. Table 5.3 shows the by-condition contrast for each verb class. In this analysis, the only class-internal difference which is significant ($p > 0.05$) is that within HITPA'EL – PI'EL. That difference is positive, indicating that the difference demonstrates a preference for theme focus contexts over agent focus contexts. Importantly, the same positive difference within PA'AL – HIF'IL is not significant. Therefore, prediction 2 is also confirmed, and HITPA'EL – PI'EL is the only class which displays a significant preference for theme focus contexts over agent focus contexts.

5.5.4 Discussion

The results indicate that, among the three template pairs realizing the causative-inchoative alternation considered in this study, only one, HITPA'EL – PI'EL, patterns as if it realizes reflexivization. Only HITPA'EL – PI'EL conditioned a significant dif-

ference between accepting theme focus and agent focus contexts, and that difference was significantly larger than the differences in the other classes. Therefore, HITPA'EL – PI'EL realizes reflexivization where the other two classes do not. Lacking evidence for reflexivization indicates that they realize decausativization. Thus, Hebrew does utilize two different strategies in the causative-inchoative alternation, reflexivization for HITPA'EL – PI'EL, and existential binding for PA'AL – HIF'IL and NIF'AL – HIF'IL.

As such, Hebrew is a semantically diverse language, utilizing both reflexivization and existential binding depending on the morphological alternation strategy applied to the verb. These results additionally confirm the findings of Amaral et al. (2023) for Brazilian Portuguese, where they demonstrated a similar pattern for Brazilian Portuguese *se*-marked inchoatives as utilizing reflexivization where unmarked inchoatives did not. Thus, the results of the three studies considered together imply a typology of languages relating the morphology of the causative-inchoative alternation to its semantics:

- (267)
- a. English-type languages: regardless of marking type, the inchoative is de-causal.
 - b. Norwegian-type languages: regardless of marking type, the inchoative is reflexive.
 - c. Hebrew-type languages: the marking determines the semantic form of the inchoative.

Importantly, (267) proposes that the morphology of a given verb form determines the semantics, i.e. knowing the morphological class of an alternating verb means knowing the semantic relationship between the two alternants. Having established the semantic diversity of Hebrew, I turn now to how that semantic diversity informs how the meaning of the templates which form causatives and inchoatives relate to one another.

5.6 Analysis

Hebrew is a semantically diverse language. Its inchoatives reflect both existential binding and reflexivization, and the difference is guided by the morphology of the verb. Specifically, HITPA'EL inchoatives have reflexive semantics, with the covert agent coidentified with the theme. This follows from its preference of contexts where the theme can reasonably be construed as the agent of its own change, demonstrated in (268).¹⁵

- (268) *ha=magevet hitkapl-a*
 the=towel REFL.fold-3SG.F.PST
 'The towel folded.'

✓ A person drops a towel over the top of a chair, which in turn folds over the top of the chair.

#/? A person intentionally and delicately folds a towel in fourths.

To contrast, PA'AL and NIF'AL inchoatives are formed by existential binding, and as such are felicitous in contexts both where the theme is feasibly the agent of its own change, as well as those where there is a salient external agent:

- (269) *ha=tmuna nif'an-a* *'al ha=kir*
 the=picture MID.lean-3SG.F.PST on the=wall
 'The picture leaned on the wall.'

✓ A person is holding a picture upright, and nonchalantly lets it go, where it falls over and hits the wall, leaning up against it.

✓ A person needs to set down a large picture, and slowly lets the picture down until it is leaning up against the wall.

¹⁵As with the idealized data presented in section 5.3, the judgments presented here indicate a preference across participants' responses.

Hebrew therefore has a language-internal difference in how it realizes the causative-inchoative alternation, and based on the template of the inchoative verb, the underlying semantics are different.

The question then remains of how the different meanings relate to one another, and additionally, whether the difference follows the markedness of the two templates which contrast. Based on the differences between the templates, there are two types of templatic contrasts which are relevant. First, because the inchoatives in Hebrew have two separate meanings based on the template in which they are realized, different inchoatives contrast based on their templatic membership. Second, because the causative and the inchoative are in different templates, they also contrast based on their templatic membership. In both cases, the VCMP requires that the verb in the more marked template should have a stronger meaning than the verb in the less marked template. I will address each of these contrasts in turn.

The templates which create inchoative verbs should contrast in their meaning such that the stronger of the two possible meanings, between existential binding and reflexivization, should attach to the more marked template. Here, I will focus on the relationship between NIF'AL and HITPA'EL. Though both of these are higher up on the markedness hierarchy, there still exists a markedness asymmetry between the two of them, where HITPA'EL is more marked than NIF'AL. Therefore, the reflexive semantics of the HITPA'EL inchoatives should have stronger truth conditions than the existential semantics of NIF'AL inchoatives.

Such should be the case with, for example, NIF'AL and HITPA'EL from (231), repeated below, both creating an inchoative from the root $\sqrt{\text{RTV}}$ related to wetness. Though the experiment did not include either of these verbs, the conclusions from the previous section apply equally to them, as members of the templatic classes which were probed.

- (231) a. *nirtav-ti* *b=a=gefem*
 MID.wet-1SG.PST in=the=rain

- b. *hitratav-ti* *b=a=gefem*
 REFL.wet-1SG.PST in=the=rain
 ‘I got wet in the rain.’ (constructed)

Based on the results here, these two differ in their meanings based on whether they are reflexive or not. By virtue of their templatic memberships and as demonstrated in their previous section, NIF’AL *nirtav* is existentially bound while HITPA’EL *hitratev* is reflexive. Both of these semantic forms operate over the same root $\sqrt{\text{RTV}}$ to form an existentially bound verb in NIF’AL and a reflexive verb in HITPA’EL.

Following Beavers & Koontz-Garboden (2020), I define the denotation of $\sqrt{\text{RTV}}$ as including a state-denoting predicate *wet'* representing the state of having a certain contextually defined degree of wetness.

$$(270) \quad \llbracket \sqrt{\text{RTV}} \rrbracket = \lambda x \lambda s [\text{wet}'(x, s)]$$

(based on Beavers & Koontz-Garboden 2020, p. 45, (48))

Beavers & Koontz-Garboden operate within a scalar model of change (Kennedy 1997; Hay et al. 1999; Kennedy & McNally 2005; Kennedy & Levin 2008), such that properties like wetness are measured along a scale, an ordered set of degrees representing the amount to which that state holds of a given entity. Change thus is scalar change, that is, an increase in the degree to which that state holds of the theme. I gloss over the formal implementation of scales and scalar change here, as it is not relevant to the specific relationship between a decausal and a reflexive inchoative, which, as I will demonstrate, crucially hinges on the identity of the covert causer.

I repeat the semantic forms of inchoatives under decausativization in (241) and under reflexivization in (245) below in (271).

- (271) a. **Causative:** $\lambda Q \lambda x \lambda y \lambda e \lambda e' \exists s [\text{effector}'(y, e') \wedge \text{cause}'(e', e) \wedge \text{become}'(e, s) \wedge Q(s, x)]$
 b. **Existential binding:** $\lambda Q \lambda x \lambda e \lambda e' \exists y \exists s [\text{effector}'(y, e') \wedge \text{cause}'(e', e) \wedge \text{become}'(e, s) \wedge Q(s, x)]$

- c. **Reflexivization:** $\lambda Q \lambda x \lambda e \lambda e' \exists s [effector'(x, e') \wedge cause'(e', e) \wedge become'(e, s) \wedge Q(s, x)]$

In the process of verb formation, the predicate represented by the root saturates the open predicate argument Q . In the case of decausativization, it forms *nirtav*, the NIF'AL verb from (254a). In the case of reflexivization, it forms *hitratev*, the HITPA'EL verb from (231b). I present these forms in (272).

- (272) a. $\llbracket nirtav \rrbracket = \lambda x \lambda e \lambda e' \exists y \exists s [effector'(y, e') \wedge cause'(e', e) \wedge become'(e, s) \wedge wet'(s, x)]$
 b. $\llbracket hitratev \rrbracket = \lambda x \lambda e \lambda e' \exists s [effector'(x, e') \wedge cause'(e', e) \wedge become'(e, s) \wedge wet'(s, x)]$

Based on the denotations in (272), the reflexive *hitratev* entails the decausal *nirtav*. Specifically, *hitratev*, being reflexive, has a specified causer, namely the theme. *nirtav*, being decausal, does not specify for a causer, but instead has that causer existentially bound. In essence, *nirtav* is compatible with any causer, and is true as long as it has something causing the change in wetness. Because existential *nirtav* is satisfied with any causer, it is equally true in the same contexts which makes the reflexive *hitratev* true, i.e. when the causer is identical to the theme. However, reflexive *hitratev* is only true when the causer is the same as the theme, and if there is a different causer, it is not true. Therefore, because *nirtav* is true in all of the cases where *hitratev* is true, *hitratev* entails *nirtav*.

Namely, the relationship between *hitratev* and *nirtav* follows the relationship between existential binding and reflexivization in (247), with the entailments associated with a form derived by existential binding being a subset of those associated with reflexivization:

$$(273) \quad \left\{ \exists x \exists e \overset{nirtav}{[effector'(e, x)]} \right\} <_s \left\{ \overset{hitratev}{\exists x \exists e [effector'(e, x)]} \right\}$$

Accordingly, *hitratev* is strictly stronger than *nirtav*. These two verbs satisfy the VCMP, as *hitratev* is in the more marked HITPA'EL template, and is strictly stronger than *nirtav* in the less marked NIF'AL template.

Second, the causative and the inchoative are formed in different templates, and therefore should have a difference in truth conditional strength following markedness lines between the templates. In this case, HITPA'EL – PI'EL realizes reflexivization and has the inchoative in the more marked HITPA'EL template, while PA'AL – HIF'IL realizes existential binding and has the causative in the more marked HIF'IL template. Therefore, a HITPA'EL inchoative should have stronger truth conditions than a PI'EL causative, as derived from the reflexive semantics of HITPA'EL, while a HIF'IL causative should have stronger truth conditions than a PA'AL inchoative, as derived from the existential semantics of PA'AL.

First, I repeat the semantic form of the causative from (271a) here.

(271a) **Causative:** $\lambda Q\lambda x\lambda y\lambda e\lambda e'\exists s[effector'(y, e') \wedge cause'(e', e) \wedge become'(e, s) \wedge Q(s, x)]$

The main entailment associated with the agent in this form is the existence of some agent, demonstrated by the inability to negate the existence of an agent with both the the PI'EL causative *mile* 'fill' and the HIF'IL causative *heziz* 'move' in (274).

(274) a. # *Yosi mile* ' et ha=kos b=mayim aval af exad lo
Yosi INTNS.fill[3SG.M.PST] ACC the=cup in=water but no.one not
garam l=a=kos l-hitmale' (b'xlal)
cause[3SG.M.PST] to=the=box INF-REFL.fill (at.all)
(intended) 'Yosi filled the cup with water, but no one caused the cup to
fill up (at all).'

b. # *Yosi heziz* et ha=kufsa aval af exad lo
Yosi CAUS.move[3SG.M.PST] ACC the=box but no.one not
garam l=a=kufsa la-zuz (b'xlal)
cause[3SG.M.PST] to=the=box INF-move (at.all)

(intended) ‘Yosi moved the box, but no one caused the box to move (at all).’

Starting with the PI’EL verb *mile’* as contrasting with its HITPA’EL inchoative *hitmale’*, the latter has reflexive semantics by virtue of its template, demonstrated by the felicity pattern in (275).

- (275) *ha=kos hitmal’-a b=mayim*
 the=cup REFL.fill-3SG.F.PST in=water
 ‘The cup filled up with water.’

✓ A woman holds a cup under a running faucet, and the cup fills up with water.

#/? A man picks up a jug of water and decants the water into the cup slowly.

As with *hitratev* above, the semantic form of *hitmale’* has the agent coidentified with the theme:

- (276) $\llbracket hitmale \rrbracket = \lambda x \lambda e \lambda e' \exists s [effector'(x, e') \wedge cause'(e', e) \wedge become'(e, s) \wedge full'(s, x)]$

Because the semantics of *hitmale’* coidentifies the theme with the agent, it is strictly stronger than its PI’EL correlate *mile’*, which just requires there to be some entity which is an agent. Schematically, the relationship between sets of entailments in (277) holds:

- (277) $\left\{ \exists x \exists e [\overset{mile'}{effector'}(e, x)] \right\} <_S \left\{ \overset{hitmale'}{\exists x \exists e [effector'(e, x)]} \right\}$
 $\left\{ \exists e [theme'(e, \iota x [effector'(e, x)])] \right\}$

Because *mile’* is in the less marked PI’EL template, and has a weaker meaning than *hitmale’* in the more marked HITPA’EL template, the VCMP holds.

To contrast, PA’AL inchoatives are formed through existential binding, as demonstrated by the felicity pattern in (278).

(278) *ha=kufsa zaz-a*
 the=box move-3SG.F.PST
 ‘The box moved.’

✓ A person gives a box a small nudge, and it slides across the floor.

✓ A person pushes hard on a box, sliding it across the floor.

As such, *zaz* has the semantic form in (279), with the covert causer existentially bound.¹⁶

(279) $\llbracket zaz \rrbracket = \lambda x \lambda e \lambda e' \exists y \exists s [effector'(y, e') \wedge cause'(e', e) \wedge become'(e, s) \wedge move'(s, x)]$

Unlike the case of HITPA'EL inchoatives, both the HIF'IL causative and the PA'AL inchoative only entail the existence of an agent, and so just on that basis, there is no strict strength relationship between the two of them. However, the two do differ in that the causer argument in the causative is not existentially bound, but an open argument. As such, the causative is able to take a specific causer where the inchoative is not, demonstrated by an infelicity with *'al yedey* ‘by’, which introduces a specific causer obliquely.

(280) *#ha=kufsa zaz-a* *'al yedey Yosi*
 the=box move-3SG.F.PST by Yosi
 (intended) ‘The box moved by Yosi.’

Intuitively, this restriction makes the HIF'IL causative more contentful, as the range of potential causers is wider. Where the inchoative will always have a non-specific causer, the causative can have a non-specific causer, but also could have a specific, named causer. Thus, I follow Jerro (2023) in assuming that the fact that the causer will be named in the causative constitutes an entailment which the causative has

¹⁶ *zaz* could alternatively have a non-causal denotation, and *heziz* is formed from it by the addition of causativization. In this case, HIF'IL *heziz* is very obviously strictly stronger than PA'AL *zaz*, as the former has entailments of causation which the latter does not. The process here would however be a process of causativization, not existential binding, which may be possible, but the results here do not adjudicate that. As such, I will continue to assume that *zaz* is causal.

that the inchoative does not. This accords with the intuition that by naming the causer, one will know more about it, and so the verb where the causer can be named has additional entailments associated with it, namely all those things true of the DP named as its causer. I schematize this relationship in (281).

$$(281) \quad \left\{ \begin{array}{c} \underline{zaz} \\ \text{Has an effector} \end{array} \right\} <_S \left\{ \begin{array}{c} \underline{heziz} \\ \text{Has an effector} \\ \text{Effector can be named} \end{array} \right\}$$

Additionally, *heziz* is in the more marked HIF'IL template, while *zaz* is in the less marked PA'AL template. Because the verb in the stronger template has the strictly stronger meaning, the VCMP holds here as well.

5.7 Conclusion

In this section, I demonstrated that the causative-inchoative alternation in Hebrew is semantically diverse. There are multiple different templatic means to realize the causative-inchoative alternation, and depending on the templates which form the alternating verbs, the meanings of the inchoatives are different. Specifically, HITPA'EL inchoatives have reflexive semantics, while PA'AL and NIF'AL inchoatives have de-causal semantics. Thus, because Hebrew is a semantically diverse language, even the causative-inchoative alternation reflects truth conditional differences between the templates which realize the alternation. Inchoatives in more marked templates, such as HITPA'EL, have stronger truth conditions than those in less marked templates, such as NIF'AL. Similarly, inchoatives in more marked templates have stronger truth conditions than their related causatives in less marked templates, and vice versa.

Importantly, I have demonstrated that the VCMP applies even for the most syntactic functions of the templates. One of the most canonical functions of the templates is realizing arity alternations like the causative-inchoative alternation, and that fact is one of the reasons why analytical approaches to the Hebrew template system treat them as reflexes of syntactic heads (e.g. Doron 2003; Kastner 2020) or operations

which change the syntactic properties of the verb (e.g. Laks 2007, 2013). However, the study of inchoative semantics here demonstrates that argument structure can only be a portion of the story, and even when the templates do change the arity of a verb, those arity changes also come with truth conditional strength differences. Therefore, throughout the three studies addressed here, I have shown that the Hebrew templates do have an effect on the meaning of the verbs which they form. In the final chapter, I will survey my results and give a few thoughts on how this approach to the Hebrew template system can extend into other markedness phenomena.

Chapter 6: Conclusion

This dissertation has argued that the Hebrew template system consists of a system of contrasting morphological forms representing an equivalent set of semantic contrasts. Namely, when two templates derive verbs for a single root, the verb in the template with greater morphological markedness will have a stronger meaning relative to the verb in the template with less morphological markedness. This relationship is governed by the Verb Complexity Marking Principle, a principle constraining lexical meaning based on a verb's morphological form.

- (282) **Verb Complexity Mapping Principle (informal):** For any two morphologically-realized lexical verb classes, a verb realized in the more marked class will have truth conditions which are a strict superset of a morphologically-related verb realized in the less marked class.

The templates thus exist in a markedness hierarchy with one another, such that verbs higher on the hierarchy are more marked than those lower in the hierarchy.

- (283) **Markedness hierarchy for the Hebrew templates:**

PA'AL << HIF'IL, PI'EL << NIF'AL << HITPA'EL

For this dissertation, I have set aside purely syntactic templates, like the two which form passive, as well as Multiple Contextual Meaning (MCM) alternations which contrast idiosyncratically, though I return to them below. Following the Verb Complexity Marking Principle, for two verbs formed off of the same root, the verb in the more marked template will have a strictly stronger meaning than the verb in the less marked template. This proposal stands in direct contrast to proposals for the function of the Hebrew template system which center syntax, and analyze the individual templates as an outgrowth of the argument structures which they associate with (e.g. Doron 2003; Arad 2005; Laks 2007; Kastner 2020).

However, the three case studies I have presented in this dissertation have investigated three different verb alternations and have demonstrated that in each case, the semantics primarily determines the template of each of the alternants. In chapter 3, I investigated a roaming motion alternation, where a manner of motion verb in PA'AL takes on a roaming meaning in HITPA'EL. This semantic effect comes with an according syntactic effect of restricting dative-marked goals. Following Kastner (2018, 2019), the roaming meaning in the HITPA'EL alternant should be linked to the syntactic restriction on datives. However, I demonstrate that HITPA'EL roaming verbs can take dative-marked arguments, and in particular dative-marked goals. Instead, I explain the distribution of datives with HITPA'EL roaming verbs through *antimaximality*, which it entails. Namely, HITPA'EL roaming verbs can take a dative-marked argument only if it does not represent a final goal. The addition of an entailment of antimaximality additionally makes the HITPA'EL roaming verb have a stronger meaning than its PA'AL correlate, satisfying the VCMP. This case study specifically showed that a templatic alternation which on its face constitutes a syntactic, argument-structural difference between its alternants instead is determined by a semantic difference, which in turn explains the argument-structural contrasts.

In chapter 4, I turned to how the Hebrew templates interact with the locative alternation, an alternation with both clear syntactic and semantic effects. Focusing specifically on the templatic distribution of loading verbs, I demonstrated that the more marked HIF'IL template generates a loading verb restricted to one of the two possible argument-structural frames, but only when it contrasts with a less marked loading verb in the PA'AL template. I derived this restriction from the VCMP requiring the more marked HIF'IL verb to have an entailment of quantized change for one of its arguments, where the PA'AL verb only entails non-quantized change. Thus, this alternation shows that even where there is an argument-structural effect associated with a templatic alternation, the specific templatic options are governed by the semantic contrast between their forms.

Finally, in chapter 5, I turned back to the canonical argument-structural function of the templates by investigating whether different realizations of the causative-inchoative alternation have different semantics. Using an experimental paradigm from Lundquist et al. (2016), I showed that HITPA'EL inchoatives have reflexive semantics, with the theme coidentified with while NIF'AL inchoatives have existential semantics, with the covert agent existentially bound (or else there is no agent at all). This templatic distinction also follows the VCMP, as the entailments associated with reflexivization are stronger than those associated with existential binding. The VCMP therefore applies even across primarily syntactic alternations.

Therefore, across these three case studies, I have demonstrated that the Hebrew templates do contribute regularly to the meaning of the verbs they form. Across a range of alternations reflecting a range of potential syntactic differences, the difference in meaning follows the markedness of the templates which constitute the alternation. Looking forward, I constrained my empirical base in this dissertation only to alternations which preserve the core idiosyncratic meaning of the verb. Specifically, I excluded the other main kind of templatic alternation in Hebrew, namely an MCM alternation from Arad (2005), where a difference in template marks a difference in idiosyncratic meaning:

- (284) a. *Yosi patax et ha=delet*
 Yosi open[3SG.M.PST] ACC the=door
 'Yosi opened the door.'
- b. *Yosi piteax delet xadaf l=a=fuk*
 Yosi INTNS.open[3SG.M.PST] door new to=the=market
ha=amerika-i
 the=america-ADJ
 'Yosi developed a new door for the American market.'

Arad (2005) treats the difference between the two alternants in an MCM alternation as listed and memorized, according with the fact that the two meanings are idiosyncratic.

The VCMP provides a framework for understanding how these two meanings might be related to one another. Though I formally define the VCMP as requiring the entailments of one verb to be a strict superset of the second, intuitively, the VCMP posits a principled connection between the markedness of the template of the verb and its meaning. Namely, a more marked template should have a stronger meaning. Extending this principle to MCM, the PI'EL verb *piteax* should have stronger truth conditions than the PA'AL verb *patax*. This relationship would be in spite of the fact that they do not have directly comparable truth conditions. Because their idiosyncratic meanings differ from one another, the truth conditions of one need not be part of the truth conditions of the other. That is, the concept of 'openness' is not necessarily part of the concept of 'developing'. Instead, the meaning of *piteax* might only have to have more entailments associated with it than the meaning of *patax*. For example, *piteax* might restrict the properties associated with its arguments more than *patax*, i.e. have a stronger individual thematic role for its subject or object.

However, in order to address phenomena such as idiosyncrasy, a more expansive approach to strength of meaning likely should be taken in order to capture those cases out of the scope of this dissertation. In those cases where there is not a shared idiosyncratic core, it becomes difficult to quantify the strength of a verb relative to another just in terms of its entailments. Namely, the unique entailments associated with a given verb or its thematic roles are so wide and so varied that it is difficult to measure them, and even further difficult to compare them due to their underlying complexity (cf. the argument in Grimshaw 2005). To that end, lexical entailment approaches have tended to focus on cases where this complexity can be simplified, either by defining a fixed set of entailments which can be compared (Dowty 1991; Ackerman & Moore 2001), or by relativizing the comparison to a constrained set of verbs, as I do here.

For these cases, pragmatic approaches to understanding contrasts in meaning may be more useful, especially where it becomes possible to quantify strength of

meaning in terms of informativity instead of entailments. Already, there is some indication that an approach where contrast derives specifically from competition between forms works to characterize how the templates contrast. Specifically, the locative alternation investigated in chapter 4 demonstrated a pattern which could indicate a just a preference for one holistic change over another in the contrasting HIF'IL verb, rather than the verb outright excluding one of the holistic readings. Namely, for *hit'in*, the 2.8% collocation with locatum objects was significantly different from 0, and so its large and significant preference for location objects might not indicate that the only meaning available for *hit'in* is holistic change of location, but rather that both meanings are available, and holistic change of location is only preferred when using *hit'in*. This preference in turn derives from the fact that *hit'in* is more marked than *ta'an*, and so is preferred when the more specific message (here, holistic change of location). The two forms, *ta'an* and *hit'in*, are thus in competition with one another, and *hit'in* wins when the more specific message is being conveyed.¹

This kind of analysis is in line with that from Ahdout & Kastner (2020) for agent entailments in NIF'AL nominalizations. For them, NIF'AL verbs can, in general, express an unaccusative reading, without an agent, and a passive reading, with an underlying agent, following from its status as a middle verb (see Chapter 2). Those NIF'AL verbs with an unaccusative reading are always able to be nominalized, as there are no simpler nominal forms which specifically have a structure which precludes a non-agentive interpretation. However, those with a passive reading are blocked when a simpler, non-derived noun exists; the underlying specifications are the same, and so the morphologically less complex form is preferred. As such, the VCMP could be collapsed into a competition-type analysis along these lines, if it turns out that, for a wider range of phenomena, informativity explains a templatic distinction better than lexical entailments.

Thus, in order for the VCMP to be able to capture cases like idiosyncrasy,

¹Thank you to Itamar Kastner for bringing this interpretation of the data to my attention.

it could be recontextualized to be compatible with a more pragmatic approach like the one above. For example, if the notion of semantic strength is recast into informativity, then the VCMP reduces to verbs in a more marked lexical class appearing in more informative contexts. Informativity is a metric for measuring strength which is uniquely quantifiable through statistical measures like Shannon entropy or pointwise mutual information and so can sidestep the issues with quantifying it using comparative numbers of entailments. Already, measures of informativity have been correlated with linguistic complexity, such as entropy predicting word length (Piantadosi et al. 2011; Mahowald et al. 2013) or, more relevantly, morphological complexity (Denlinger et al. 2024). The strict specificity of lexical entailments for verb meaning would emerge as a sub-case of this approach, where comparing uses of two verbs which share a core idiosyncratic meaning leads to informativity corresponding to strictly stronger or weaker meanings.

Future work could therefore expand the VCMP by investigating how these different measures of informativity correlate or predict templatic membership. However, as it is now, this dissertation represents a first foray into understanding what that connection might be, validated across a range of case studies and a variety of semantic classes of verbs. The case studies here dived into exactly how the semantics of the root and the semantics of the template interact in order to create the range of meanings a template and root can take on, thus constituting a systematic study of the effect root meaning can have on the meaning of the verb. The results from across all three case studies have also disentangled the argument structure from the morphological template, indicating that independent principles can determine both, and furthermore interact with one another. In many ways, the VCMP as argued for here constitutes an additional branch in the range of principles which constrain what a verb can mean. By centering the lexical entailments associated with each verb, I have therefore also folded verbal class marking into theories of verb meaning, thus additionally extending the utility of that approach. In addition, the case studies I surveyed here demonstrated the extent to which root classes matter in determining a

verb's meaning in Hebrew.

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